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EXTRAPOLATING SURVIVAL WITH APPLICATIONS TO HEALTH TECHNOLOGY ASSESSMENT

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Extrapolating Survival with Applications to Health Technology Assessment

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A wise old owl lived in an oak;
The more he saw the less he spoke;
The less he spoke the more he heard.
Why can't we all be like that wise old bird?

— *The Oxford Dictionary of Nursery Rhymes*,
edited by Iona and Peter Opie (1977), No. 394

Popular Science Summary

Methuselah is the longest-lived person in the Bible, recorded as having lived 969 years[1]. Today, his name is synonymous with extreme longevity. In demographic literature, for example, the "Methuselah effect"[2] refers to a type of measurement error in which some individuals appear to live impossibly long because their deaths cannot be correctly linked to official death records. My thesis focuses on how to make better predictions about how many years patients will live in order to avoid such unrealistic "Methuselah-like" results.

When researchers try to predict how long patients with a particular disease are likely to live, one useful approach is to think of the risk of dying as coming from two separate sources: (i) the baseline risk of dying that every individual has simply from being human, also known as the general population mortality rate, and (ii) the extra risk of dying from the disease itself. This approach is known as "relative survival extrapolation". Although previous studies have assessed the performance of relative survival extrapolation in estimating life expectancy, its ability to predict survival over a specified time horizon rather than lifetime horizon and its application within multistate models remain understudied.

In my thesis, I evaluated the performance of relative survival extrapolation using different parametric survival models. Using data from the Swedish Cancer Register, I found that modelling the baseline and extra risks of dying separately, rather than combining them, provides more accurate survival predictions. However, this approach made little difference when predicting survival over a 10-year horizon; its advantage became more evident in lifetime predictions.

Another major contribution of this thesis is the extension of relative survival extrapolation into a multistate framework. Unlike standard survival analysis, which typically considers two states (for example, diagnosis and death), this framework allows for multiple health states, such as diagnosis, disease progression, and death, with patients either staying or transitioning among these states over time. Specifically, this multistate approach was applied to examine the quality-adjusted life years (QALYs) of patients with chronic myeloid leukaemia (CML), a blood cancer that can now be managed well with tyrosine kinase inhibitors. QALYs combine the quantity and quality of life into a single measure. Results showed that while CML patients currently lose very few life years compared to the general population, they on average still lose more QALYs, indicating that their quality of life remains an important area for improvement.

Finally, I examined the economic impact of CML on Sweden's healthcare system in light of these updated models. The number of people living with CML is expected to nearly double between 2015 and 2030. However, treatment costs are declining, and this is expected to help reduce the overall economic burden.

[1]Biblica. Holy Bible, New International Version. 2011. Genesis 5:27: "Altogether, Methuselah lived a total of 969 years, and then he died." Available from: https://www.biblica.com/.

[2]Black et al. The Methuselah Effect: The Pernicious Impact of Unreported Deaths on Old-Age Mortality Estimates. *Demography.* 2017 Dec;54(6);2001-2024.

This thesis pushes methods forward to prevent those "Methuselah-like" problems in estimations of life years, helping scientists to predict patients' survival more accurately. It supports clinicians in giving patients estimates of their individual survival and QALYs, and guides decision-makers in allocating limited healthcare budgets for maximum effectiveness.

Abstract

Survival extrapolation is fundamental to health economic modelling, addressing the task of projecting long-term outcomes from short-term survival data. Clinical studies often have limited follow-up periods, yet accurate projections of survival curves are essential in order to estimate restricted mean survival time, life expectancy, and quality-adjusted life years for health technology assessment (HTA). These projections allow decision-makers to compare the costs and effectiveness of new and existing interventions, helping them decide whether to adopt new healthcare technologies.

To evaluate the accuracy of survival extrapolation methods, we systematically compared standard parametric models (SPMs) and flexible parametric models (FPMs) in Study I. Using data from the Swedish Cancer Register, this study compared the all-cause survival framework (ASF) with the relative survival framework (RSF) to predict survival at 10-year and lifetime horizons. Results showed that FPMs generally outperformed SPMs in predicting 10-year survival. When extrapolating to a lifetime horizon, RSF models provided more accurate predictions than ASF models, especially when using FPMs. Notably, ASF models tended to overestimate survival from limited follow-up data, whereas RSF models showed a trend toward underestimation.

In Study II, we developed a novel multistate framework integrating relative survival extrapolation with mixed time scales for cost-effectiveness analysis. We demonstrated this approach using an irreversible illness-death model applied to clinical trial data, extending and comparing it with previous cost-effectiveness analyses. This framework enables the incorporation of external long-term mortality data into survival extrapolation from short-term follow-up data within a multistate model.

The multistate framework was applied to model the natural history of chronic myeloid leukaemia (CML) treatment in Study III. We estimated life expectancy and quality-adjusted life expectancy for patients with chronic-phase CML (CP-CML) diagnosed in Sweden between 2007 and 2017, comparing these with the general population. Results showed proportional losses in life expectancy of 9–15% and in quality-adjusted life expectancy of 29–33% across age-sex strata, indicating modestly low losses in life expectancy but greater losses in quality-adjusted life expectancy among patients with CP-CML.

The natural history model from Study III was extended to a cost-of-illness study through estimation and projection of the economic burden for CML in Sweden from 2015 to 2030 in Study IV. Results showed that prevalent cases are estimated to nearly double from 1205 (95% CI: 1014–1397) in 2015 to 2120 (95% CI: 1916–2325) by 2030, total direct healthcare expenditures declined from USD 40.04 million (95% CI: 33.70–46.40) to USD 30.67 million (95% CI: 27.70–33.64). This trend indicates that decreasing treatment costs may mitigate the economic burden on the Swedish healthcare system.

In conclusion, in this thesis I have worked collaboratively to systematically evaluate and

advance survival extrapolation methods for HTA. A key methodological innovation is the integration of relative survival extrapolation with multistate modelling, combining short-term survival data with long-term external information. Together with collaborators, I applied these methods in two CML studies to demonstrate how such advanced statistical modelling can provide practical insights to support clinical understanding and inform health policy.

List of Scientific Papers

This thesis is based on the following scientific articles, which are referred to by Roman numerals throughout and presented in full at the end.

I. Enoch Yi-Tung Chen, Yuliya Leontyeva, Chia-Ni Lin, Jung-Der Wang, Mark S. Clements, and Paul W. Dickman. **Comparing Survival Extrapolation within All-Cause and Relative Survival Frameworks by Standard Parametric Models and Flexible Parametric Spline Models Using the Swedish Cancer Registry**. *Medical Decision Making.* 2024;44(3):269-282.

II. Enoch Yi-Tung Chen, Paul W. Dickman, and Mark S. Clements. **A Multistate Model Incorporating Relative Survival Extrapolation and Mixed Time Scales for Health Technology Assessment**. *PharmacoEconomics.* 2025;43:297-310.

III. Enoch Yi-Tung Chen, Torsten Dahlén, Leif Stenke, Magnus Björkholm, Shuang Hao, Paul W. Dickman, and Mark S. Clements. **Loss in Overall and Quality-Adjusted Life Expectancy for Patients with Chronic-Phase Chronic Myeloid Leukemia**. *European Journal of Haematology.* 2025;114(2):334–342.

IV. Enoch Yi-Tung Chen, Paul W. Dickman, Fabrizio Di Mari, Torsten Dahlén, Leif Stenke, Magnus Björkholm, Mark S. Clements, and Shuang Hao. **Empirical and Projected Economic Burden of Chronic Myeloid Leukaemia in Sweden from 2015 to 2030: a Population-Based Study**. *Submitted manuscript.*

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List of Abbreviations

1L TKI first-line TKI

2L TKI second-line TKI

3L+ TKI third-line or later TKI

ABL1 Abelson murine leukaemia 1

alloSCT allogeneic stem cell transplantation

AP accelerated phase

APC age-period-cohort

ASF all-cause survival framework

ATC Anatomical Therapeutic Chemical

BCR breakpoint cluster region

BP blastic phase

CDF cumulative distribution function

CEA cost-effectiveness analysis

CEAC cost-effectiveness acceptability curve

CI confidence intervals

CML chronic myeloid leukaemia

COI cost of illness

CP-CML chronic-phase chronic myeloid leukaemia

CUA cost-utility analysis

EEA European Economic Area

EQ-5D EuroQol 5 dimensions

EU European Union

FC fludarabine and cyclophosphamide

FPM flexible parametric model

HEOR health economic and outcomes research

HMD Human Mortality Database

HRQoL health-related quality of life

HSUV health state utility value

HTA health technology assessment

HUI3 Health Utilities Index Mark 3

ICD International Classification of Diseases

ICER incremental cost-effectiveness ratio

K-M Kaplan-Meier

LE life expectancy

LLE loss in life expectancy

LQALE loss in quality-adjusted life expectancy

NICE National Institute for Health and Care Excellence

NMB net monetary benefit

PDF probability density function

PH proportional hazards

PIAMOD Prevalence, Incidence, Analysis MODel

PLLE proportion of loss in life expectancy

PLQALE proportion of loss in quality-adjusted life expectancy

PO proportional odds

PSA probabilistic sensitivity analysis

QALE quality-adjusted life expectancy

QALY quality-adjusted life year

RFC rituximab plus fludarabine and cyclophosphamide

RMST restricted mean survival time

RSF relative survival framework

SE standard error

SEER Surveillance, Epidemiology, and End Results

SF-6D Short Form 6 dimensions

SPM standard parametric model

TFR treatment-free remission

TKI tyrosine kinase inhibitor

TLV Dental and Pharmaceutical Benefits Agency

WTP willingness-to-pay

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Chapter 1 Introduction

Survival extrapolation is a cornerstone of health economic modelling. Although clinical studies, particularly clinical trials, are often limited to short follow-up periods, projecting restricted mean survival time (RMST), life expectancy (LE), and quality-adjusted life year (QALY) is often required for economic evaluation. In health technology assessment (HTA), survival extrapolation is used to estimate the long-term effectiveness of new versus standard interventions to assess whether a health technology represents good value for money.

Other measures such as incidence, prevalence, and mortality rates may also be needed for monitoring disease dynamics and planning for resource allocation. These projections are used to estimate future disease burden on both society and the healthcare system. When predicting the economic burden, they serve as the foundation: if they are inaccurate, any cost estimates built on them will also be unreliable. They help answer key questions in health economics, such as: *How much will it cost?*

Prediction is, by nature, full of uncertainty. Yet, decisions in HTA cannot wait for lifetime data to become available; they must be made based on the best possible projections from current data. The inherent challenge, therefore, is not to eliminate uncertainty entirely, which is an impossible task, but to manage it through robust and transparent methods built upon reasonable assumptions.

This thesis contributes to advancing this field by evaluating survival extrapolation methods and extending HTA modelling approaches by incorporating relative survival extrapolation. The applications focus on estimating LE and quality-adjusted life expectancy (QALE) for patients with chronic myeloid leukaemia (CML) and comparing these to those of the general population, as well as projecting the economic burden of the disease in Sweden from 2015 to 2030.

The following chapters of this thesis are structured as follows: Chapter 2 introduces the methodological foundations for the approaches used in Studies I to IV. Chapter 3 outlines the main research aims. Chapter 4 describes the data sources, including Sweden's national registers, the Human Mortality Database, clinical trials, published literature, and official healthcare price lists. Chapter 5 discusses ethical considerations and lists the approvals that covered the studies. Chapter 6 describes the methodological contributions, including an assessment of survival extrapolation, extensions of multistate models to incorporate relative survival and mixed time scales, and the development of a natural history model for CML treatment. This chapter also defines all key quantities estimated in the studies. Chapter 7 provides an overview of the four studies, presented one by one. Chapter 8 summarises the main findings, and Chapter 9 offers suggestions for future research.

Chapter 2 Background

2.1 Survival analysis

2.1.1 Survival, hazard, cumulative hazard functions

Survival analysis comprises statistical methods for modelling the distribution of a non-negative random variable T that records the time until the occurrence of one or more events of interest [1]. Inference is often based on three inter-related functions:

1. Survival function, $S(t)$: the probability of remaining event-free up to time t;
2. Hazard function, $h(t)$: the instantaneous rate at time t;
3. Cumulative hazard function, $H(t)$: the cumulative hazard at time t, defined as the integral of the hazard function from time 0 to t.

The survival function, $S(t)$, gives the probability of surviving up to time t, $P(T > t)$. $S(t)$ is a monotone non-increasing function, defined as:

$$S(t) = P(T > t) = 1 - F(t), \tag{2.1}$$

where $F(t)$ is the cumulative distribution function (CDF), which is the integral of the probability density function (PDF), $f(t)$, to time t:

$$F(t) = P(T \leq t) = \int_0^t f(u)\, du. \tag{2.2}$$

The hazard function, $h(t)$, is defined as the instantaneous rate of an event at time t given surviving up to time t. It is a non-negative function and has units of events per person-time. When the event time T follows a distribution that is not absolutely continuous (for example, discrete time), the hazard function can be expressed using the left-hand limit of the survival function:

$$\begin{aligned} h(t) &= \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} \\ &= \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t)}{\Delta t} \cdot \frac{1}{S(t-)} \\ &= \lim_{\Delta t \to 0} \frac{F(t + \Delta t) - F(t)}{\Delta t} \cdot \frac{1}{S(t-)} \\ &= \frac{f(t)}{S(t-)}, \end{aligned} \tag{2.3}$$

where the left-hand limit of the survival function at time t is defined as: $S(t-) = \lim_{\varepsilon \to 0^+} S(t-\varepsilon)$. If T is absolutely continuous, then $P(T \geq t) = P(T > t)$, and the survival function is continuous so that $S(t) = S(t-)$. In such cases, the hazard function can be written as:

$$h(t) \;=\; \frac{f(t)}{S(t)} \;=\; -\frac{d}{dt} \log S(t). \tag{2.4}$$

The cumulative hazard function, $H(t)$, is defined as the accumulated hazard up to time t. It is a monotonically non-decreasing function, defined by:

$$H(t) = \int_0^t h(u)du. \tag{2.5}$$

In summary, the survival, cumulative distribution, hazard, and cumulative hazard functions are inter-related as follows:

$$S(t) = 1 - F(t) = \exp[-H(t)] = \exp[-\int_0^t h(u)du]. \tag{2.6}$$

Summary measures of the survival function such as the RMST and the LE also provide clinically interpretable quantities related to survival. The RMST up to a finite time point τ, denoted $\mu(\tau)$, represents the mean survival time within the interval $[0, \tau]$, defined as:

$$\mu(\tau) = \int_0^\tau S(u)\, du.$$

LE represents as the mean survival time an individual is expected to live. LE is a case of RMST as the horizon τ extends to infinity, defined by:

$$LE = \int_0^\infty S(u)\, du. \tag{2.7}$$

In practice, the upper limit ∞ is replaced by $t_{\max}$, the time at which $S(t)$ reaches zero (that is, when the entire study population has died). Because most studies do not follow their participants until $S(t) = 0$, the estimation of LE generally requires extrapolating survival beyond the observed data.

Both RMST and LE can be generalised to account for delayed entry at time t_0. For an individual who has survived to t_0, the conditional RMST from t_0 to τ is defined as:

$$\mu(\tau \mid t_0) = \int_{t_0}^\tau \frac{S(u)}{S(t_0)}\, du,$$

and the conditional LE is given by:

$$LE \mid t_0 = \int_{t_0}^\infty \frac{S(u)}{S(t_0)}\, du,$$

where $\frac{S(u)}{S(t_0)}$ denotes the probability of survival to time u, conditional on survival up to t_0. Such generalisations are often relevant in many clinical studies, as they provide RMST and LE estimates for individuals who have survived up to a specific age or time point, such as estimating LE given surviving to age 50 years for cancer patients.

Figure 2.1 illustrates $h(t)$, $H(t)$, $S(t)$, $\mu(\tau)$, and LE; see also Table 2.3 for a comparison of these quantities.

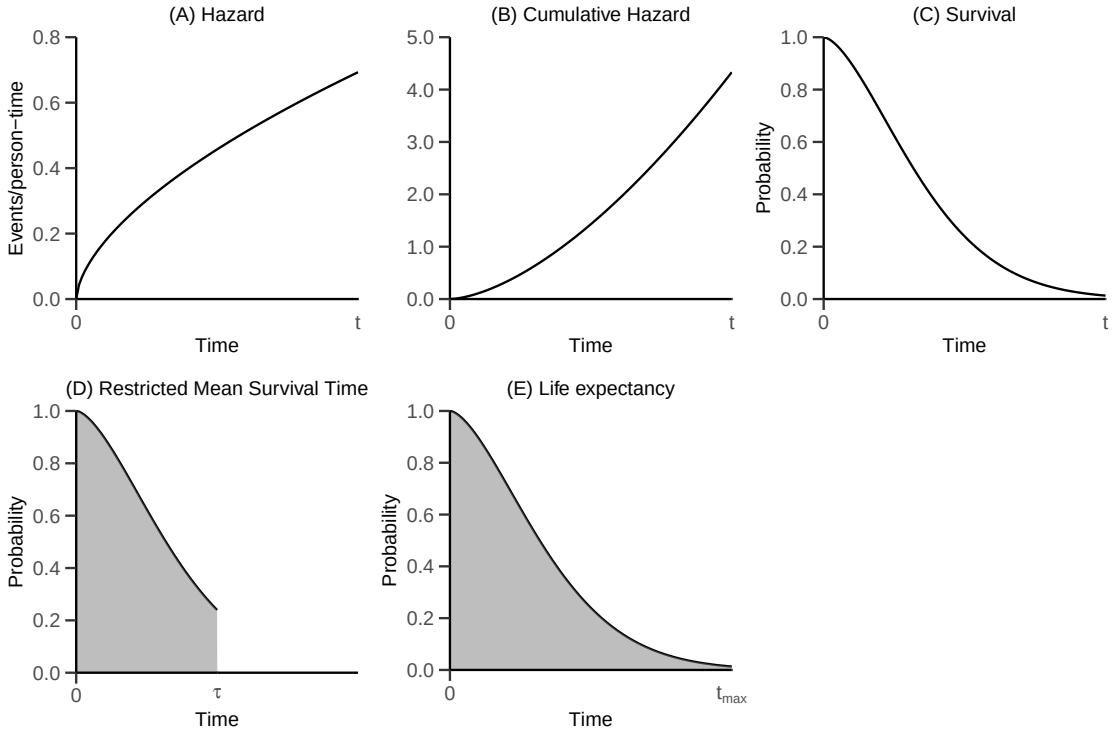


Figure 2.1. Examples of (A) hazard function, $h(t)$, (B) cumulative hazard function, $H(t)$, (C) survival function, $S(t)$, (D) restricted mean survival time, $\mu(t)$, and (E) life expectancy, LE. Survival data were simulated using a Weibull distribution with shape parameter 1.6 and scale parameter 4.

2.1.2 Censoring and truncation

Censoring and truncation are key features that distinguish survival analysis from other statistical methods [1, 2]. Censoring affects whether an individual's event time is fully observed or not, while truncation determines whether an individual is included in the study at all.

Censoring refers to situations where the exact event time is not observed for some individuals, so the true event time is only partially known. Let T_i denote the true event time and C_i the censoring time for an individual i. Three common types of censoring include:

- **Right-censoring**: The event has not occurred by the end of the follow-up period, at time C_i. We only know that $T_i > C_i$. This typically includes administrative censoring at the study end or loss to follow-up.
- **Left-censoring**: The event has already occurred before the start of follow-up at time C_i, but the true event time T_i is unknown. Hence, we only observe $T_i < C_i$. For example, in studies where cancer onset is the event of interest, only the diagnosis time is typically recorded and treated as a proxy for the onset time. In such cases, the true onset time is left-censored because it occurred before the recorded diagnosis but at an unknown time [3].
- **Interval-censoring**: The event is known to have occurred within an interval $(L_i, R_i]$, so the true event time T_i falls in the time interval: $L_i < T_i \leq R_i$. This occurs, for

example, in settings with periodic clinical assessments, such as when a diagnosis is made at time R_i, but the actual onset of disease occurred sometime after the last assessment at time L_i.

Truncation refers to situations where individuals are only included in the study if their event or observation time falls within a certain range.

- **Left-truncation**: An individual is included in the study only if they survive beyond a truncation time U_i; in other words, the individual's observed event time is conditional on $T_i > U_i$. For example, in studies using age as the time scale, individuals enter the risk set only after attaining a specified age. This is also known as delayed entry time, as individuals must survive at least to age U_i to be included in the study.
- **Right-truncation**: An individual is included only if their event time is less than a truncation time V_i; that is, we observe individuals conditional on $T_i < V_i$. This is less common but can occur in retrospective studies where only individuals who experienced the event before a certain time are included.

Figure 2.2 provides an illustration of each type of censoring and truncation.

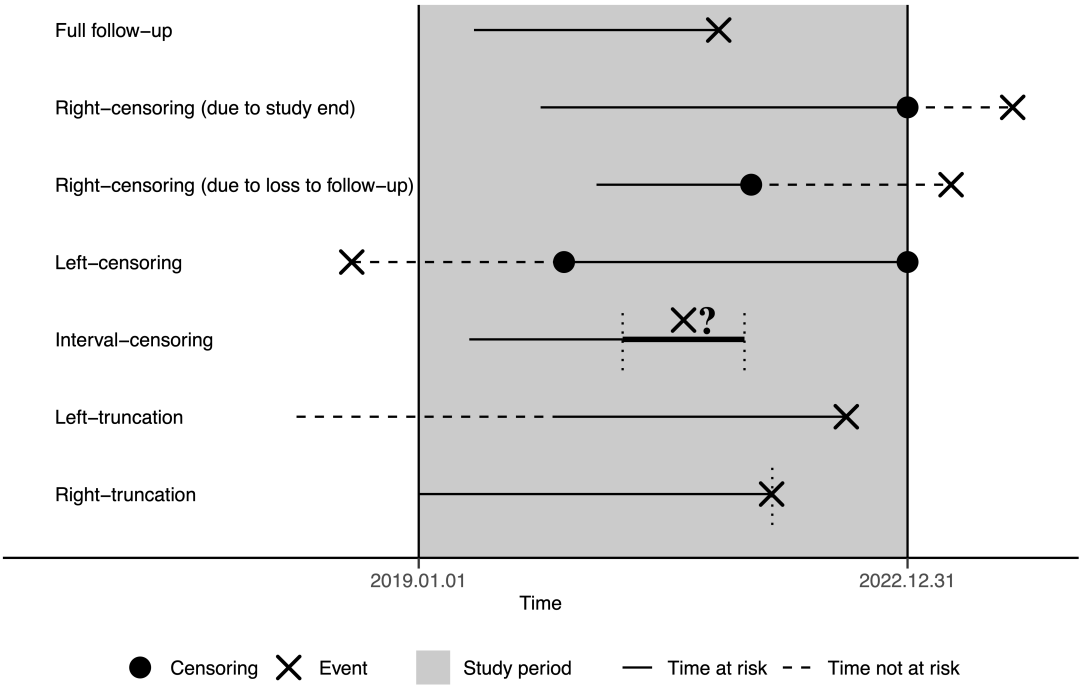


Figure 2.2. Illustration of censoring and truncation. Each horizontal line represents one hypothetical individual. Solid dots represent censoring; crosses represent event. The grey area is the study period starting from January 1, 2019 to December 31, 2022. Solid horizontal lines indicate time at risk. Dashed horizontal lines indicate time not at risk.

Throughout this thesis, I only consider right-censoring and left-truncation. Right-censoring in the studies is treated as non-informative, meaning that the censoring mechanism is independent of the event of interest. Left-truncation (delayed entry) is assumed independent

of the future event time, conditional on survival to the entry age. This delayed entry time is essential when age and/or calendar time is the time scale: for example, in Studies I–V, expected hazards were estimated using individuals' attained age and calendar year.

2.1.3 Estimation

In survival analysis, survival models can be estimated non-parametrically, semi-parametrically, and parametrically. Non-parametric approaches, such as the Kaplan-Meier (K-M) estimator [4] for survival functions, do not require specific assumptions on the distribution of the survival times. Semi-parametric approaches include both an unspecified function and a parametric component. For example, the Cox proportional hazards (PH) model does not specify the baseline hazard function but estimates hazard ratios for covariates. Parametric approaches require specifying the underlying distribution of the survival times. This PhD thesis focuses heavily on parametric models that are used to fit survival data using the maximum likelihood method [5]. Below, I will introduce the likelihood function both with and without considering censoring for survival times. Additionally, I will introduce two parametric model families: standard parametric models (SPMs) and flexible parametric models (FPMs). Variances (and hence standard errors and confidence intervals (CI)) for the estimated parameters are obtained using the Delta method [5].

2.1.4 Likelihood

Without considering censoring, let T be a random variable with density $f_\theta(\cdot)$, where θ denotes the unknown parameter(s). Let $t_1, \ldots, t_n$ denote the observed (uncensored) event times for n individuals. We assume the event times are independent and identically distributed:

$$T_i \sim f_\theta(t), \qquad i = 1, \ldots, n.$$

The contribution to the likelihood from each uncensored observation is $f_\theta(t_i)$. The likelihood function, $L(\theta \mid t)$, is defined as follows:

$$\begin{aligned} L(\theta \mid t) &= \prod_{i=1}^{n} f_\theta(t_i) \\ &= \prod_{i=1}^{n} h_\theta(t_i)\, S_\theta(t_i). \end{aligned}$$

Denote the logarithm of $L(\theta \mid t)$, called the log-likelihood function, $\log L(\theta \mid t)$, by

$$\begin{aligned} \log L(\theta \mid t) &= \sum_{i=1}^{n} \log f_\theta(t_i) \\ &= \sum_{i=1}^{n} \Big[\log h_\theta(t_i) + \log S_\theta(t_i) \Big]. \end{aligned}$$

The parameter(s), θ, is (are) estimated by maximising the log-likelihood function, yielding the estimate(s) $\hat{\theta}$.

In practice, survival data often contain right-censored observations, since not every individual experiences the event of interest before the study ends. As a result, right-censoring must be taken into account in the likelihood function. To consider this issue, an event indicator is defined as:

$$\delta_i = \begin{cases} 1, & \text{if the event is observed at } t_i, \\ 0, & \text{if the observation is right-censored at } t_i, \end{cases} \qquad i = 1, \ldots, n.$$

Each individual then contributes $f_\theta(t_i)^{\delta_i} S_\theta(t_i)^{1-\delta_i}$ to the likelihood function, giving

$$\begin{aligned} L(\theta \mid t, \delta) &= \prod_{i=1}^{n} f_\theta(t_i)^{\delta_i} S_\theta(t_i)^{1-\delta_i} \\ &= \prod_{i=1}^{n} \left[\frac{f_\theta(t_i)}{S_\theta(t_i)}\right]^{\delta_i} S_\theta(t_i) \\ &= \prod_{i=1}^{n} h_\theta(t_i)^{\delta_i}\, S_\theta(t_i). \end{aligned}$$

Hence, the log-likelihood function becomes

$$\log L(\theta \mid t, \delta) = \sum_{i=1}^{n} \Big[\delta_i \log h_\theta(t_i) + \log S_\theta(t_i)\Big], \tag{2.8}$$

and $\hat{\theta}$ is still obtained by maximising Equation(2.8).

2.1.5 Standard parametric models

Standard parametric models specify the probability distribution for the continuous survival time T, and their parameters are estimated using the maximum likelihood method. In HTA, this typically involves fitting a set of SPMs, each with a specific functional form for the survival distribution.

Following the recommendations in the National Institute for Health and Care Excellence (NICE) Technical Support Document 14 [6], six commonly used parametric models should be considered for extrapolating survival: exponential, Weibull, Gompertz, log-logistic, log-normal, and generalised gamma distributions. Table 2.1 summarises the functional forms for each model, including the probability density function, $f(t)$, cumulative distribution function, $F(t)$, hazard function, $h(t)$, and survival function, $S(t)$. The expressions and notations are primarily based on Collett (2023) [2].

Properties of hazard functions. The hazard functions of these distributions differ in shape. The exponential distribution assumes a constant hazard. The Weibull and Gompertz distributions support monotonically increasing or decreasing hazards (or a constant hazard

Table 2.1. Probability density function, $f(t)$, cumulative distribution function, $F(t)$, hazard function, $h(t)$, cumulative hazard function, $H(t)$, survival function, $S(t)$, and parameters for the exponential, Weibull, Gompertz, log-logistic, log-normal, and generalised gamma distributions.

	Exponential	**Weibull**	**Gompertz**	**Log-logistic**	**Log-normal**	**Generalised gamma**
$f(t)$	$\lambda e^{-\lambda t}$	$\lambda\gamma t^{\gamma-1}e^{-\lambda t^{\gamma}}$	$\lambda e^{\theta t}\exp\left[\frac{\lambda}{\theta}(1-e^{\theta t})\right]$	$\frac{e^{\theta}\kappa t^{\kappa-1}}{(1+e^{\theta}t^{\kappa})^2}$	$\frac{1}{\sigma\sqrt{2\pi}}t^{-1}\exp\left[-\frac{(\log t-\mu)^2}{2\sigma^2}\right]$	$\frac{\theta\lambda^{\rho}t^{\rho-1}\exp\left(-(\lambda t)^{\theta}\right)}{\Gamma(\rho/\theta)}$
$F(t)$	$1-e^{-\lambda t}$	$1-e^{-\lambda t^{\gamma}}$	$1-\exp\left[-\frac{\lambda}{\theta}(e^{\theta t}-1)\right]$	$\frac{e^{\theta}t^{\kappa}}{1+e^{\theta}t^{\kappa}}$	$\Phi\left(\frac{\log t-\mu}{\sigma}\right)$	$\frac{\Gamma_{(\lambda t)^{\theta}}(\rho/\theta)}{\Gamma(\rho/\theta)}$
$h(t)$	λ	$\lambda\gamma t^{\gamma-1}$	$\lambda e^{\theta t}$	$\frac{e^{\theta}\kappa t^{\kappa-1}}{1+e^{\theta}t^{\kappa}}$	$\frac{f(t)}{S(t)}$	$\frac{f(t)}{S(t)}$
$H(t)$	λt	λt^{γ}	$\frac{\lambda}{\theta}(e^{\theta t}-1)$	$\log\left(1+e^{\theta}t^{\kappa}\right)$	$-\log\left[1-\Phi\left[\frac{\log t-\mu}{\sigma}\right]\right]$	$-\log\left[1-\frac{\Gamma_{(\lambda t)^{\theta}}(\rho/\theta)}{\Gamma(\rho/\theta)}\right]$
$S(t)$	$e^{-\lambda t}$	$e^{-\lambda t^{\gamma}}$	$\exp\left[-\frac{\lambda}{\theta}(e^{\theta t}-1)\right]$	$\frac{1}{1+e^{\theta}t^{\kappa}}$	$1-\Phi\left(\frac{\log t-\mu}{\sigma}\right)$	$1-\frac{\Gamma_{(\lambda t)^{\theta}}(\rho/\theta)}{\Gamma(\rho/\theta)}$
Parameters	$\lambda>0$ (rate)	$\lambda>0$ (scale), $\gamma>0$ (shape)	$\lambda>0$ (scale), $\theta\in\mathbb{R}$ (shape)	$\theta\in\mathbb{R}$ (scale), $\kappa>0$ (shape)	$\mu\in\mathbb{R}$ (log-mean), $\sigma>0$ (log-SD)	$\lambda>0$, $\rho>0$, $\theta>0$ (shape)
Shapes of $h(t)$	Constant	Constant, monotonically increasing/decreasing	Constant, monotonically increasing/decreasing	Monotonically decreasing, unimodal	Unimodal	Constant, monotonically increasing/decreasing, unimodal, bathtub-shaped

For the log-normal and generalised gamma distributions, the hazard function, $h(t)$, do not have closed-form expressions in elementary functions; hence, it is expressed as $h(t)=f(t)/S(t)$. The gamma function is defined as $\Gamma(\rho)=\int_0^{\infty}x^{\rho-1}e^{-x}\,dx$, and the lower incomplete gamma function as $\Gamma_{(\lambda t)^{\theta}}(\rho)=\int_0^{(\lambda t)^{\theta}}u^{\rho-1}e^{-u}\,du$. The function $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution, given by $\Phi(x)=\int_{-\infty}^{x}\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{t^2}{2}\right)dt$. SD, standard deviation.

as a limiting case). The log-logistic distribution allows both monotonically decreasing and unimodal (arc-shaped) hazards, whereas the log-normal distribution permits only unimodal (arc-shaped) hazards. The generalised gamma distribution is flexible to generate all of these patterns as well as bathtub-shaped hazards. Moreover, the log-hazard of a Weibull model is linear in $\log t$, while the log-hazard of a Gompertz model is linear in t.

These distributions are related through special cases. The exponential distribution is a special case of the Weibull distribution when $\gamma = 1$, and of the Gompertz distribution when $\theta = 0$. The generalised gamma distribution reduces to the Weibull when $\rho = \theta$, to the exponential when $\rho = \theta = 1$, to the gamma when $\theta = 1$, and to the log-normal distribution when $\theta \to 0$.

2.1.6 Flexible parametric models

The hazard function for many diseases often shows complex patterns. For many cancer types, the hazard (mortality) rises shortly after diagnosis, falls, and may climb again later in the long-term follow-up. The SPMs described in Section 2.1.5 may not be adequately flexible to capture such behaviour. To address this limitation, FPMs, also referred to as spline models in this thesis, have been developed. A widely used class of FPMs was introduced by Royston and Parmar [7], commonly referred to as the Royston–Parmar model.

This approach fits survival data with a spline function of log time, denoted as $s(\log t \mid \boldsymbol{\gamma})$, on the following scales: log cumulative hazard, log cumulative odds, probit, or log hazard scales. Below, I summarise the link functions for the first three scales (log cumulative hazard, log cumulative odds, and probit) and provide detailed descriptions of each model in subsequent sections.

Denote $g(\cdot)$ a link function, and a covariate vector, $\boldsymbol{x} = (x_1, \ldots, x_k)$, with their corresponding coefficients, $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_k)^T$:

- Spline-based PH model:

$$\begin{aligned} g\Big[S(t \mid \boldsymbol{x})\Big] &= \log\Big[-\log S(t \mid \boldsymbol{x})\Big] \\ &= \log H(t \mid \boldsymbol{x}) \\ &= \log H_0(t) + \boldsymbol{x}\boldsymbol{\beta} \\ &= s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta}. \end{aligned} \tag{2.9}$$

- Spline-based proportional odds (PO) model:

$$\begin{aligned} g\Big[S(t \mid \boldsymbol{x})\Big] &= \text{logit}\Big[1 - S(t \mid \boldsymbol{x})\Big] \\ &= \text{logit}\Big[1 - S_0(t)\Big] + \boldsymbol{x}\boldsymbol{\beta} \\ &= s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta}, \end{aligned} \tag{2.10}$$

  where $\text{logit}\Big[1 - S_0(t)\Big] = \log\Big[\frac{1-S_0(t)}{S_0(t)}\Big]$.

- Spline-based probit model:

$$\begin{aligned} g\Big[S(t \mid \boldsymbol{x})\Big] &= -\Phi^{-1}\Big[S(t \mid \boldsymbol{x})\Big] \\ &= -\Phi^{-1}\Big[S_0(t)\Big] + \boldsymbol{x}\boldsymbol{\beta} \\ &= s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta}, \end{aligned} \tag{2.11}$$

  where $\Phi^{-1}(\cdot)$ is the inverse cumulative distribution function of the standard normal distribution [8].

Spline functions. FPMs incorporate spline functions to flexibly model the baseline function, using any of the link functions described above. Unlike the Cox PH model [9], which does not specify the baseline hazard function, FPMs explicitly model it. To motivate the FPMs, I begin by defining a cubic spline function:

$$s\left(x \mid \boldsymbol{\gamma}, \boldsymbol{\theta}\right) = \gamma_0 + \gamma_1 x + \gamma_2 x^2 + \gamma_3 x^3 + \sum_{p=1}^{K} \theta_i \left(x - k_p\right)_+^3, \tag{2.12}$$

where K is the number of knots at $k_1 < ... < k_K$, and the total number of parameters is $K + 4$. Here, the "+" notation is defined as:

$$u_+ = \begin{cases} u, & \text{if } u > 0 \\ 0, & \text{if } u \leq 0. \end{cases}$$

Different software implementations of FPMs employ distinct spline basis functions. Two common choices are **restricted cubic splines**, typically expressed in a truncated–power basis and constrained to be linear beyond the boundary knots (as in `Stata`'s `stpm2`), and **natural cubic splines**, often constructed by projecting cubic B-splines also with boundary constraints (as in `R`'s `rstpm2`).

- **Restricted cubic splines:** The `Stata` command `stpm2` implements restricted cubic splines using the basis functions defined below:

$$s\left(x \mid \boldsymbol{\gamma}\right) = \gamma_0 + \gamma_1 x + \sum_{p=2}^{K-1} \gamma_p v_p(x), \tag{2.13}$$

  where K is the number of knots and $v_p(x)$ denotes the pth basis function, defined as:

$$v_p(x) = (x - k_p)_+^3 - \frac{k_K - k_p}{k_K - k_1}(x - k_1)_+^3 - \frac{k_p - k_1}{k_K - k_1}(x - k_K)_+^3 \quad \text{for } p = 2, \ldots, K-1,$$

  where k_1 and k_K are the boundary knots, k_p the pth internal knot.
  The spline variables, $v_p(x)$, can be highly correlated, and `stpm2` inherently orthogonalises them by using Gram–Schmidt orthogonalisation [10]. Internal knots are positioned at empirical percentiles of the distribution of uncensored log event times [10],

while boundary knots k_1 and k_K are placed at the minimum and maximum uncensored log event times [8, 11]. For example, with five knots (three internal plus two boundary knots), the internal knots are positioned at the 25th, 50th, and 75th percentiles, and the boundary knots are at the 0th and 100th percentiles of uncensored log event times. The spline function is constrained to be linear outside the boundary knots (that is, before the first knot and after the last knot).

- **Natural cubic splines:** The `R` package `rstpm2` [12] models time using a natural cubic spline basis, constructed from B-splines [13, 14]. Let $B_{i,k}(x)$ denote the ith B-spline basis function of degree k. Using the recursive formula [15], B-splines are defined as:

$$B_{i,0}(x) = \begin{cases} 1, & \text{for } t_i \le x < t_{i+1}, \\ 0, & \text{otherwise,} \end{cases}$$
$$B_{i,k}(x) = \frac{x - t_i}{t_{i+k} - t_i} B_{i,k-1}(x) + \frac{t_{i+k+1} - x}{t_{i+k+1} - t_{i+1}} B_{i+1,k-1}(x),$$

where $t_1 < t_2 < \cdots < t_k$ are the ordered k knots. In `rstpm2`, linearity is imposed beyond the boundary knots. These constraints are implemented by projecting the spline basis onto the null space of the constraint matrix. When `stpm2` in `Stata` and `rstpm2` in `R` use identical knot locations, the resulting fitted curves and parameter estimates are expected to be nearly identical.

In this thesis, I used `stpm2` with **restricted cubic splines** for FPMs in Study I and `rstpm2` with **natural cubic splines** in Studies II–IV.

Spline-based proportional-hazards models. Among the three modelling scales, the PH scale is the most commonly used in FPMs. Following Royston and Lambert's definitions [8], the Weibull distribution is a special case among SPMs (Table 2.1) where the log cumulative hazard is a linear function of log time:

$$\log H_0(t) = \log \lambda + \gamma \log t. \tag{2.14}$$

This function implies a log-linear relationship between the baseline cumulative hazard and time on the log–log scale, allowing the hazard to increase or decrease monotonically over time depending on the value of γ, the shape parameter. Now I include a covariate vector, $\boldsymbol{x}$, with their coefficients, $\boldsymbol{\beta}$, into Equation (2.14):

$$\begin{aligned} g\big[S(t \mid \boldsymbol{x})\big] &= \log H(t \mid \boldsymbol{x}) \\ &= \log H_0(t) + \boldsymbol{x}\boldsymbol{\beta} \\ &= \log \lambda + \gamma \log t + \boldsymbol{x}\boldsymbol{\beta}. \end{aligned} \tag{2.15}$$

The specification in Equation (2.15) implies a PH form, as $\boldsymbol{x}\boldsymbol{\beta}$ does not depend on t. By differentiating the cumulative hazard in Equation (2.15) with respect to t, the hazard

function is then obtained:

$$\begin{aligned} h(t \mid \boldsymbol{x}) &= \frac{\partial}{\partial t} H(t \mid \boldsymbol{x}) \\ &= \frac{\partial}{\partial t}\Big[\lambda t^{\gamma} \exp(\boldsymbol{x}\boldsymbol{\beta})\Big] \\ &= \lambda\gamma\, t^{\gamma-1} \exp(\boldsymbol{x}\boldsymbol{\beta}) \\ &= h_0(t) \exp(\boldsymbol{x}\boldsymbol{\beta}), \end{aligned} \tag{2.16}$$

where $h_0(t)$ is the baseline hazard function with a Weibull distribution.

To introduce more flexibility into Equation (2.15), the linear term $\gamma \log t$ can be generalised using a spline function of log time. This allows the log cumulative hazard to follow a spline-modelled shape rather than a pre-specified parametric form, shown as:

$$\log H(t \mid \boldsymbol{x}) = s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta}, \tag{2.17}$$

where $s(\log t \mid \boldsymbol{\gamma})$ denotes the spline function of log time with a vector of parameters $\boldsymbol{\gamma}$. It is important to note that this formulation remains a PH model, although the baseline hazard is now modelled using a spline function. Beyond the final knot, the spline is constrained to be linear in log time, causing the model to behave asymptotically like a Weibull distribution on the hazard scale [16], which can be a useful property for extrapolating survival.

To accommodate non-proportional hazards, time-dependent effects can be introduced by including interactions between covariates and functions of time. In the FPM framework, this is achieved by extending Equation (2.17) as follows:

$$\log H(t \mid \boldsymbol{x}) = s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_i)\boldsymbol{x}_i \tag{2.18}$$

where D is the number of covariates with time-dependent effects, and $s(\log t \mid \boldsymbol{\gamma}_i)$ represents the spline function for the ith time-dependent effect. This extension results in a non-proportional hazards model that captures how covariate effects change over time.

Spline-based proportional-odds models. To relax the PH assumption while retaining spline-based baseline functions, one may apply alternative link functions. These include the PO and probit scales, as introduced by Royston and Lambert [8].

Among these two scales, the PO model is a natural alternative to the PH model. It assumes PO of failure rather than PH. The log-logistic distribution is a common parametric example of a PO model [7], in which the log cumulative odds of failure is linear in $\log t$. This motivates modelling the survival function using a logit link on the PO scale.

The PO assumption states that the odds of failure at time t, conditional on a vector of covariates $\boldsymbol{x}$, is proportional to the baseline odds of failure:

$$\frac{1 - S(t \mid \boldsymbol{x})}{S(t \mid \boldsymbol{x})} = \frac{1 - S_0(t)}{S_0(t)} \cdot \exp(\boldsymbol{x}\boldsymbol{\beta}).$$

Taking the logarithm of both sides yields the log cumulative odds with covariates:

$$\begin{aligned} \log\Big[\frac{1-S(t \mid \boldsymbol{x})}{S(t \mid \boldsymbol{x})}\Big] &= \log\Big[\frac{1-S_0(t)}{S_0(t)} \cdot \exp(\boldsymbol{x}\boldsymbol{\beta})\Big] \\ \implies \operatorname{logit}\Big[1-S(t \mid \boldsymbol{x})\Big] &= \operatorname{logit}\Big[1-S_0(t)\Big] + \boldsymbol{x}\boldsymbol{\beta}, \end{aligned} \tag{2.19}$$

where $\operatorname{logit}\Big[1-S_0(t)\Big] = \log\Big[\frac{1-S_0(t)}{S_0(t)}\Big]$.

To specify a parametric form for the baseline function, we assume that the log cumulative odds of the baseline survival function is linear in log time:

$$\operatorname{logit}\Big[1-S_0(t)\Big] = \gamma_0 + \gamma_1 \log t,$$

where γ_0 is the intercept and $\gamma_1 \log t$ represents the linear baseline term. Substituting the function above into Equation (2.19) yields:

$$\operatorname{logit}\Big[1-S(t \mid \boldsymbol{x})\Big] = \gamma_0 + \gamma_1 \log t + \boldsymbol{x}\boldsymbol{\beta}.$$

Then, modelling the baseline term with a spline function of log time, $s(\log t \mid \boldsymbol{\gamma})$, yields:

$$\operatorname{logit}\Big[1-S(t \mid \boldsymbol{x})\Big] = s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta}. \tag{2.20}$$

Similar to Equation (2.18), time-dependent effects of covariates are incorporated by modelling the interactions between each covariate with a spline function of log time:

$$\operatorname{logit}\Big[1-S(t \mid \boldsymbol{x})\Big] = s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_i)\, x_i. \tag{2.21}$$

Equations (2.20) and (2.21) together define an FPM on the PO scale, which uses a logit link function to model the survival function with spline-based baseline functions with time-dependent effects. A key property of the PO model is that the hazard ratio can converge to one as time approaches infinity. This is a useful assumption in survival studies evaluating treatments whose benefits may disappear over time, such as treatment effect waning [17]. The PO model naturally allows for time-varying hazard ratios that typically decrease over time.

Spline-based probit models. Spline-based models using a probit link function extends the lognormal distribution. The probit link function is defined by $g(x) = -\Phi^{-1}(x)$, where $\Phi^{-1}(\cdot)$ is the inverse cumulative distribution function of the standard normal distribution [7]. A survival model with a probit transformation can be expressed as:

$$\begin{aligned} g\Big[S(t \mid \boldsymbol{x})\Big] &= -\Phi^{-1}\Big[S(t \mid \boldsymbol{x})\Big] \\ &= \gamma_0 + \gamma_1 \log t + \boldsymbol{x}\boldsymbol{\beta}. \end{aligned}$$

This formulation generalises the lognormal model by allowing covariate effects $\boldsymbol{\beta}$ to enter additively on the probit scale.

To increase flexibility, the intercept and linear term, γ_0 and $\gamma_1 \log t$, can be replaced by a spline function of $\log t$, yielding

$$-\Phi^{-1}\Big[S(t \mid \boldsymbol{x})\Big] = s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta}. \tag{2.22}$$

Time-dependent effects are incorporated through additional spline terms between covariates and time:

$$-\Phi^{-1}\Big[S(t \mid \boldsymbol{x})\Big] = s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_i)\, x_i. \tag{2.23}$$

Together, Equations (2.22) and (2.23) define an FPM on the probit scale.

Summary of link functions. For each modelling scale, the transformed survival function is written as a spline-modelled baseline function in $\log t$ and with optional spline interactions between covariates and $\log t$ to allow covariate effects to vary over time:

$$\left.\begin{aligned} \text{Spline PH:} \quad & \log\Big[-\log S(t \mid \boldsymbol{x})\Big] \\ \text{Spline PO:} \quad & \operatorname{logit}\Big[1 - S(t \mid \boldsymbol{x})\Big] \\ \text{Spline probit:} \quad & -\Phi^{-1}\Big[S(t \mid \boldsymbol{x})\Big] \end{aligned}\right\} = s(\log t \mid \boldsymbol{\gamma}) + \boldsymbol{x}\boldsymbol{\beta} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_i)\, x_i,$$

where $s(\log t \mid \boldsymbol{\gamma})$ is the spline function of the baseline function, $\boldsymbol{x}\boldsymbol{\beta}$ represents proportional covariate effects, and $\sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_i)\, x_i$ represents D time-dependent effects for each covariate x_i.

Table 2.2 presents the link functions, baseline survival and density functions, and key properties of FPMs under various modelling scales.

2.2 Multistate models

Multistate models extend survival analysis by allowing each individual to either stay or transition among a finite set of states over time. These states are mutually exclusive, meaning a person can only be in one state at a time, and are classified as either transient or absorbing [18]. A transient state can be left for another state, whereas an absorbing state cannot be left once entered, such as the death state. At the start of a study, individuals begin in one or more possible initial states. These are different from intermediate states, which may be visited after the initial state(s) but before reaching an absorbing state. In this thesis, for illustrative purposes, only multistate models with a single initial state are considered, and all applications in Studies II–IV involve models with one initial state.

Table 2.2. Link functions, baseline survival functions, baseline probability density functions, and properties of flexible parametric models under different modelling scales.

	Proportional Hazards	**Proportional Odds**	**Probit**
Link function, $g\big[S(t)\big]$	$\log\big[-\log S(t)\big]$	$\operatorname{logit}\big[1-S(t)\big]$	$\Phi^{-1}\big[1-S(t)\big]$
Baseline survival function, $S_0(t)$	$\exp\Big\{-\exp\big[s(\log t \mid \boldsymbol{\gamma})\big]\Big\}$	$\Big\{1+\exp\big[s(\log t \mid \boldsymbol{\gamma})\big]\Big\}^{-1}$	$\Phi\big[-s(\log t \mid \boldsymbol{\gamma})\big]$
Baseline PDF, $f_0(t)$	$\dfrac{\mathrm{d}s}{\mathrm{d}t}\exp\Big\{s(\log t \mid \boldsymbol{\gamma})-\exp\big[s(\log t \mid \boldsymbol{\gamma})\big]\Big\}$	$\dfrac{\mathrm{d}s}{\mathrm{d}t}\exp\big[s(\log t \mid \boldsymbol{\gamma})\big]\Big\{1+\exp\big[s(\log t \mid \boldsymbol{\gamma})\big]\Big\}^{-2}$	$\dfrac{\mathrm{d}}{\mathrm{d}t}\Phi\big[s(\log t \mid \boldsymbol{\gamma})\big]$
Equivalent parametric distribution*	Weibull	Log–logistic	Log–normal

Let $s(\log t \mid \boldsymbol{\gamma})$ be the spline function of $\log t$, and let $\mathrm{d}s/\mathrm{d}t = s'(\log t)/t$ be its derivative. *When the degree of freedom is 1 (that is, no internal knot), the spline becomes linear in $\log t$, reducing each model to the corresponding standard parametric model. Baseline cumulative distribution and hazard functions can be computed as $F_0(t) = 1 - S_0(t)$ and $h_0(t) = f_0(t)/S_0(t)$, respectively. PDF, probability density function.

Notation. To express this framework mathematically, I adopt the notations of Fiocco et al. [19]. Let $\{Y(t) : t \geq 0\}$ be a stochastic process that records an individual's state at time t in a finite state space $\mathcal{S} = \{1, \ldots, M, \ldots, L\}$, where the first M states are transient states, and the remaining $(L - M)$ states are absorbing. Denote by $\mathcal{H}_{s-} = \{Y(u) : 0 \leq u < s\}$ the history generated by the process up to, but not including, time s. For $t \geq s$, the transition probability from state i at time s to state j at time t is defined as:

$$P_{ij}(s,t) = P\Big[Y(t) = j \mid Y(s) = i,\ \mathcal{H}_{s-}\Big], \qquad i, j \in \mathcal{S}.$$

Assuming the Markov property, the future depends only on the present state, so the conditioning on the past $\mathcal{H}_{s-}$ can then be dropped out:

$$P_{ij}(s,t) = P\Big[Y(t) = j \mid Y(s) = i\Big].$$

The transition-specific hazard, also known as the cause-specific intensity, for moving from state i to state j at time t is

$$h_{ij}(t) = \lim_{\Delta t \to 0} \frac{P\Big[Y(t + \Delta t) = j \mid Y(t) = i\Big]}{\Delta t},$$

which represents the instantaneous risk of leaving state i for state j at time t. Integrating $h_{ij}(t)$ up to time t yields the cumulative transition-specific hazard:

$$H_{ij}(t) = \int_0^t h_{ij}(u)\, du.$$

For any $i \in \mathcal{S}$, the state-occupancy probability at time t is defined as:

$$P_i(t) = P\Big[Y(t) = i\Big],$$

which can be interpreted as the proportion of individuals expected to be in state i at time t.

Integrating $P_i(t)$ up to a time point τ gives the sojourn time:

$$T_i(\tau) = \int_0^\tau P_i(u)\, du,$$

which represents the mean total time an individual is expected to spend in state i during the time interval $[0, \tau]$.

The RMST up to time τ is defined as the total time spent in the transient states:

$$\mu(\tau) = \int_0^\tau \Big[\sum_{i \in \mathcal{T}} P_i(u)\Big]\, du,$$

where $\mathcal{T} = \{1, \ldots, M\}$ denotes the set of transient (non-absorbing) states.

Integrating up to the upper limit t_{max} yields LE:

$$LE = \int_0^{t_{max}} \Big[\sum_{i \in \mathcal{T}} P_i(u)\Big]\, du,$$

which equals the area under the overall survival curve, as Equation (2.7).

In summary, standard survival analysis (described in Section 2.1.1) can be interpreted as a simple two-state model that distinguishes between "alive" and "dead" states. Multistate models, introduced in this section, generalise this framework to accommodate transitions among states. Table 2.3 summarises and compares the quantities of interest in both frameworks.

2.2.1 Time scales

The choice of time scale is crucial in survival analysis, as it significantly impacts the interpretation and modelling of event occurrences. Commonly used time scales in survival analysis include time since study entry, attained age, and attained calendar time. In multistate models, two common modelling assumptions regarding the time scale for transitions are the Markov (clock-forward) and semi-Markov (clock-reset) frameworks [18].

- Markov (clock-forward): Under this assumption, time t continually moves forward from the individual's entry into the initial state. Despite transitions through intermediate states, the clock does not reset, and time remains measured from initial state entry. This applies to time since study entry, attained age, and calendar time. While age and calendar time do not start from zero at study entry, they differ from time since study entry only by a constant offset and maintain the clock-forward property.
- Semi-Markov (clock-reset): The semi-Markov assumption, also known as Markov-renewal, resets the clock at each transition into a new state. Thus, for hazard functions $h_{ij}(t)$, the time t explicitly defines the duration since entering state i.

Figure 2.3 illustrates four commonly used time scales: (i) calendar time, (ii) age, (iii) time since study entry, and (iv) time since entry into the current state. The first three follow the Markov (clock-forward) assumption, whereas the fourth corresponds to the semi-Markov (clock-reset) approach.

2.2.2 Estimation

Point estimates. Point estimates of transition probabilities and state-occupancy probabilities are obtained through Monte Carlo simulation based on the inversion method by Bender et al. [20]. While standard approaches such as the Aalen–Johansen estimator or Kolmogorov forward differential equations are typically used for estimating both transition probabilities and state-occupancy probabilities in multistate models [21], Monte Carlo simulation is employed here to generate predictions when mixed time scales exist. A brief summary of this approach is described below.

Table 2.3. Quantities in standard survival analysis versus multistate models.

Quantities	**Units**	**Standard survival analysis**	**Multistate models**
Rates	Time^{-1} (events per person–time)	Hazard, $h(t)$	Transition-specific hazard from state i to j, $h_{ij}(t)$
Cumulative rates	Time^{0} (dimensionless)	Cumulative hazard, $H(t) = \int_0^t h(u)\,du$	Cumulative transition-specific hazard from state i to j, $H_{ij}(t) = \int_0^t h_{ij}(u)\,du$
Probabilities	Time^{0} (dimensionless)	Survival, $S(t) = \exp[-H(t)]$	State-occupancy probability for state j, $P_j(t) = P[X(t) = j]$
Times	Time^{1} (for example, years)	Restricted mean survival time, $\mu(\tau) = \int_0^\tau S(u)\,du$; Life expectancy, $LE = \int_0^\infty S(u)\,du$	Sojourn time for state j, $T_j(\tau) = \int_0^\tau P_j(u)\,du$; Restricted mean survival time, $\mu(\tau) = \int_0^\tau \left[\sum_{j\in\mathcal{T}} P_j(u)\right] du$; Life expectancy, $LE = \int_0^{t_{\max}} \left[\sum_{j\in\mathcal{T}} P_j(u)\right] du$, where $\mathcal{T} = \{1, \ldots, M\}$ denotes the set of transient states.

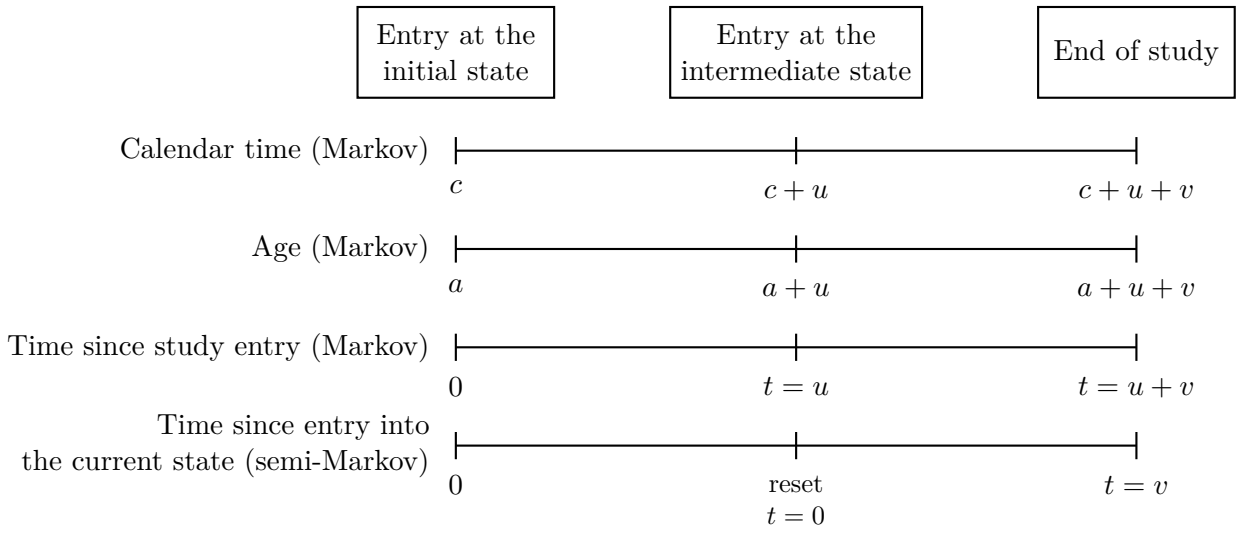


Figure 2.3. Illustration of time scales in a multistate model. The top three lines show Markov (clock-forward) time scales with different starting points: (i) calendar time, (ii) age, and (iii) time since study entry. The fourth line shows a semi-Markov (clock-reset) time scale, that is, time since entry into the current state. Here, c is the calendar time at study entry, a is the age at study entry, u is the time spent in the initial state before transition to the intermediate state, and v is the time spent in the intermediate state before the end of study.

Let T be a non-negative random variable with a CDF, $F(t)$, which can be any arbitrary distribution functions, such as SPMs (Section 2.1.5) or FPMs (Section 2.1.6). Draw a random number from a uniform distribution with the interval from 0 to 1, $u \sim \mathrm{U}(0, 1)$, and solve

$$F(t) = u \quad \Longrightarrow \quad t = F^{-1}(u), \tag{2.24}$$

so that the survival time t can be obtained. To invert the CDF, which is a function of cumulative hazard function (Equation (2.6)), one needs to integrate the corresponding hazard function. If the baseline cumulative hazard function lacks a closed form expression, numerical integration can be applied to solve for t. If a closed form exists but cannot be analytically inverted, the iterative root finding approach can be applied to solve for t [22].

In a multistate setting, each individual trajectory is simulated sequentially through the model until reaching an absorbing state or the maximum follow-up time, $t_{\max}$. An individual is at risk for a set of possible transitions from their current state. This process can be considered as a series of competing-risks settings.

For instance, for an individual starting in state i at time t_i, the simulation proceeds as follows:

1. For each possible destination state $j \in S$, define the cumulative hazard function for the transition from state i to j, $H_{ij}(t)$, where $j = 2, \ldots, L$. If $S = \emptyset$, indicating that the individual has reached an absorbing state, then the simulation stops.
2. Generate all the transition times $\mathbf{T} = \{t_{i1}, t_{i2}, \ldots, t_{iL}, t_{\max}\}$ using the inversion method

for each transition from state i.

3. Identify the smallest time among $\mathbf{T}$, denoted t_{ik}. The transition to state k then occurs at this time. If $t_{ik} = t_{\max}$, then the simulation stops.
4. Update $i = k$ and $t_i = t_{ik}$, and return to step 1. Repeat this process until either an absorbing state is reached or the time is $t_{\max}$.

Repeating this algorithm for a large Monte Carlo sample of size n produces an empirical distribution of simulated individuals. Point estimates of transition probabilities, state-occupancy probabilities, and related functionals can then be estimated at any time point by averaging across these n individuals' trajectories.

Confidence intervals. The simulation-based approach [19, 23] is also extended to quantify uncertainty by estimating variances and, hence, standard errors and CI, as follows:

1. A set of model parameters is drawn from a multivariate normal distribution, where the mean vector is given by the parameter point estimates from the original model fit, $\hat{\boldsymbol{\beta}}$, and the variance by the corresponding variance-covariance matrix, $\boldsymbol{V}$.
2. Using this new set of parameters, a Monte Carlo simulation of size n is performed to generate individual trajectories. The quantities of interest, such as transition probabilities and state-occupancy probabilities, are then calculated.
3. Steps 1 and 2 are repeated a total of m times, which yields an empirical distribution of m replicated estimates for each quantity of interest.
4. The 95% CI for the quantities of interest are constructed from their empirical distributions using one of two common methods [24]:

    - **Normal approximation:** The standard deviation of the m replicated estimates is used as the Monte Carlo estimate of the standard error (SE). The 95% CI is then calculated as the point estimate $\pm z_{0.975} \times \text{SE}$, where $z_{0.975} \approx 1.96$ under the standard normal distribution.
    - **Percentile method:** The 2.5th and 97.5th percentiles of the empirical distribution are used as the lower and upper bounds of the 95% CI. This approach does not assume normality.

    In this thesis, the **percentile method** was used to construct 95% CI for all quantities derived from the multistate model.

2.2.3 Illness–death model

A typical example of multistate models is the illness–death model (Figure 2.4), a three-state model frequently used to describe disease progression [25]. Arrows from one state to another represent transitions. For illustrative purposes, only an irreversible illness-death model with a semi-Markov assumption (clock-reset) is considered here.

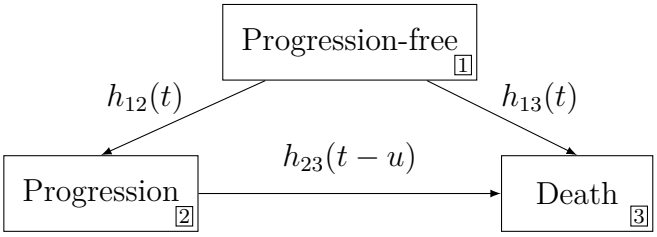


Figure 2.4. An irreversible illness-death model with a semi-Markov assumption (clock-reset), where t is time since study entry, and u is time at progression. Image adapted from Study II: Chen EYT, Dickman PW, Clements MS. A Multistate Model Incorporating Relative Survival Extrapolation and Mixed Time Scales for Health Technology Assessment. *PharmacoEconomics*. 2025;43:297-310. Licensed under CC BY–NC 4.0.

- **State 1 (progression-free)** is the initial state. All individuals start here at $t = 0$ with $P_1(0) = 1$. $h_{12}(t)$ is the instantaneous risk of progression at time t, and $h_{13}(t)$ is the hazard of dying without progression. $h_{12}(t)$ and $h_{13}(t)$ are competing risks, so the first of these two events to occur determines the next state. The state-occupancy probability $P_1(t)$ is the probability of patients expected to remain progression-free at time t, and the sojourn time $T_1(\tau) = \int_0^\tau P_1(u)\,du$ represents the mean total time spent in the progression-free state up to τ years since study entry.
- **State 2 (progression)** is the intermediate state. The hazard $h_{23}(t-u)$ is the transition from progression to death, and $P_2(t)$ is the probability of being alive with progression at time t. The sojourn time $T_2(\tau)$ represents the mean total time lived with progression up to τ years after study entry.
- **State 3 (death)** is the absorbing state; once entered, no further transitions from this state are possible.

Figure 2.5 shows examples of (A-B) transition-specific hazard functions, (C-D) cumulative hazard functions, (E) RMST, and (F) LE from an irreversible illness-death model with a semi-Markov assumption. Because the states are mutually exclusive (no individual can occupy more than one state at a time) and exhaustive (every individual must be in one of the defined states), $P_1(t) + P_2(t) + P_3(t) = 1$ for all t. Area-under-the-curve summary measures of the two living states, $P_1(t) + P_2(t)$, yield the RMST up to time τ, $\mu(\tau)$, and the LE, respectively:

$$\mu(\tau) = \int_0^\tau \Big[P_1(u) + P_2(u)\Big]\,du;$$

$$\text{LE} = \int_0^{t_{\max}} \Big[P_1(u) + P_2(u)\Big]\,du.$$

2.3 Net survival

Net survival is the survival probability after disease diagnosis in a hypothetical world where the disease under study is the only possible cause of death [26]. There are two common

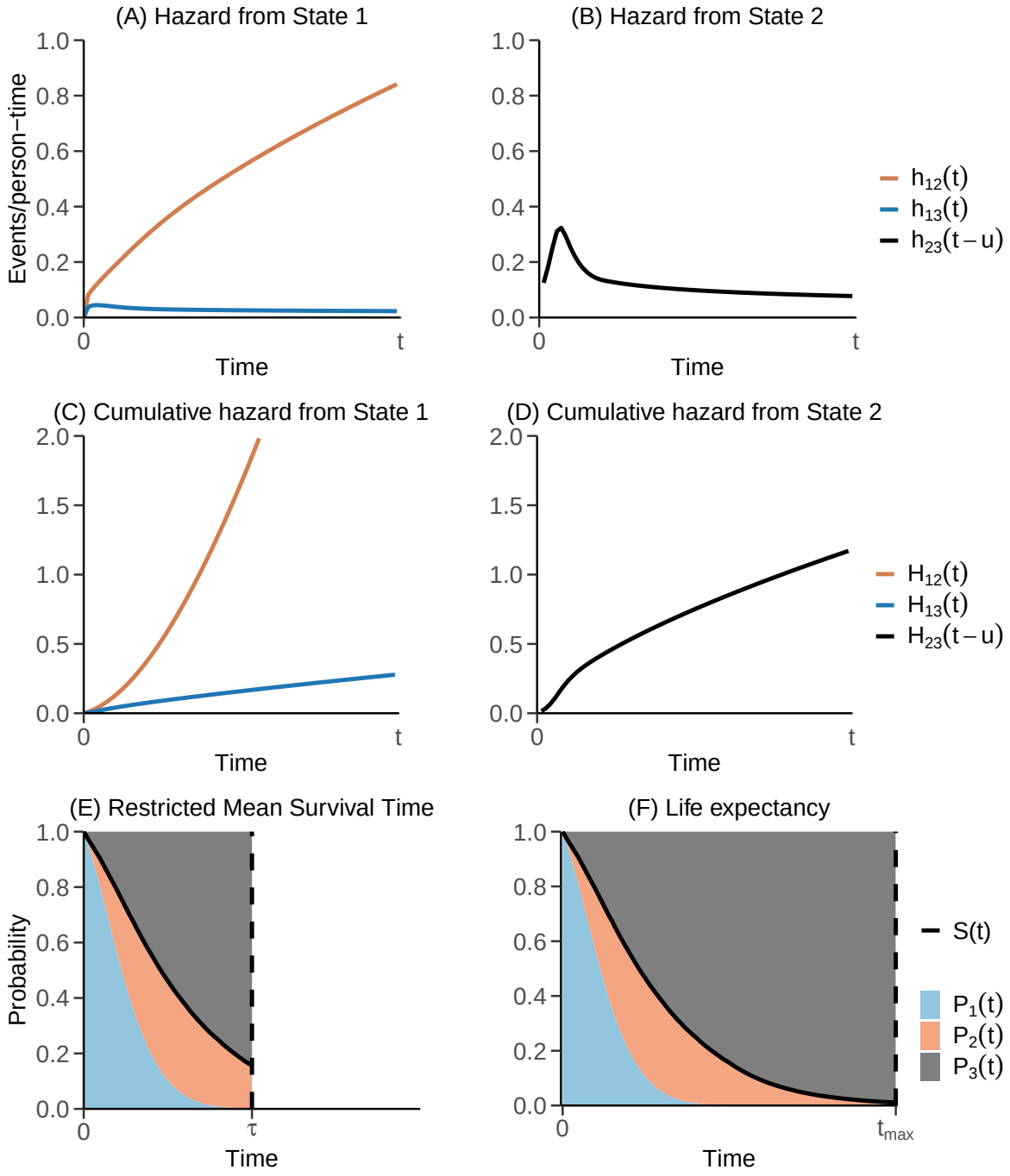


Figure 2.5. Examples of (A–B) hazard functions, (C–D) cumulative hazard functions, (E) restricted mean survival time, and (F) life expectancy from an irreversible illness-death model with a semi-Markov assumption (clock-reset), where t is time since study entry, and u is time at progression. $h_{ij}(t)$ denotes transition-specific hazard functions, $H_{ij}(t)$ cumulative hazard functions, and $P_i(t)$ state-occupancy probability functions. The overall survival function is defined as the sum of state-occupancy probability functions for the transient states: $S(t) = P_1(t) + P_2(t)$.

frameworks for estimating net survival: the cause-specific survival framework and the relative survival framework (RSF). Under the cause-specific framework, cause of death must be explicitly classified as either death due to the disease, or death due to other causes. This classification typically relies on cause-of-death registries or electronic health records. In contrast, the RSF does not require explicit information on cause of death. Instead, relative survival is commonly estimated as the ratio of all-cause survival (overall survival) among a patient cohort to the expected survival of a comparable population without the disease of interest. This thesis in particular focuses on measures related to net survival within the RSF. A brief introduction to this framework is given below.

2.3.1 Relative survival framework

Relative survival, denoted as $R(t)$, is defined as the ratio between the all-cause survival, $S(t)$, of a patient cohort and the expected survival, $S^*(t)$, from a comparable population, such as the general population matched on age, sex, and calendar year, and assumed to be free of the disease of interest [3], expressed as:

$$R(t) = \frac{S(t)}{S^*(t)}. \tag{2.25}$$

The expected survival, $S^*(t)$, is typically derived from general population mortality rates (also referred to as background mortality in this thesis). A widely used resource for these data is the Human Mortality Database (HMD) [27], which includes mortality statistics from multiple countries.

The hazard analogue of relative survival is the excess hazard, $\lambda(t)$, representing the difference between the all-cause hazard from the patient cohort, $h(t)$, and the expected hazard from the general population, $h^*(t)$, matched by age, sex, calendar year:

$$\lambda(t) = h(t) - h^*(t). \tag{2.26}$$

By integrating with respect to t, the cumulative excess hazard is defined as:

$$\Lambda(t) = H(t) - H^*(t), \tag{2.27}$$

where $\Lambda(t)$ denotes the cumulative excess hazard, $H(t)$ the cumulative all-cause hazard, and $H^*(t)$ the cumulative expected hazard.

Several non-parametric approaches have been proposed to estimate net survival within an RSF, including the Ederer I [28], Ederer II [28], Hakulinen [29], and Pohar-Perme [30] estimators. However, this thesis mainly focuses on parametric approaches, since survival extrapolation and covariate adjustment are the core of HTA applications.

2.3.2 Parametric relative survival models

Parametric relative survival models, also called parametric excess hazard models, allow for adjusting for covariates and extrapolating survival beyond the observed follow-up period. Let $R(t \mid \mathbf{x})$ be the relative survival function and let $g[\cdot]$ denote a link function. A general relative survival is defined as

$$g\Big[R(t \mid \mathbf{x})\Big] \;=\; g\Big[R_0(t)\Big] \;+\; \mathbf{x}\boldsymbol{\beta},$$

where $\boldsymbol{x}$ denotes a vector of covariates with corresponding coefficients $\boldsymbol{\beta}$, and $R_0(t)$ represents the baseline relative survival function.

Under a PH assumption, the equation above can be shown on the (excess) hazard scale:

$$\lambda(t \mid \boldsymbol{x}) = \lambda_0(t)\, \exp(\boldsymbol{x}\boldsymbol{\beta}),$$

where $\lambda(t \mid \boldsymbol{x})$ is the excess hazard function, and $\lambda_0(t)$ the baseline excess hazard function.

Both SPMs described in Section 2.1.5 and FPMs described in Section 2.1.6 have been implemented in a relative survival model, for example, the `flexsurv` package [31] and the `rstpm2` package [12] in `R`, and the `stpm2`/`stpm3` packages [10] and the `merlin` package [32] in `Stata`.

The implementation began by incorporating covariates into Equation (2.27), resulting in the cumulative excess hazard as follows:

$$\Lambda(t \mid \mathbf{x}) \;=\; H(t \mid \mathbf{x}) - H^*(t \mid \mathbf{x}'),$$

where $\mathbf{x}$ is a covariate vector of patients' characteristics, and $\mathbf{x}'$ denotes the stratification factors used to obtain expected mortality from the general population mortality rates.

By introducing spline functions for modelling the baseline excess hazard, the model on the log cumulative excess hazard scale is:

$$\log \Lambda(t \mid \mathbf{x}) \;=\; s\Big(\log t \mid \boldsymbol{\gamma}\Big) \;+\; \mathbf{x}\boldsymbol{\beta}, \tag{2.28}$$

where $s\Big(\log t \mid \boldsymbol{\gamma}\Big)$ is a spline function of log time with parameters $\boldsymbol{\gamma}$. Similar to Equation (2.18), time-dependent effects can also be accommodated by incorporating the spline function with covariates, yielding:

$$\log \Lambda(t \mid \mathbf{x}) = s\Big(\log t \mid \boldsymbol{\gamma}\Big) \;+\; \mathbf{x}\boldsymbol{\beta} \;+\; \sum_{i=1}^{D} s\Big(\log t \mid \boldsymbol{\gamma}_i\Big)\, x_i, \tag{2.29}$$

where D is the number of covariates with time-dependent effects, and each $s(\log t \mid \boldsymbol{\gamma}_i)x_i$ represents the interactions between the spline function of log time and the corresponding covariate x_i. This extension allows the modelling of time-dependent effects for covariates in a parametric relative survival model.

Likelihood. The maximum likelihood described in Section 2.1.4 is also applied to relative survival. The general log-likelihood function for a relative survival model can be expressed as:

$$\begin{aligned}\log L(\theta \mid t, \delta) &= \sum_{i=1}^{n} \left[\delta_i \log(h_\theta(t_i)) + \log(S_\theta(t_i))\right] \\ &= \sum_{i=1}^{n} \left[\delta_i \log\Big(\lambda_\theta(t_i) + h^*(t_i)\Big) + \log S^*(t_i) + \log R_\theta(t_i)\right],\end{aligned}$$

where δ_i is the event indicator, $h_\theta(t_i) = \lambda_\theta(t_i) + h^*(t_i)$ the all-cause hazard, $S_\theta(t_i) = S^*(t_i)R_\theta(t_i)$ the all-cause survival, and $R_\theta(t_i)$ the relative survival. Since the expected survival is independent from any parameters θ, it can be omitted from the log-likelihood function, yielding

$$\log L(\theta \mid t, \delta) = \sum_{i=1}^{n} \left[\delta_i \log\Big(\lambda_\theta(t_i) + h^*(t_i)\Big) + \log R_\theta(t_i)\right],$$

where the expected hazard, $h^*(t_i)$, is assumed to be known (from the general population mortality rates) for every individual i, so it can be directly incorporated at the event times. The estimate(s) $\hat{\theta}$ is (are) then obtained by maximising this log-likelihood function.

2.4 Health technology assessment

2.4.1 An overview

HTA provides a systematic framework for evaluating medical interventions to support decision-making in healthcare. As defined by O'Rourke et al., "HTA is a multidisciplinary process that uses explicit methods to determine the value of a health technology at different points in its lifecycle. The purpose is to inform decision-making in order to promote an equitable, efficient, and high-quality health system" [33].

The scope of HTA has expanded substantially over the past decades. Initially focused on clinical effectiveness, HTA has evolved into a comprehensive evaluation framework that includes both clinical effectiveness and cost-effectiveness to systematically consider economic, organisational, social, and ethical dimensions [34].

HTA relies heavily on decision modelling to synthesise evidence and extrapolate outcomes beyond follow-up [35]. This is critical, as policy and reimbursement decisions must often be made before long-term data on clinical outcomes, resource use, and costs become available across all relevant populations and settings. Cost-effectiveness analysis (CEA) serves as the key determinant for technology appraisal and reimbursement decisions in many healthcare systems.

2.4.2 Cost-effectiveness analysis

CEA forms the analytical foundation of HTA by systematically comparing both the costs and health outcomes of medical interventions, while explicitly accounting for the uncertainty in

these estimates [36]. In many HTA bodies, particularly among high-income countries, the QALY is the preferred effectiveness measure, as it combines gains in both length and quality of life into a single and comparable metric, allowing comparisons across different interventions and disease areas. In this thesis, CEA that uses QALYs as the measure of effectiveness is considered equivalent to cost-utility analysis (CUA).

QALYs can be estimated through two main approaches. The first employs a multistate modelling framework [35], which multiplies the duration spent in each health state by its corresponding health state utility value (HSUV), summing across all transient states over the specified time horizon. Alternatively, QALYs may be derived by integrating the survival function multiplied with a longitudinal quality-of-life function over time [37].

When comparing two interventions (for example, A versus B), the results of a CEA can fall into several scenarios where the choice is clear [34]. These scenarios are illustrated in Figure 2.6 and include:

i. **A dominates B:** Intervention A yields greater QALYs at lower costs than intervention B.
ii. **A is dominated by B:** Intervention A provides fewer QALYs at higher costs than intervention B.
iii. **Less costly for the same effectiveness:** Interventions A and B generate the same QALYs, but A costs less; A is cost-saving.
iv. **More costly for the same effectiveness:** If intervention A is more expensive than B but produces the same QALYs, it is considered inefficient and should be rejected.
v. **Same costs with greater effectiveness:** If two interventions have the same cost, the one that generates more QALYs is preferred.
vi. **Same costs with poorer effectiveness:** If two interventions have the same cost, the one that generates fewer QALYs should be rejected.

However, the most common scenario in HTA involves a new intervention that is both more effective and more costly. In this situation, the key summary statistic for this purpose is the incremental cost-effectiveness ratio (ICER), which represents the difference in mean costs divided by the difference in mean QALYs between two interventions. It quantifies the additional cost required to gain one additional QALY with one intervention compared to another:

$$\text{ICER} = \frac{\Delta\overline{\text{Cost}}}{\Delta\overline{\text{QALY}}} = \frac{\overline{\text{Cost}}_A - \overline{\text{Cost}}_B}{\overline{\text{QALY}}_A - \overline{\text{QALY}}_B}. \tag{2.30}$$

This metric provides a framework for evaluating whether the incremental health benefits of an intervention justify the additional costs. However, point estimates of ICERs alone provide only partial guidance. The assessment of its uncertainty is equally important. To quantify this, probabilistic sensitivity analysis (PSA) is conducted by sampling from the joint distribution of model parameters and propagating this uncertainty through the model [35].

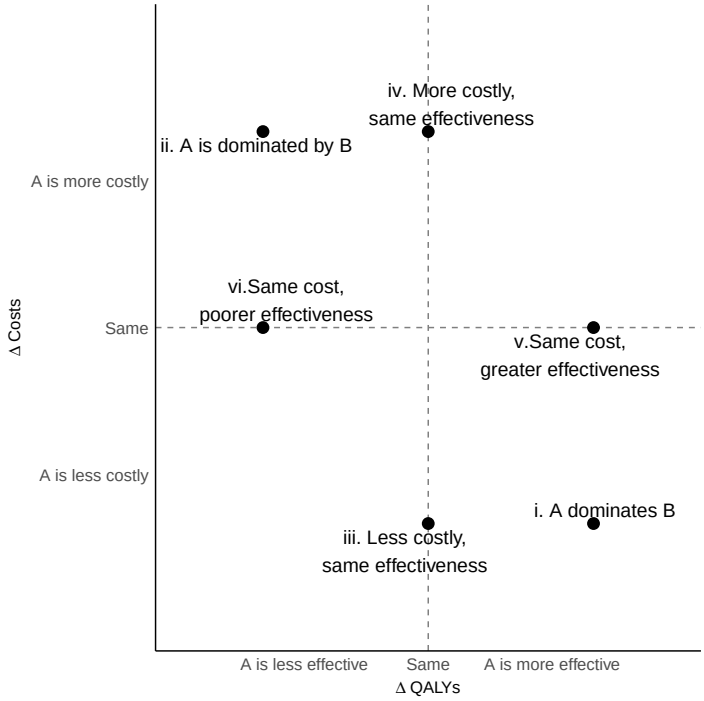


Figure 2.6. Scenarios in cost-effectiveness analysis where the choice is clear. The cost-effectiveness plane shows six scenarios when comparing intervention A to intervention B. The position of each scenario is determined by its incremental cost (ΔCost) and incremental effectiveness (ΔQALYs), indicating whether A is dominant, dominated by B, or otherwise preferred/rejected. QALYs, quality-adjusted life years.

Results of PSA are typically summarised using cost-effectiveness acceptability curves, which show the probability that an intervention is cost-effective across a range of willingness-to-pay (WTP) thresholds in order to support decision making under uncertainty.

2.4.3 Why survival extrapolation?

Survival extrapolation is the backbone of CEA, as HTAs frequently require projections beyond the observed duration of follow-up. Lifetime horizon is frequently adopted in HTA to ensure that all relevant costs and health outcomes are captured. However, clinical studies, including both clinical trials and population-based register studies, often provide limited follow-up data. For example, clinical trials typically offer two to five years of follow-up, while register-based studies, despite potentially longer observation periods, may still be insufficient for providing complete lifetime survival; that is, the survival function of the study population reaches zero. As a result, survival must be extrapolated beyond the follow-up period to support economic evaluations using a lifetime horizon.

The methods for survival extrapolation have evolved considerably. Traditional approaches rely on parametric survival models such as the Weibull, Gompertz, and log-logistic distributions (described in Section 2.1.5), each making different assumptions about how hazard rates behave over time. More sophisticated methods include spline models (FPMs described in Sec-

tion 2.1.6) that can capture complex hazard shapes, and mixture-cure models that explicitly account for the possibility of cure [6, 38, 39]. More recently, attention has turned to models using external data sources to address uncertainty in longer-term survival projections [40, 41].

Accurate survival extrapolation is critical. The extrapolated part of the survival curve may account for the majority of projected QALYs and thus has a substantial impact on ICER estimates. The implications for reimbursement decisions are therefore considerable. For this reason, HTA agencies require transparent justification of all extrapolation methods and extensive sensitivity analyses to explore the robustness of cost-effectiveness results under varying survival extrapolation alternatives [6].

2.4.4 Health state utility values

HSUVs play an essential role in health economic and outcomes research (HEOR). HSUVs quantify the value of health states on a scale with zero representing death, one representing full health, and negative values representing states considered worse than death. In this thesis, the terms health state and disease state are used interchangeably.

HSUVs are typically derived from generic preference-based measures, also known as multi-attribute utility scales. The most widely used instruments [42] are the Health Utilities Index Mark 3 (HUI3) [43], EuroQol 5 dimensions (EQ-5D) from the EuroQoL Group [44], and Short Form 6 dimensions (SF-6D) [45]. Each instrument captures health across multiple domains using structured descriptive systems. [34]. For instance, the EQ-5D assesses (i) mobility, (ii) self-care, (iii) usual activities, (iv) pain/discomfort, and (v) anxiety/depression with response options varying between the 3-level (EQ-5D-3L [44]) and 5-level (EQ-5D-5L [46]) versions. These questionnaires summarise health across several domains and convert each response profile to a single utility score through county-specific value sets using techniques such as time trade-off or standard gamble [42].

When HSUVs are not directly available from clinical studies, they are often obtained from literature [42]. However, these indirect approaches introduce additional uncertainty that should be captured through sensitivity analyses and parameter distributions. Within a multistate model, each health state is assigned an HSUV (ideally adjusted for age and comorbidities using population norms for example) and multiplied sojourn time in the state. Integration over all transient states, excluding the absorbing state (death), yields the total expected QALYs, corresponding to the area under the quality-adjusted survival curve. To reflect parameter uncertainty in PSA, HSUVs are typically sampled from beta distributions [35].

2.4.5 Uncertainty and generalisability

Uncertainty. Briggs and colleague [47] highlight that models serve not only to synthesise clinical, epidemiological, and economic evidence quantitatively, but also to provide qualitative insights that support decision-making. Building on this perspective, the strength of a model

lies not merely in its ability to generate point estimates, but in its capacity to systematically examine and clearly report the uncertainty surrounding outcomes and decisions.

The authors categorise uncertainty for decision-making into four main types:

- **Stochastic uncertainty** (or Monte Carlo error/first-order uncertainty), which arises from random variation in individual outcomes.
- **Parameter uncertainty** (or second-order uncertainty), reflecting uncertainty in model parameters estimated from data.
- **Heterogeneity**, which arises from the variability in characteristics or outcomes across patient subgroups.
- **Structural uncertainty** (or model uncertainty), referring to uncertainty arising from assumptions about model structure, such as the choice between individual-based versus cohort-based state-transition models, and survival extrapolation using different survival frameworks (all-cause survival framework (ASF) versus RSF).

Generalisability. In this thesis, generalisability refers to the extent to which model results from one setting can be applied to inform decisions in other contexts, for instance, across patient populations, healthcare systems, or time periods [48]. This definition focuses on the transferability of findings, rather than the adaptability of the model structure or inputs to another setting. This transparency allows other researchers and decision-makers to assess whether the results are applicable to their own situations and, if necessary, adjust the estimates accordingly [49].

This thesis considers both uncertainty and generalisability throughout all studies. A detailed discussion of each type of uncertainty, its implications for interpretation, and considerations for generalisability is presented for each study in Chapter 7.

2.5 Cost-of-illness studies

Cost of illness (COI) studies quantify the economic burden of a disease, potentially including direct medical costs, non-medical costs, productivity losses, and intangible costs [50], depending on the study perspective and scope [51]. However, COI analyses are inherently descriptive and do not support decision-making by comparing the costs and effectiveness of alternative health technologies. As a result, they fall outside the scope of typical HTA frameworks [34]. Nevertheless, COI studies provide contextual evidence that supports broader healthcare decision-making processes by quantifying economic burden and informing resource allocation priorities [52].

Challenges with survival extrapolation. COI studies face similar methodological challenges as those in HTA when projecting economic burden beyond observed follow-up time. Because survival data require extrapolation to project outside of the observed period

using parametric models, long-term cost predictions must account for evolving treatment practices, disease progression, and changing healthcare utilisation patterns [52].

Quantifying disease burden to support health technology assessment. COI studies provide estimates of economic burden that help prioritise conditions for formal HTA evaluation [50] and supply essential input parameters for health economic models [53]. When new interventions demonstrate modest incremental effectiveness but target diseases with substantial societal costs, COI evidence quantifies the magnitude of the economic burden of disease. It also complements CEA by highlighting the potential population-level impact of reimbursement or public health policy decisions [52, 54], such as cancer screening [53].

2.6 Chronic myeloid leukaemia

CML is a slowly progressing haematological malignancy characterised by a specific genetic abnormality: the Philadelphia chromosome (Ph+), which results from a reciprocal translocation between chromosomes 9 and 22, where part of the Abelson murine leukaemia 1 (ABL1) gene on chromosome 9 fuses with part of the breakpoint cluster region (BCR) gene on chromosome 22 [55]. The translocation creates the BCR::ABL1 oncogenic fusion gene, which produces a protein with continuously active tyrosine kinase activity.

The disease is often discovered during routine blood tests, as about half of patients in the chronic phase have no symptoms at diagnosis. The blood test typically shows peripheral leucocytosis (an abnormally high number of white blood cells), including immature forms. Anaemia (a low red blood cell count) and either thrombocytosis (an increased platelet count) or thrombocytopenia (a reduced platelet count) may also be observed [56]. As the disease progresses, symptoms may develop due to extramedullary haemopoiesis (blood cell production outside the bone marrow), leading to a markedly enlarged spleen and liver, and bone marrow failure (suppression of normal blood cell production), resulting in severe anaemia, thrombocytopenia, and granulocytopenia (a low granulocyte count) [57]. If left untreated, CML inevitably progresses from a chronic phase into an acute fatal blast crisis. The prognosis for patients with CML has significantly improved since the introduction of tyrosine kinase inhibitors (TKIs), a targeted oral therapy that inhibits BCR::ABL1 kinase activity [58].

The prevalence of CML has increased substantially due to improved survival outcomes with TKI treatment. Gunnarsson et al. estimated that prevalence increased from 5.7 per 100,000 individuals in 2000 to 11.9 per 100,000 in 2012 in Sweden [59]. This improved prognosis was further supported by results from the Swedish Cancer Register data communicated by Bower et al. [60], suggesting that LE among CML patients has recently approached that of the general population in Sweden.

The expiration of patents for first-generation TKIs, notably imatinib in 2016, led to significant reductions in treatment costs. Generic imatinib's cost was approximately 96% lower by 2017 compared to branded prices in 2013 [61]. Imatinib remains a common first-line treat-

ment due to both its clinical and cost effectiveness, together with a manageable tolerability. Second-generation TKIs, such as dasatinib and nilotinib, have become increasingly prevalent in first-line therapy due to superior molecular response rates and reduced progression risk. However, they have not shown distinct overall survival benefit compared to imatinib [62].

A recent significant advancement in CML management is the concept of treatment-free remission (TFR), first demonstrated by the Stop Imatinib (STIM) trial [63]. TFR refers to the successful discontinuation of TKI therapy in patients who achieve and maintain a sustained deep molecular response [64]. A population-based study in Sweden [65] reported that approximately 23% of patients diagnosed with chronic-phase chronic myeloid leukaemia (CP-CML) discontinued TKI treatment after achieving a deep molecular response, and about 49% of them had re-initiated TKI treatment within two years. The criteria for TFR eligibility, optimal monitoring, and predictors of successful discontinuation remain active research areas, emphasising ongoing challenges and opportunities for further improving patient outcomes and cost-effectiveness in CML management.

Beyond survival and TFR, CML research continues to focus on addressing quality-of-life issues, optimising sequencing of different generation TKIs, and managing side effects that can significantly impact patient adherence and treatment efficacy [64]. Foulon et al. [66] conducted a cross-sectional study using the EQ-5D-3L questionnaire to assess the health status values of CML patients in France, providing valuable insights into the patient-reported outcomes and quality of life impacts associated with this condition. Additionally, the economic implications of evolving treatment strategies, including introductions of newer-generation TKIs and patent expirations for second-generation TKIs, are pivotal to healthcare policy and resource allocation decisions [67].

Chapter 3 Research Aims

The primary aims of this thesis are to evaluate methods for survival extrapolation using data from the Swedish Cancer Register and to develop approaches that incorporate relative survival extrapolation into multistate models for applications in HTA and other HEOR. These methods are applied to quantify losses in LE and QALE for patients with CML in Sweden and to estimate the economic burden of the disease. Figure 3.1 below provides an overview of the study aims, the types of analyses, and their relationships.

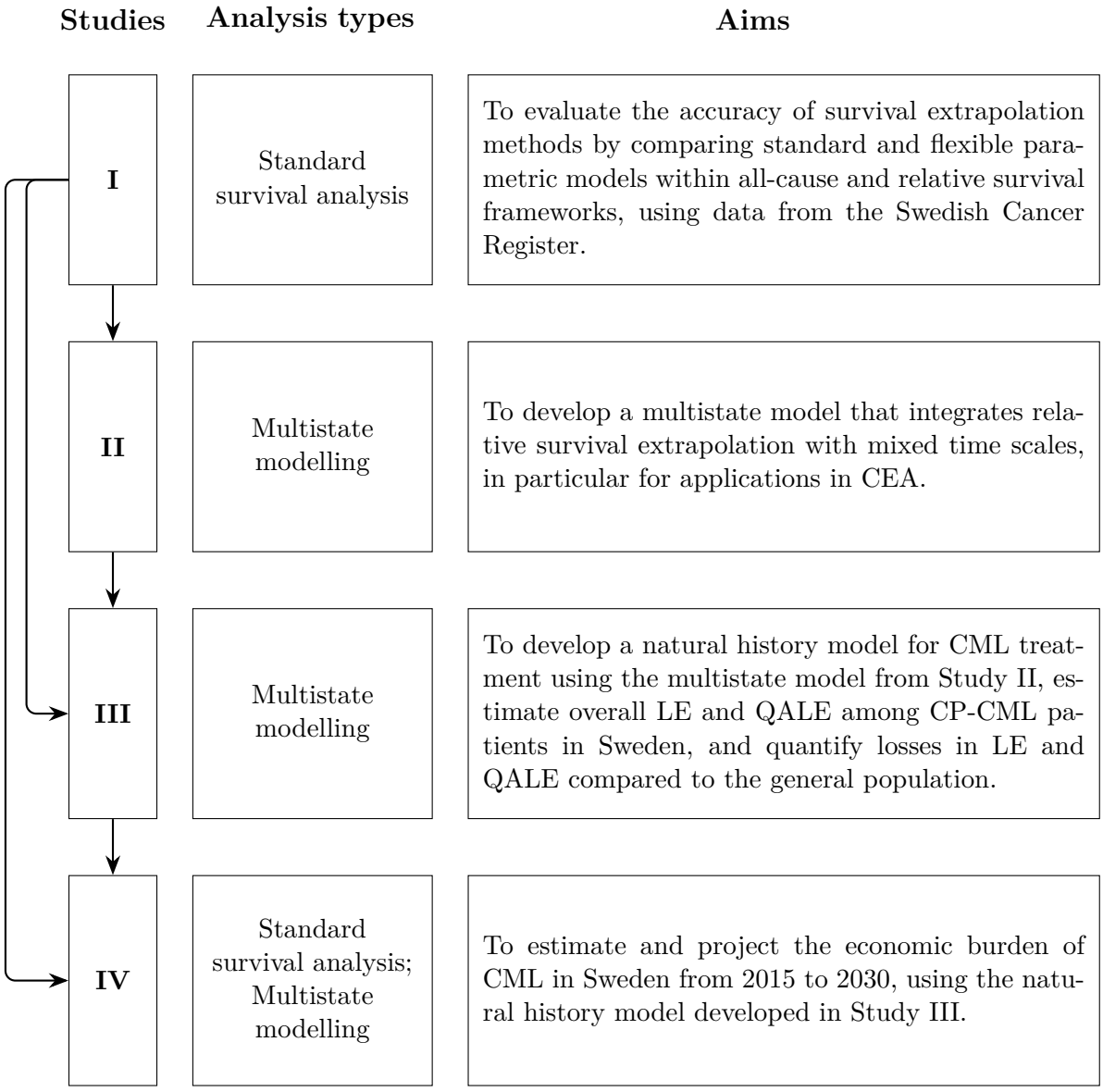


Figure 3.1. Overview of the aims and analysis types for Studies I–IV and their relationships. Arrows indicate how one study informs another. Standard survival analysis refers to modelling time to a single event, which differs from multistate modelling that models transitions between multiple states. CEA, cost-effectiveness analysis; CML, chronic myeloid leukaemia; LE, life expectancy; QALE, quality-adjusted life expectancy; CP-CML, chronic-phase chronic myeloid leukaemia.

Chapter 4 Materials

4.1 National registers

National Cancer Register. The National Cancer Register, hereafter referred to as the Swedish Cancer Register in this thesis, has recorded all incident malignancies nationwide since 1958 through mandatory reporting from both clinicians and pathologists. It has been shown to capture 96% of all malignant tumours diagnosed in Sweden with high diagnostic accuracy, as approximately 99% of cases are morphologically verified [68]. It contains information such as the personal identity number, sex, place of residence, reporting hospital or clinic, date of diagnosis, clinical and morphological classification, and the extent of the tumour at diagnosis [69]. Each case is coded using successive revisions of the International Classification of Diseases (ICD), with continuous updates to maintain data quality and completeness, thereby enabling long-term follow-up of cancer trends and outcomes.

In this thesis, data from the Swedish Cancer Register were linked with the Cause of Death Register (described below) to obtain death information for cancer patients, and the Total Population Register (described below) to obtain emigration information.

National Quality Register for Chronic Myeloid Leukaemia. The National Quality Register for CML, also known as the Swedish CML register and hereafter referred to as such, was established in 2002 and achieves approximately 98% national coverage of adult CML cases [56]. It provides richer clinical records (than the Swedish Cancer Register), including disease phase at diagnosis, TKI therapy, molecular monitoring, TFR periods, outcomes, and information on diagnostics, waiting time, treatment, and follow-up.

National Patient Register. The National Patient Register contains data on all completed inpatient stays since 1964 (nationwide coverage from 1987) and specialist outpatient visits since 2001 [70]. A recent report found generally good diagnostic validity, with a median positive predictive value of 84% for diagnoses and nearly 100% for surgical procedures [71]. The register includes admission and discharge dates, primary and secondary diagnoses (ICD codes), medical procedures, and reporting hospital or clinic, enabling comprehensive tracking of healthcare utilisation patterns.

Prescribed Drug Register. Since July 2005, the Prescribed Drug Register has recorded every prescription dispensed at Swedish pharmacies, including Anatomical Therapeutic Chemical (ATC) codes, dispensing dates, prescribed and dispensed quantities, and costs [72]. This enables individual-level tracking of TKI utilisation patterns, treatment adherence, and pharmaceutical expenditures.

Total Population Register The Swedish Total Population Register [73], maintained by Statistics Sweden, is a nationwide database covering all individuals registered as residents in Sweden, with data available from 1968 onwards. It contains information on births, deaths, marital status, immigration, emigration, addresses, and family relationships. Each individual is uniquely identified by a personal identity number, enabling reliable linkage to other registers.

Cause of Death Register. The Cause of Death Register [74] provides nearly 100% coverage of all deaths in Sweden since 1952, including the underlying cause of death coded according to ICD and exact dates of death.

4.2 Human Mortality Database.

The HMD [27] is an open-access resource that provides mortality and population data for various countries. In this thesis, general population mortality rates were retrieved from this source and used as a proxy for expected hazard (that is, the risk of dying from other causes) in estimating expected survival, such as for relative survival or expected LE calculations. For Sweden, the mortality data provided in the HMD originate from Statistics Sweden.

4.3 Data from clinical trials and health technology appraisals

Data from the CLL-8 randomised clinical trial [75, 76] (ClinicalTrials.gov ID: NCT00281918) were used as an illustrative example in Study II. This trial, formally titled, "Fludarabine and Cyclophosphamide With or Without Rituximab in Patients With Previously Untreated Chronic B-Cell Lymphocytic Leukaemia", compared first-line treatment with rituximab plus fludarabine and cyclophosphamide (RFC) versus fludarabine and cyclophosphamide (FC). Conducted across 190 centres in 11 countries, the trial provided the primary clinical evidence for the NICE Technology Appraisal TA174 [77, 78].

The multistate-formatted dataset of the CLL-8 trial was originally prepared by Williams et al. [79]. It was reconstructed by digitising the published K-M curves for overall survival and progression-free survival from the original trial publication [75], using the method developed by Guyot et al. [80]. In addition, survival curves from the publication with extended follow-up by Fischer et al. [76] were also digitised to enable comparisons between extrapolated and observed long-term survival. As the dataset was reconstructed from published aggregate data, it did not contain identifiable individual-level information.

4.4 Data from literature

HSUVs for CML patients were obtained from published studies by Foulon et al. [66] and Szabo et al. [81], while HSUVs for the Swedish general population were retrieved from Teni

et al. [82]. Treatment patterns for CML were mainly based on the report by Ohm et al. [83]. Yearly cost per patient estimates for care during the accelerated phase (AP)/blastic phase (BP), as well as for allogeneic stem cell transplantation (alloSCT), were derived from a cost study of acute myeloid leukaemia using Swedish registers by Hernlund et al. [84].

4.5 Data from official healthcare pricing lists

Drug prices during 2019–2025 were extracted from the Dental and Pharmaceutical Benefits Agency (TLV)'s Price and Decision Database [85], which maintains comprehensive records of reimbursed pharmaceutical prices in Sweden. Laboratory test prices were obtained from the National Price List for Laboratory Medical Services by Karolinska University Hospital [86] and the Extra-regional Pricing List for European Union (EU)/European Economic Area (EEA) and Switzerland [87], representing standardised pricing across healthcare regions in Sweden.

Chapter 5 Ethical considerations

Study I, III, and IV were based on individual-level data from national health and population registers. Table 5.1 provides an overview of the ethical approvals covering these studies. All ethical permissions were granted by the Swedish Ethical Review Authority (Etikprövningsmyndigheten) in accordance with the requirements for research involving sensitive personal data.

According to the Swedish Ethical Review Act (2003:460) [88], individual informed consent is not mandatory for register-based studies that use routinely collected data from registers. Nevertheless, ethical review remains essential to ensure that such studies are conducted in a way that minimises risks to individuals and safeguards privacy.

Study II did not require ethical review, as no individual-level data were accessed. Instead, survival data were digitised from published K-M curves reported by Hallek et al. [75] and Fischer et al. [76], both of which are publicly available in the publications.

All statistical analyses were conducted within the secure computing environment at the Department of Medical Epidemiology and Biostatistics, Karolinska Institutet. Access to this environment is strictly limited to authorised personnel, and all outputs are reviewed and reported in aggregate form to ensure data confidentiality and prevent any risk of identifying individuals. As a researcher, I acknowledge the responsibility to handle sensitive data with care, ensuring compliance not only with legal standards but also with the ethical principles of respect for persons and protection of privacy.

Table 5.1. Ethical approvals applicable to Study I, III, and IV.

Diary Number	Principal Investigator	Summary description	Studies
2017/641-31/1	Paul Dickman	Ethical approval for the development and application of statistical methods for studying cancer patient survival. The general aim is to develop and apply statistical methods for estimating and modelling cancer survival using registry data. The ultimate goal is to improve understanding of factors influencing morbidity and mortality among cancer patients, and thereby improve future prognoses.	I, III, and IV
2020-06544	Paul Dickman	Amendment to 2017/641-31/1. This application expands and clarifies the study objectives, including the investigation of differences in cancer survival between cancer types.	I, III, and IV
2020-05425	Torsten Dahlén	Ethical approval for population-based registry studies of patients diagnosed with CML, based on data from the Swedish CML register.	III and IV
2021-02472	Paul Dickman	Amendment to 2017/641-31/1. This application clarifies the research questions and specifies the data sources to be used: the Swedish CML register with several population-based registers: the Swedish Cancer Register, National Patient Register, Prescribed Drug Register, Cause of Death Register, Medical Birth Register, Total Population Register, Population and Housing Censuses, and the Longitudinal Integration Database for Health Insurance and Labour Market Studies.	III and IV

Chapter 6 Methods

6.1 Evaluating survival extrapolation

To assess survival extrapolation, extrapolated survival estimates from parametric models are compared with observed estimates from the K-M estimator. The observed K-M curve serves as the "gold standard" with complete follow-up, while the extrapolated curve is generated by right-censoring follow-up at specified time points and projecting survival probabilities to later times (for example, 10 years or lifetime). Extrapolations were conducted using SPMs (Section 2.1.5) and FPMs (Section 2.1.6), each fitted under both all-cause and relative survival frameworks.

As illustrated in Figure 6.1, let $S_{\text{obs}}(t)$ denote the observed K-M survival curve and $S_{\text{ext}}(t)$ the extrapolated survival curve, which was fitted to data up to the last observed time, t_{last}, and projected to a future time τ (or t_{max}, where survival reaches zero). The difference between $S_{\text{ext}}(t)$ and $S_{\text{obs}}(t)$ is used to evaluate the accuracy of survival extrapolation, with smaller differences indicating higher accuracy.

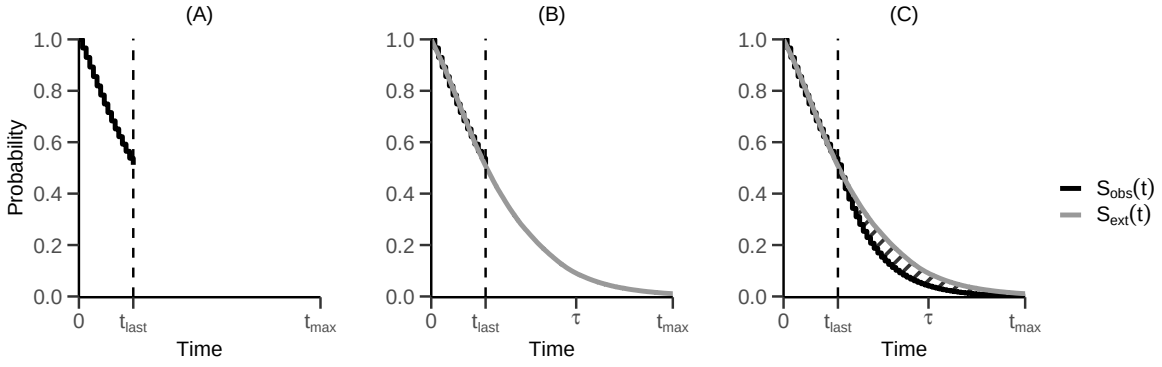


Figure 6.1. Evaluating survival extrapolation. (A) The step function shows the observed K-M survival curve, $S_{\text{obs}}(t)$, up to the last follow-up time t_{last}. (B) The grey curve represents the extrapolated survival curve, $S_{\text{ext}}(t)$, fitted to the observed data until t_{last} and extrapolated to a future time point τ (or t_{max}). (C) Now the observed survival is also plotted until τ (or t_{max}). The shaded area between $S_{\text{obs}}(t)$ and $S_{\text{ext}}(t)$ is used to assess the model's extrapolation performance.

To quantify extrapolation accuracy (see Figure 6.1(C)), three approaches were employed:

1. **Area under the survival curve.** The RMST or LE is computed from both observed and extrapolated curves. The difference in area for RMST is

$$\Delta\mu(\tau) = \int_0^\tau S_{\text{ext}}(u)\,du - \int_0^\tau S_{\text{obs}}(u)\,du,$$

where τ is a specified time point. For LE,

$$\Delta LE = \int_0^{t_{\max}} S_{\text{ext}}(u)\, du - \int_0^{t_{\max}} S_{\text{obs}}(u)\, du,$$

where $t_{\max}$ is the time point at which both S_{ext} and $S_{\text{obs}}(u)$ reach zero.

2. **Survival probability at times.** Extrapolated survival probabilities are compared to observed probabilities at times (for example, 5-year or 10-year survival). For any time τ,

$$\Delta S(\tau) = S_{\text{ext}}(\tau) - S_{\text{obs}}(\tau),$$

which measures the difference between the observed and extrapolated survival probabilities at time τ.

3. **Visual assessment.** Extrapolated survival curves (smooth lines) are overlaid on observed K-M curves (step functions) to assess model fit and the plausibility on survival extrapolation. Alternatively, comparing observed and modelled hazard functions also provides visual assessment beyond t_{last}.

6.2 Multistate models

6.2.1 Extended multistate model with relative survival extrapolation and mixed time scales

Building on the illness–death model (Figure 2.4) introduced in Section 2.2, an extension is developed to allow for relative survival extrapolation by expressing each transition hazard on its most appropriate time scale. This approach enables prediction from mixed time scales, where transition hazards are modelled with different time scales. Figure 6.2 illustrates this suggested extension. Notation follows that defined in Section 2.2, where the transition-specific hazard from state i to state j at time t is denoted by $h_{ij}(t)$. In the context below, $\lambda_{ij}(t)$ or $h^*_{ij}(t)$ is used to distinguish excess and expected hazard functions, respectively.

The death state is divided into two mutually exclusive states: excess death (death due to the disease) and expected death (death due to other causes). This classification reflects the concept of net survival described in Section 2.3. The extended model therefore consists of four states: (1) progression-free, (2) progression, (3) excess death, and (4) expected death, with five possible transitions. All patients enter the model from state 1 (progression-free) and may transition to the progression, excess death, or expected death states. From the progression state, patients may transition to the excess or expected death states. A semi-Markov approach is adopted here: the hazard for excess death depends on the time since progression, whereas the hazard for expected death depends on attained age and calendar year. Each transition is specified below along with its statistical model.

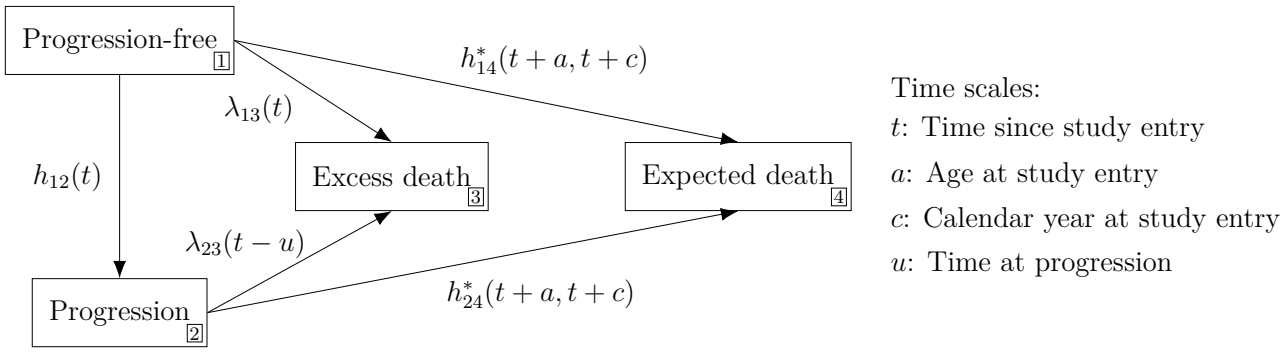


Figure 6.2. An irreversible illness–death model incorporating a relative survival framework, with five transitions: $h_{12}(t)$, progression-free → progression; $\lambda_{13}(t)$, progression free → excess death; $h^*_{14}(t+a, t+c)$, progression free → expected death; $\lambda_{23}(t-u)$, progression → excess death; and $h^*_{24}(t+a, t+c)$, progression → expected death, with a semi-Markov approach. Image adapted from Study II: Chen EYT, Dickman PW, Clements MS. A Multistate Model Incorporating Relative Survival Extrapolation and Mixed Time Scales for Health Technology Assessment. *PharmacoEconomics*. 2025;43:297-310. Licensed under CC BY–NC 4.0.

Transition 1→2: modelling the incidence rate of progression-free to progression. The hazard for transitioning from progression–free to progression, $h_{12}(t)$, is estimated using an FPM on the log cumulative hazard scale:

$$\begin{aligned}\log H_{12}\left(t \mid \boldsymbol{x}_{12}\right) &= \log H_{12,0}(t) + \boldsymbol{x}_{12}\boldsymbol{\beta}_{12} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_{12,i})\boldsymbol{x}_{12,i} \\ &= s(\log t \mid \boldsymbol{\gamma}_{12,0}) + \boldsymbol{x}_{\mathbf{12}}\boldsymbol{\beta}_{12} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_{12,i})\boldsymbol{x}_{12,i},\end{aligned}$$

where the log cumulative hazard function, $H_{12,0}(t)$, is modelled with a spline function, $\boldsymbol{x}_{12}$ the vector of patient covariates, $\boldsymbol{\beta}_{12}$ the corresponding coefficients, and D is the number of covariates with time-dependent effects with each $s(\log t \mid \boldsymbol{\gamma}_{12,i})$ modelling the time interaction for covariate $\boldsymbol{x}_{12,i}$.

Transition 1→3: modelling the mortality rate from progression-free to excess death. Within an RSF, the all-cause hazard from the progression-free state to death can be partitioned into the excess and the expected hazards, yielding:

$$h_{13,14}(t, t+a, t+c \mid \boldsymbol{x}_{13}, \boldsymbol{x}_{14}) = \lambda_{13}(t \mid \boldsymbol{x}_{\mathbf{13}}) + h^*_{14}(t+a, t+c \mid \boldsymbol{x}_{\mathbf{14}}), \tag{6.1}$$

where $\lambda_{13}(t \mid \boldsymbol{x}_{\mathbf{13}})$ denotes the excess hazard due to death from the disease, and $h_{14}(t+a, t+c \mid \boldsymbol{x}_{\mathbf{14}})$ represents the expected hazard due to death from other causes obtained from general population mortality rates. The expected hazard is stratified by demographic characteristics, $\boldsymbol{x}_{\mathbf{14}}$ (typically age, sex, and calendar year), modelled with attained age $(t+a)$ and attained calendar year $(t+c)$ as the time scales, while the time scale for the excess hazard is time since study entry, t.

Equation (6.1) can be expressed on the cumulative hazard scale, as follows:

$$H_{13,14}(t, t+a, t+c \mid \boldsymbol{x}_{13}, \boldsymbol{x}_{14}) = \Lambda_{13}(t \mid \boldsymbol{x_{13}}) + H^*_{14}(t+a, t+c \mid \boldsymbol{x_{14}}). \tag{6.2}$$

The excess hazard is modelled using an FPM on the log cumulative excess hazard scale (see in Section 2.3.2):

$$\begin{aligned}\log \Lambda_{13}(t \mid \boldsymbol{x_{13}}) &= \log \Lambda_{13,0}(t) + \boldsymbol{x}_{13}\boldsymbol{\beta}_{13} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_{13,i})\boldsymbol{x}_{13,i} \\ &= s(\log t \mid \boldsymbol{\gamma}_{13,0}) + \boldsymbol{x}_{13}\boldsymbol{\beta}_{13} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_{13,i})\boldsymbol{x}_{13,i},\end{aligned}$$

where $\log \Lambda_{13,0}(t \mid \boldsymbol{x_{13}})$ represents the baseline log cumulative excess hazard modelled with splines, $s(\log t \mid \boldsymbol{\gamma})$, and $\boldsymbol{\beta}_{13}$ contains the regression coefficients for covariates $\boldsymbol{x_{13}}$. Time-dependent effects are incorporated by including D interactions between log time and the covariates.

Transition 1→4: modelling the mortality rate from progression-free to expected death. The cumulative hazard from the progression-free state to the expected death state is specified using a piecewise-constant hazard model (also known as a piecewise exponential model):

$$H^*_{14}(t+a,\, t+c \mid \boldsymbol{x}_{14}) = \sum_{u=0}^{K-1} h^*\big(\lfloor a+\tau_u \rfloor,\ \lfloor c+\tau_u \rfloor \mid \boldsymbol{x}_{14}\big) \cdot (\tau_{u+1} - \tau_u), \tag{6.3}$$

where:

- t is time since study entry;
- a and c are age and calendar year at study entry, respectively;
- $\tau_0 = 0 < \tau_1 < \cdots < \tau_K = t$ are the ordered time points where either $\lfloor a+\tau \rfloor$ or $\lfloor c+\tau \rfloor$ changes, where the floor function $\lfloor u \rfloor$ denotes the greatest integer less than or equal to u;
- $h^*(\lfloor a+\tau_u \rfloor,\ \lfloor c+\tau_u \rfloor \mid \boldsymbol{x}_{14})$ is the constant hazard within the Lexis cell defined by age and calendar year at time τ_u;
- $\tau_{u+1} - \tau_u$, also denoted as Δ_u in Table 6.1 , is the time spent in the Lexis cell starting at τ_u.

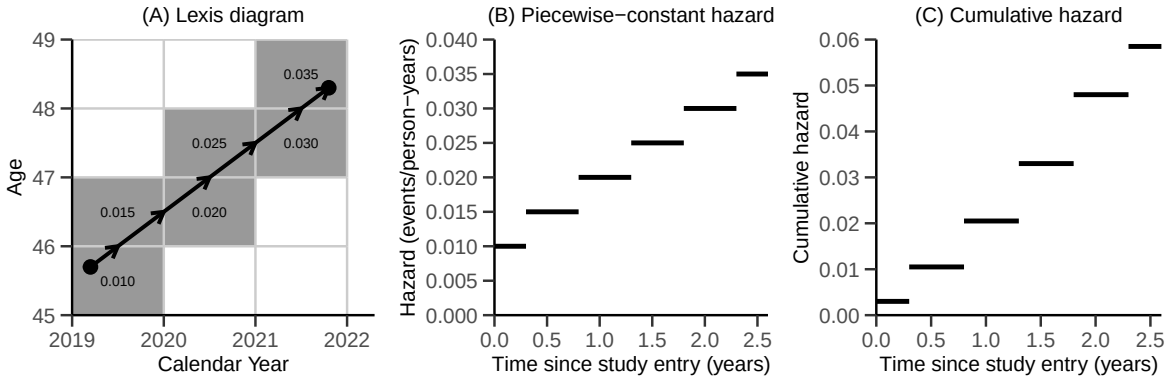
Table 6.1 presents an example of split time intervals and corresponding piecewise-constant hazards, h^*, for an individual who enters the study at age $a = 45.7$ and calendar year $c = 2019.2$, followed for $t = 2.6$ years. The follow-up time was split into six intervals ($u = 0, \ldots, 5$) according to changes in integer age or calendar year. Figure 6.3 demonstrates how the background mortality rates increase with increasing age and calendar year, where Panel (A) Lexis diagram shows the trajectory of the individual across Lexis cells, where each segment

of the diagonal line corresponds to a row in Table 6.1. Panel (B) and (C) show the piecewise constant hazard function and the cumulative hazard function over time since study entry.

Table 6.1. Example of splitting survival data at specified times for a hypothetical individual entering the study at age 45.7 years and calendar year 2019.2, followed for 2.6 years.

u	$[\tau_u, \tau_{u+1})$	Δ_u	Age at τ_u	Year at τ_u	$h^*(u)$	$H^*(u)$
0	[0.0, 0.3)	0.3	45.7	2019.2	0.010	0.0030
1	[0.3, 0.8)	0.5	46.0	2019.5	0.015	0.0105
2	[0.8, 1.3)	0.5	46.5	2020.0	0.020	0.0205
3	[1.3, 1.8)	0.5	47.0	2020.5	0.025	0.0330
4	[1.8, 2.3)	0.5	47.5	2021.0	0.030	0.0480
5	[2.3, 2.6)	0.3	48.0	2021.5	0.035	0.0585

Figure 6.3. Example of a Lexis diagram and its corresponding piecewise-constant hazard function and cumulative hazard function: (A) Lexis diagram, (B) piecewise-constant hazard, and (C) cumulative hazard for an individual entering the study at age 45.7 years and calendar year 2019.2, followed for 2.6 years. The two dots in Panel (A) are (2019.2, and 45.7) and (2021.8, 48.3).



Transition 2→3: modelling the mortality rate from progression to excess death. Once individuals enter the progression state, the only subsequent possible transitions are to the excess or expected deaths, as this illness-death model is, by definition, irreversible. The all-cause hazard from the progression state to death can be decomposed into the excess and expected hazards, yielding:

$$h_{23,24}(t-u;\, t+a,\, t+c \mid \boldsymbol{x}_{23}, \boldsymbol{x}_{24}) = \lambda_{23}(t-u \mid \boldsymbol{x}_{23}) + h^*_{24}(t+a,\, t+c \mid \boldsymbol{x}_{24}). \tag{6.4}$$

On the cumulative hazard scale, this relationship becomes:

$$H_{23,24}(t-u;\, t+a,\, t+c \mid \boldsymbol{x}_{23}, \boldsymbol{x}_{24}) = \Lambda_{23}(t-u \mid \boldsymbol{x}_{23}) + H^*_{24}(t+a,\, t+c \mid \boldsymbol{x}_{24}). \tag{6.5}$$

The excess hazard, $\lambda_{23}(t-u \mid \boldsymbol{x}_{23})$, is modelled using an FPM on the log cumulative excess hazard scale. A semi-Markov approach (clock-reset) (see Section 2.2.1) is adopted here, where the time scale is the time since progression, denoted $(t-u)$, where u is the time at progression.

$$\begin{aligned}\log \Lambda_{23}(t-u \mid \boldsymbol{x_{23}}) &= \log \Lambda_{23,0}(t-u) + \boldsymbol{x}_{23}\boldsymbol{\beta}_{23} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_{23,i})\boldsymbol{x}_{23,i}, \\ &= s(\log t \mid \boldsymbol{\gamma}_{23,0}) + \boldsymbol{x}_{23}\boldsymbol{\beta}_{23} + \sum_{i=1}^{D} s(\log t \mid \boldsymbol{\gamma}_{23,i})\boldsymbol{x}_{23,i},\end{aligned}$$

where $\log \Lambda_{23,0}(t-u)$ denotes the log baseline cumulative excess hazard modelled using splines, $s(\log t \mid \boldsymbol{\gamma})$, and $\boldsymbol{\beta}_{23}$ is the vector of coefficients with covariates $\boldsymbol{x}_{23}$. Time-dependent effects are included by including D interaction terms between log time and the corresponding covariates.

Transition 2→4: modelling the mortality rate from progression to expected death. The cumulative hazard from the progression state to the expected death state is also specified by a piecewise-constant hazard model, similar to transition 1→4 described above:

$$H^*_{24}(t+a,\, t+c \mid \boldsymbol{x}_{24}) = \sum_{u=0}^{K-1} h^*\big(\lfloor a+\tau_u \rfloor,\ \lfloor c+\tau_u \rfloor \mid \boldsymbol{x}_{24}\big) \cdot (\tau_{u+1} - \tau_u), \tag{6.6}$$

where:

- t is time since study entry;
- a, and c are age and calendar year at study entry, respectively;
- $\tau_0 = 0 < \tau_1 < \cdots < \tau_K = t$ are the ordered time points where either $\lfloor a+\tau \rfloor$ or $\lfloor c+\tau \rfloor$ changes;
- $h^*(\lfloor a+\tau_u \rfloor,\ \lfloor c+\tau_u \rfloor \mid \boldsymbol{x}_{24})$ is the constant hazard within the Lexis cell defined by age and calendar year at time τ_u;
- $\tau_{u+1} - \tau_u$ is the time spent in the Lexis cell starting at τ_u.

Estimation. The inversion method for simulating event times, introduced in Section 2.2.2, extends naturally to the proposed extended multistate model. It facilitates the estimation of transition probabilities, state-occupancy probabilities, and other quantities of interest through simulations based on the fitted transition-specific hazards.

Figure 6.4 summarises the key quantities of interest for an irreversible illness–death model with relative survival extrapolation and mixed time scales. Panels (A-B) show transition-specific hazard functions. Panels (C-D) show the cumulative hazard functions. Panel (E)

shows RMST, and Panel (F) shows LE. Because the four states are exhaustive and mutually exclusive (see definitions in Section 2.2.3), the state-occupancy probabilities still satisfy

$$P_1(t) + P_2(t) + P_3(t) + P_4(t) \;=\; 1 \quad \text{for all} \quad t,$$

where $P_1(t)$ is the probability of being in the progression-free state, P_2 the progression state(t), $P_3(t)$ the excess death state, $P_4(t)$ the expected death state. The extended multistate model enables relative survival extrapolation to be easily carried out by simulating event times from relative survival models and background mortality rates. Partitioning the death state into excess and expected states further separates state-occupancy probabilities into death due to the disease and other causes, facilitating interpretation of net survival in a multistate setting.

Area-under-the-curve summary measures for the two living states, $P_1(t) + P_2(t)$, yield the RMST up to time τ, $\mu(\tau)$, and the LE, respectively:

$$\mu(\tau) = \int_0^{\tau} \Big[P_1(u) + P_2(u)\Big]\, du; \tag{6.7}$$

$$\text{LE} = \int_0^{t_{\max}} \Big[P_1(u) + P_2(u)\Big] du. \tag{6.8}$$

6.2.2 Natural history model for chronic myeloid leukaemia treatment

A multistate model was developed to represent the natural history of CML treatment and disease progression. The extended multistate framework incorporating relative survival extrapolation (see Section 6.2.1) was employed to predict survival beyond the observed follow-up. This approach avoids unreasonable projections that may arise when extrapolating survival within an ASF, particularly in long-term hazards where expected mortality contributes a substantial portion of the all-cause hazard. Given the improved survival of CML patients since the introduction of TKIs, accounting for background mortality was considered essential to generate realistic survival predictions. By modelling the excess and expected hazards separately in a multistate setting, the model aims to generate more credible long-term estimates.

The model includes the following states: first-line TKI (1L TKI), second-line TKI (2L TKI), third-line or later TKI (3L+ TKI), TFR, progression, alloSCT, and excess and expected death. All patients diagnosed with CP-CML enter the model in the 1L TKI state, reflecting the standard practice of initiating TKI therapy after diagnosis. Patients may transition between defined health states over time. Clinical pathways such as returning from TFR to 1L TKI therapy (that is, 1L TKI $\rightarrow$ TFR $\rightarrow$ 1L TKI) are observed in practice. However, such re-initiations are modelled as transitions to distinct states rather than revisiting previous ones. In other words, each state in the model is defined by both treatment line and treatment sequence; for example, patients re-initiating 1L TKI after TFR enter a separate 1L TKI state that is distinct from the initial 1L TKI state. As a result, while the model's structure

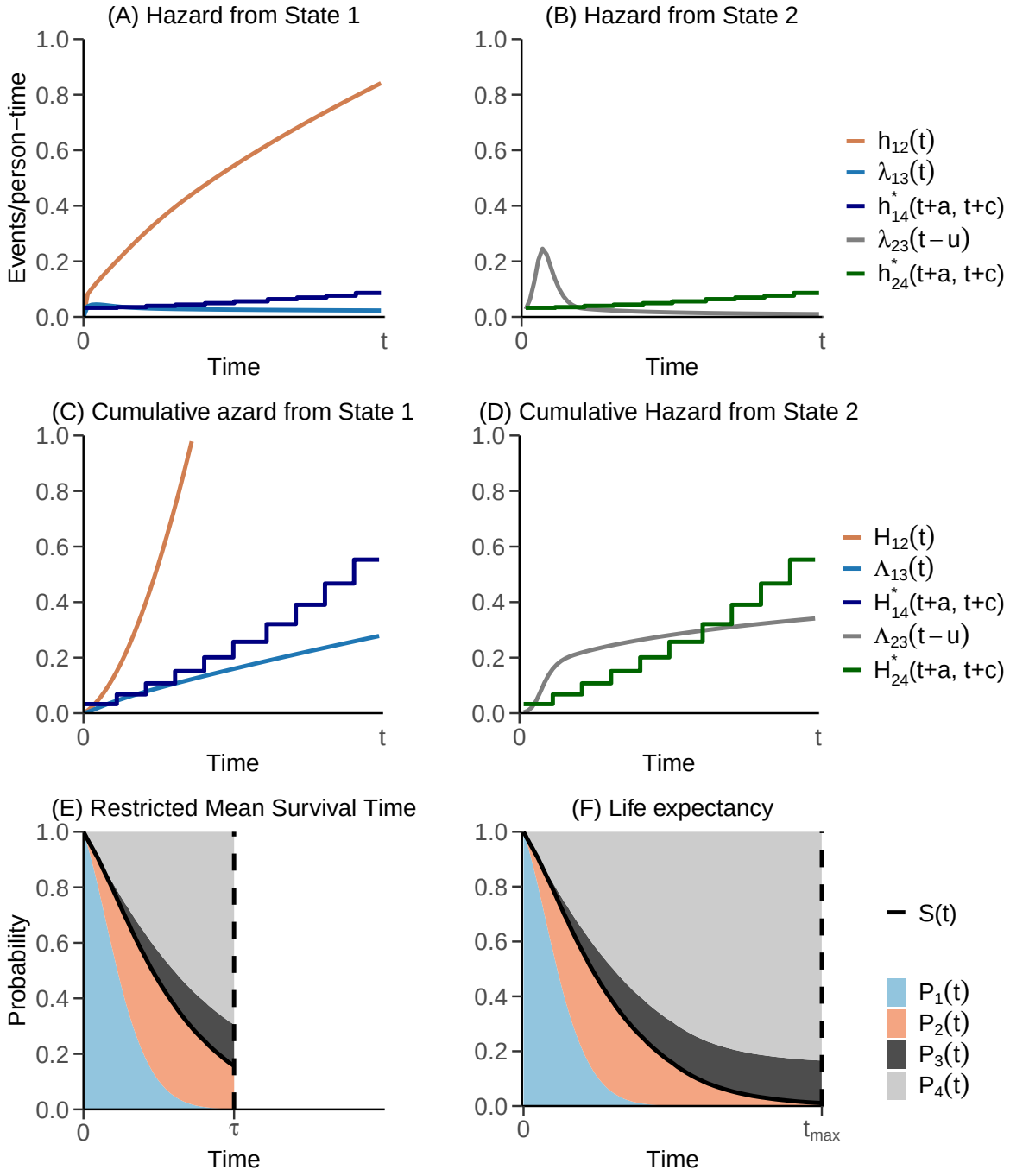


Figure 6.4. Examples of (A–B) hazard functions, (C–D) cumulative hazard functions, (E) restricted mean survival time, and (F) life expectancy from an irreversible illness-death model with relative survival extrapolation and mixed time scales, where $h_{ij}(t)$ denotes transition-specific hazard functions, $H_{ij}(t)$ cumulative hazard functions, and $P_i(t)$ state-occupancy probability functions. Here, a denotes age at study entry, c calendar year at study entry, and u time at progression. The overall survival function is defined as the sum of state-occupancy probability functions for the transient states: $S(t) = P_1(t) + P_2(t)$.

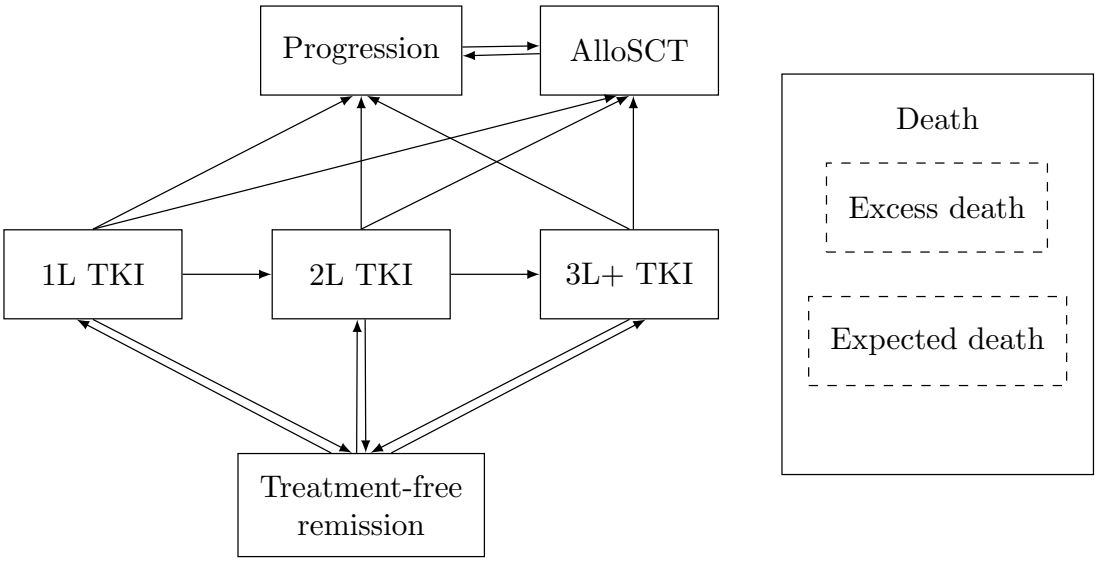


Figure 6.5. A multistate natural history model for chronic myeloid leukaemia treatment. Transitions are assumed from every living state to the excess or expected death state (arrows not shown). 1 L, first-line; 2 L, second-line; 3 L+, third-line and later; TKI, tyrosine kinase inhibitor; AlloSCT, allogeneic stem cell transplantation. Image retrieved from Study III: Chen EYT et al. Loss in Overall and Quality-Adjusted Life Expectancy for Patients with Chronic-Phase Chronic Myeloid Leukemia. *European Journal of Haematology.* 2025;114(2):334–342. Licensed under CC BY–NC 4.0.

allows double-headed arrows to illustrate potential recurrent transitions, each state is only visited once during the modelling process. This approach ensures the clinical relevance of the multistate model while allowing for statistical modelling of patient transitions. Full details of the modelling approach are provided in the Supplementary Materials of Study III.

6.3 Quantities of interest

6.3.1 Life expectancy and loss in life expectancy

Life expectancy. As illustrated in Section 2.1.1, the RMST up to a time point τ is defined as:

$$\mu(\tau) = \int_0^\tau S(u)\, du,$$

where $S(u)$ denotes the probability of being alive at time u. LE extends this definition to the time $t_{\max}$ when survival reaches zero. In other words, the area under the all-cause survival curve represents this quantity:

$$LE = \int_0^{t_{max}} S(u)\, du.$$

In most clinical trials and population-based studies, follow-up ends before every patient experiences the event of interest, such as death. As a result, LE cannot be calculated directly from the area under the observed survival curve. The survival curve must then be extrapolated

from the right-censored survival data. A common solution is to fit a parametric survival model to the observed survival data, and extrapolate it until the survival curve reaches zero, as illustrated in Figure 6.1 in Section 6.1. Hakama and Hakulinen [89] recommended performing survival extrapolation within an RSF, as shown by Equation (2.25):

$$\begin{aligned} LE(\boldsymbol{x}) &= \int_0^{t_{max}} S(u \mid \boldsymbol{x})\, du \\ &= \int_0^{t_{max}} S^*(u \mid \boldsymbol{x'})\, R(u \mid \boldsymbol{x})\, du, \end{aligned}$$

where $\boldsymbol{x}$ denotes a vector of patient characteristics, and $\boldsymbol{x'}$ includes covariates used for stratification in general population mortality rates, such as age, sex, and calendar year. This formulation enables separate extrapolations of expected survival, $S^*(u \mid \boldsymbol{x'})$, typically derived from general population mortality rates matched by $\boldsymbol{x'}$, and relative survival, $R(u \mid \boldsymbol{x})$, which is the survival analogue of the excess mortality due to the disease. Andersson et al. [90] extended this approach by fitting flexible parametric relative survival models to estimate $R(u \mid \boldsymbol{x})$, showing improved accuracy in estimating LE compared to directly extrapolating $S(t)$ in an ASF. Figure 6.6(A) illustrates the estimation of LE as the area under the all-cause survival curve.

To estimate LE within a multistate framework, the relative survival extrapolation approach has been extended to a multistate model, described in Section 6.2.1. Consider a multistate model consists of L states, of which M are transient (non-absorbing) states, and the remaining $(L - M)$ states are absorbing (excess and expected death). The all-cause survival probability equals to the sum of the probabilities of being in any of the transient states. Hence, LE can be expressed as the integrated sum of the state occupancy probabilities over time:

$$\begin{aligned} LE(\boldsymbol{x}) &= \int_{u=0}^{t_{\max}} \sum_{i=1}^{M} P_i(u \mid \boldsymbol{x})\, du \\ &= \sum_{i=1}^{M} T_i(\boldsymbol{x}), \end{aligned}$$

where $P_i(u \mid \boldsymbol{x})$ is the probability of occupying state i at time u, $t_{\max}$ the time horizon when survival reaches zero (that is, $\lim_{u \to t_{max}} \sum_{i=1}^{M} P_i(u \mid \boldsymbol{x}) = 0$), and $T_i(\boldsymbol{x}) = \int_0^{t_{\max}} P_i(u \mid \boldsymbol{x})\, du$ the sojourn time spent in state i for lifetime.

This formulation generalises the definition of LE as the area under the survival curve within a multistate framework. Transition hazards can likewise be extrapolated within the RSF, thereby separating excess hazard (disease-specific) from expected hazard (background mortality). By simulating individual patient trajectory through the multistate model (described in Section 6.2.1), one can estimate LE from a multistate model within an RSF. Figure 6.7(A) illustrates estimating LE in a multistate framework.

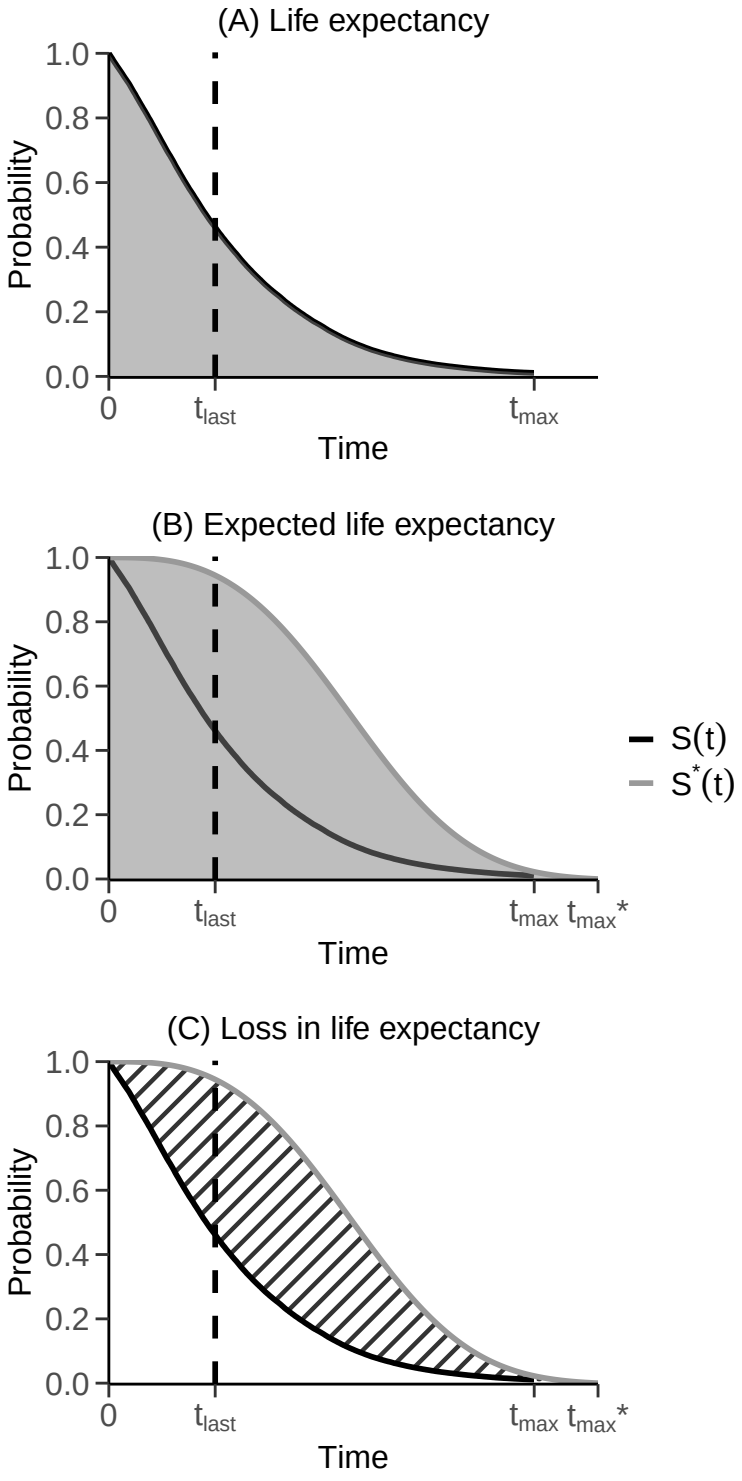


Figure 6.6. Illustration of estimating loss in life expectancy. Panel (A) shows the all-cause survival, $S(t)$, estimated from a patient cohort following a disease diagnosis; the area under this curve corresponds to the life expectancy for the cohort. Panel (B) presents the expected survival, $S^*(t)$, for a comparable population without the disease; the area under this curve reflects the life expectancy of that disease-free population. Panel (C) illustrates the loss in life expectancy (striped area), calculated as the difference between the areas under the curves $S^*(t)$ and $S(t)$. The point t_{last} marks the end of observed follow-up, beyond which $S(t)$ and $S^*(t)$ are extrapolated until t_{max} and t^*_{max}, respectively, until both survival functions reach zero.

Loss in life expectancy. Loss in life expectancy (LLE) quantifies the number of life years lost due to a disease [16, 89], providing a meaningful measure of disease burden and prognosis. LLE is defined as the difference between the LE for a comparable population without the disease of interest, such as the age-, sex- and calendar year-matched general population, denoted LE*, and the LE for the patient cohort, denoted LE, as follows:

$$\begin{aligned} LLE(\boldsymbol{x}) &= LE^*(\boldsymbol{x}) - LE(\boldsymbol{x}) \\ &= \int_0^{t^*_{max}} S^*(u \mid \boldsymbol{x'})\, du - \int_0^{t_{max}} S(u \mid \boldsymbol{x})\, du, \\ &= \int_0^{t^*_{max}} S^*(u \mid \boldsymbol{x'})\, du - \int_0^{t_{max}} S^*(u \mid \boldsymbol{x'})\, R(u \mid \boldsymbol{x})\, du, \end{aligned} \tag{6.9}$$

where $S^*(u \mid \boldsymbol{x'})$ denotes the expected survival of a disease-free population, $S(u \mid \boldsymbol{x})$ the all-cause survival of the patient cohort, $R(u \mid \boldsymbol{x})$ the relative survival estimated from a fitted parametric relative survival model, where $t_{\max}$ and t^*_{max} are the times at which the all-cause and expected survival curves reach zero, respectively. In practice, survival curves may be extrapolated to $\max(t_{\max}, t^*_{\max})$, the time point at which both curves approach zero. LLE Figure 6.6(A) to (C) illustrates LLE as the area between the all-cause and expected survival curves.

In a multistate framework, patients can either stay or transition among multiple health states. This contrasts with standard survival analysis, where individuals are assumed to remain in a single state until the event occurs. LLE is defined as:

$$\begin{aligned} LLE(\boldsymbol{x}) &= LE^*(\boldsymbol{x'}) - LE(\boldsymbol{x}) \\ &= \int_0^{t^*_{max}} S^*(u \mid \boldsymbol{x'})\, du - \int_{u=0}^{t_{\max}} \sum_{i=1}^{M} P_i(u \mid \boldsymbol{x})\, du \\ &= \int_0^{t^*_{max}} S^*(u \mid \boldsymbol{x'})\, du - \sum_{i=1}^{M} T_i(\boldsymbol{x}), \end{aligned} \tag{6.10}$$

where $P_i(u \mid \boldsymbol{x})$ denotes the probability of being in transient (non-absorbing) state i at time u, and $T_i(\boldsymbol{x}) = \int_0^{t_{\max}} P_i(u \mid \boldsymbol{x})\, du$ is the sojourn time spent in state i over the lifetime horizon. The sum $\sum_{i=1}^{M} T_i(\boldsymbol{x})$ equals the LE estimated from the multistate model. Figure 6.7 shows the LE as the area under the survival curve for a patient cohort in Panel (A), the expected LE as the area under the expected survival curve in Panel (B), and the resulting LLE as the difference in Panel (C).

Regardless of estimating LLE in a standard survival analysis or a multistate framework, the expected LE part is derived by extrapolating the expected survival using the same approach. Expected survival, $S^*(u \mid \boldsymbol{x'})$, is typically calculated from general population mortality rates by matching age, sex, and calendar year. For extrapolation beyond observed data, assumptions about future mortality trends must be specified. For example, common approaches contain assuming that mortality rates remain constant at the level of the last observed year for future years [90] or incorporating projections from demographic statistics [60].

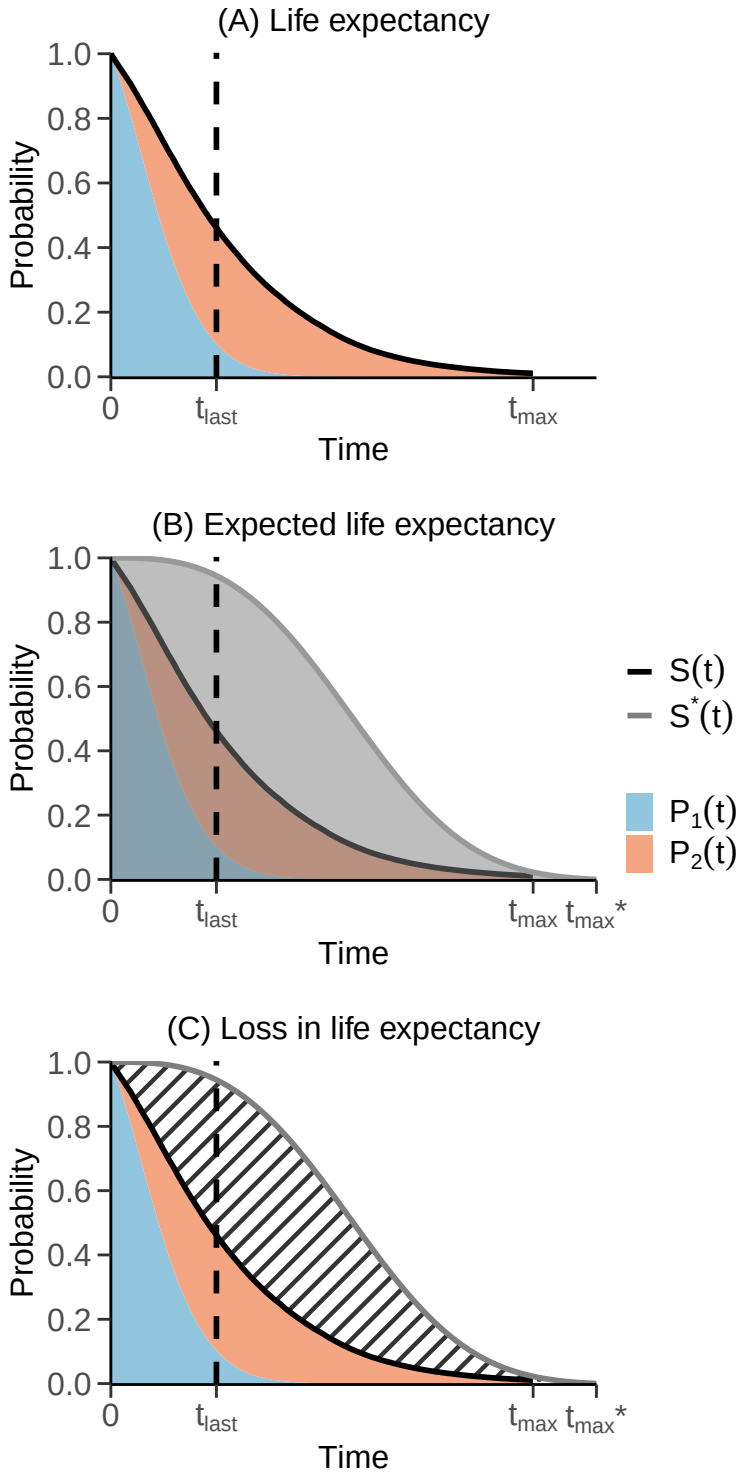


Figure 6.7. Illustration of estimating loss in life expectancy using a multistate framework (example of an irreversible illness-death model). Panel (A) shows the all-cause survival, $S(t)$, estimated from a patient cohort following disease diagnosis, calculated as the sum of the state occupancy probabilities, $P_1(t) + P_2(t)$; the area under this curve corresponds to the life expectancy of the patient cohort. Panel (B) presents the expected survival, $S^*(t)$, for a comparable disease-free population, representing the life expectancy of a population without the disease. Panel (C) illustrates the loss in life expectancy (striped area), calculated as the difference between the areas under the curves $S^*(t)$ and $S(t)$. The point t_{last} indicates the end of observed follow-up, beyond which $S(t)$ and $S^*(t)$ are extrapolated until $t_{\max}$ and $t^*_{\max}$, respectively, until both survival functions reach zero.

Proportion of loss in life expectancy. While LLE quantifies the absolute number of life years lost due to the disease, it is also informative to present the loss relative to what would have been expected in the absence of disease. The proportion of loss in life expectancy (PLLE) serves this purpose to rescale the loss to a proportion, defined as:

$$\begin{aligned} PLLE(\boldsymbol{x}) &= \frac{LLE(\boldsymbol{x})}{LE^*(\boldsymbol{x}')} \\ &= \frac{LE^*(\boldsymbol{x}') - LE(\boldsymbol{x})}{LE^*(\boldsymbol{x}')} \\ &= 1 - \frac{\int_0^{t_{max}} S(u \mid \boldsymbol{x})\,\mathrm{d}u}{\int_0^{t^*_{max}} S^*(u \mid \boldsymbol{x}')\,\mathrm{d}u} \end{aligned} \tag{6.11}$$

where $LE^*(\boldsymbol{x}')$ is the expected LE of a disease-free population, and $LE(\boldsymbol{x})$ is the LE of a patient cohort.

This proportional measure provides an alternative interpretation to the absolute loss of life years (LLE): for instance, a PLLE of 20% indicates that the disease reduces LE by 20% relative to a disease-free population.

6.3.2 Quality-adjusted life expectancy and loss in quality-adjusted life expectancy

Quality-adjusted life expectancy. QALE, measured in QALYs, extends the concept of LE by weighting each unit of survival time by the HSUV of each state. In a multistate setting with M transient states, let $w_i(t \mid \boldsymbol{x}) \in [0, 1]$ denote the utility of state i for an individual with covariates $\boldsymbol{x}$ at time t. In practice, $w_i(t)$ is typically modelled as a piecewise-constant function, such as one value for the first year in state i and another for subsequent years, and may be further modified by attained age, sex, or other covariates. In most applications, $w_i(t \mid \boldsymbol{x}) = 1$ corresponds to perfect health, and $w_i(t \mid \boldsymbol{x}) = 0$ represents death. Utilities are usually derived from validated instruments, such as EQ-5D [44], or collected longitudinally in the study; if these data are not available, published utility values may be obtained from literature (see Section 2.4.4 for details). QALE is defined as:

$$QALE(\boldsymbol{x}) \;=\; \int_{u=0}^{t_{\max}} \sum_{i=1}^{M} w_i(u \mid \boldsymbol{x})\, P_i(u \mid \boldsymbol{x})\, du, \tag{6.12}$$

where $P_i(u \mid \boldsymbol{x})$ is the state occupancy probability for state i at time u, and $t_{\max}$ is the time horizon when survival reaches zero.

In this thesis, the estimation of QALE is done in three steps:

1. estimate and extrapolate each transition-specific hazard between states;
2. specify utility weights $w_i(u \mid \boldsymbol{x})$ for each state;
3. simulate individual trajectories to obtain the weighted state-occupancy probabilities, $w_i(u \mid \boldsymbol{x})P_i(u \mid \boldsymbol{x})$. Numerical integration yields the QALE for each individual.

When utilities are assumed to remain constant over time within each state (that is, $w_i(u \mid \boldsymbol{x}) = w_i(\boldsymbol{x})$ for all u), the integration reduces to a weighted sum of sojourn times:

$$\begin{aligned} QALE(\boldsymbol{x}) &= \int_{u=0}^{t_{\max}} \sum_{i=1}^{M} w_i(\boldsymbol{x})\, P_i(u \mid \boldsymbol{x})\, du \\ &= \sum_{i=1}^{M} w_i(\boldsymbol{x}) \cdot T_i(\boldsymbol{x}), \end{aligned}$$

where $T_i(\boldsymbol{x})$ is the sojourn time spent in state i over the lifetime horizon.

When utilities vary over time due to treatment effects or age-related comorbidity effects, it is preferable to model $w_i(t \mid \boldsymbol{x})$ explicitly. For instance, following the multiplier method by Brazier et al. [91], let $w_i^{\text{ref}}(t)$ be the disease-specific HSUV for state i, measured at reference age a_0 using data from a patient population from Country X. Let $g(a) \in [0, 1]$ be the age-specific HSUV for the general population in Country Y at attained age a, corresponding to the expected utility for an age-matched individual in that population. This enables the application of utility estimates derived in Country X to a target population in Country Y with a different age distribution using the multiplier, defined as:

$$m_i(t) = \frac{w_i^{\text{ref}}(t)}{g(a_0)}.$$

Using the multiplier method, the age-specific HSUV for state i for patients at age a_1 in Country Y is given by:

$$w_i(t \mid a_1) = m_i(t) \cdot g(a_1).$$

This age-adjusted HSUV can then be incorporated into the QALE calculation in Equation (6.12), where the utility $w_i(u \mid \boldsymbol{x})$ now explicitly accounts for both disease-specific effects and age-related comorbidity through the multiplier method.

Loss in quality-adjusted life expectancy. Loss in quality-adjusted life expectancy (LQALE) extends the concept of LLE by incorporating HSUVs. It quantifies the QALYs lost due to a disease, accounting for both mortality and morbidity impacts.

Let $w^*(u)$ denote the utility value for the general population at time u, typically derived from health surveys or published literature on the general population's health-related quality of life (HRQoL). The QALE for the general population is denoted as $QALE^*$, defined as:

$$QALE^*(\boldsymbol{x'}) = \int_0^{t^*_{max}} w^*(u \mid \boldsymbol{x'})\, S^*(u \mid \boldsymbol{x'})\, du, \tag{6.13}$$

where $S^*(u \mid \boldsymbol{x'})$ is the expected survival of a disease-free population, typically derived from the general population mortality rates matched by age, sex, and calendar year. The LQALE is then defined as:

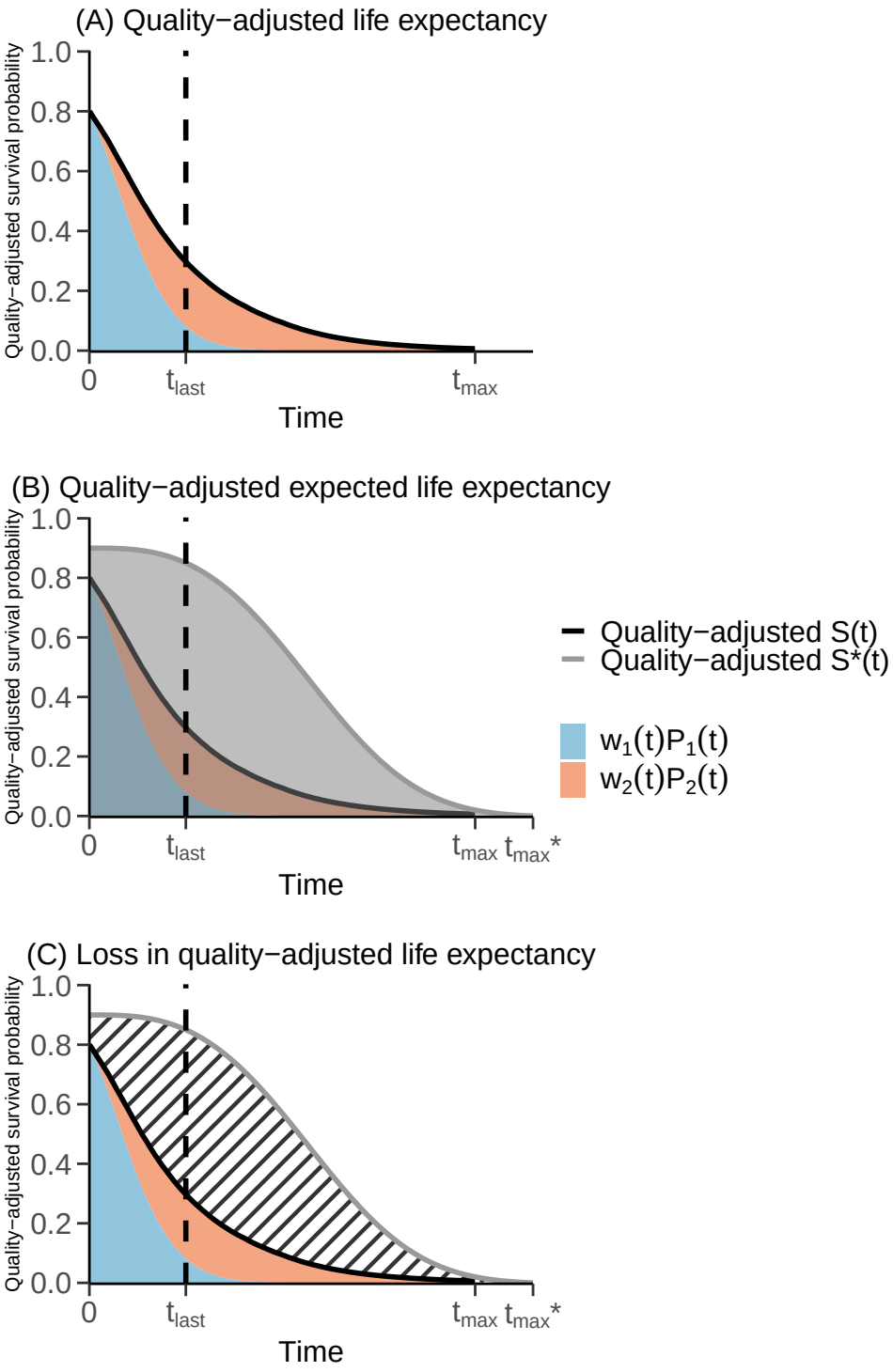


Figure 6.8. Illustration of estimating loss in quality-adjusted life expectancy (QALE) using a multistate framework (irreversible illness-death model). Panel (A) shows quality-adjusted survival, $w_1(t)P_1(t) + w_2(t)P_2(t)$, based on utility weights $w_i(t)$ and state occupancy probabilities $P_i(t)$ estimated from a patient cohort. The area under the curve corresponds to the QALE of a disease-free population. Panel (B) presents the expected quality-adjusted survival for a comparable disease-free population. The area under the curve is considered as the QALE of that population. Panel (C) shows the loss in QALE (striped area), calculated as the difference between the areas under the quality-adjusted $S^*(t)$ and $S(t)$. The point t_{last} marks the end of observed follow-up, beyond which both curves are extrapolated to t_{max} and t^*_{max}, respectively, until both survival functions reach zero.

$$
\begin{aligned}
LQALE(\boldsymbol{x}) &= QALE^*(\boldsymbol{x'}) - QALE(\boldsymbol{x}) \\
&= \int_0^{t^*_{max}} w^*(u \mid \boldsymbol{x'})\, S^*(u \mid \boldsymbol{x'})\, du - \int_{u=0}^{t_{\max}} \sum_{i=1}^{M} w_i(u \mid \boldsymbol{x})\, P_i(u \mid \boldsymbol{x})\, du,
\end{aligned} \tag{6.14}
$$

where $w_i(u \mid \boldsymbol{x})$ represents the HSUV for state i at time u, and $P_i(u \mid \boldsymbol{x})$ is the corresponding state occupancy probability estimated from the patient cohort. The first integral represents the expected QALE from the general population, while the second represents the expected QALE for an individual with the disease, based on the multistate model. Figure 6.8 illustrates that the QALE as the area under the quality-adjusted survival curve for a patient cohort in Panel (A), the expected QALE as the area under the quality-adjusted expected survival curve in Panel (B), and the resulting LQALE as the difference in Panel (C).

Proportion of loss in quality-adjusted life expectancy. The proportion of loss in quality-adjusted life expectancy (PLQALE) represents the loss as a proportion of the expected QALE in the absence of disease, and is defined as:

$$
\begin{aligned}
PLQALE(\boldsymbol{x}) &= \frac{LQALE(\boldsymbol{x})}{QALE^*(\boldsymbol{x'})} \\
&= \frac{QALE^*(\boldsymbol{x'}) - QALE(\boldsymbol{x})}{QALE^*(\boldsymbol{x'})} \\
&= 1 - \frac{\int_{u=0}^{t_{\max}} \sum_{i=1}^{M} w_i(u \mid \boldsymbol{x})\, P_i(u \mid \boldsymbol{x})\, du}{\int_0^{t^*_{max}} w^*(u \mid \boldsymbol{x'})\, S^*(u \mid \boldsymbol{x'})\, du},
\end{aligned} \tag{6.15}
$$

where $QALE(\boldsymbol{x})$ is the QALE for the patient cohort and $QALE^*(\boldsymbol{x'})$ the expected QALE from a disease-free population as defined in Equation (6.13). For instance, a PLQALE of 35% for a patient cohort indicates a 35% reduction in QALE compared to the general population matched by age, sex, and calendar year.

6.3.3 Incremental cost-effectiveness ratio

The ICER quantifies the additional cost required to gain one extra effectiveness when comparing two interventions A and B, yielding:

$$
\text{ICER} = \frac{\Delta\overline{\text{Cost}}}{\Delta\overline{\text{QALY}}} = \frac{\overline{\text{Cost}}_A - \overline{\text{Cost}}_B}{\overline{\text{QALY}}_A - \overline{\text{QALY}}_B}, \tag{6.16}
$$

where $\Delta\overline{\text{Cost}} = \overline{\text{Cost}}_A - \overline{\text{Cost}}_B$ denotes the difference in mean total costs, and $\Delta\overline{\text{QALY}} = \overline{\text{QALY}}_A - \overline{\text{QALY}}_B$ denotes the difference in mean total QALYs. This corresponds to Equation (2.30). See Section 2.4.2 for a detailed introduction to the use of ICER in cost-effectiveness analysis.

Willingness-to-pay threshold The WTP threshold, γ, also known as the cost-effectiveness threshold, specifies the maximum amount one is willing to pay for one additional QALY. An intervention is considered cost-effective if ICER $\leq \gamma$. In this case, intervention A is considered more cost-effective than B. In contrast, if ICER $> \gamma$, it is not.

Probabilistic sensitivity analysis PSA aims to capture and demonstrate uncertainty in model inputs (for example costs, HSUVs, and parametric survival models). Typically a large number of Monte Carlo simulations, such as 1 000 to 10 000 iterations, is performed [35]. In each iteration, sample all parameters from their distributions, recalculate costs and QALYs for interventions A and B, and compute an ICER. This process generates $(\Delta\overline{\text{QALY}}, \Delta\overline{\text{Cost}})$ points on the cost-effectiveness plane, revealing uncertainty around the ICER for the base case analysis. Figure 6.9(A) gives an example of PSA with 1000 Monte Carlo simulations with a WTP threshold of USD 20 000 per QALY gained.

Cost-effectiveness acceptability curve The cost-effectiveness acceptability curve (CEAC) uses PSA results to show, for each WTP threshold γ, the probability that intervention A is cost-effective compared to B. This is equivalent to the probability that the net monetary benefit (NMB) [35] of A relative to B is ≥ 0, where the incremental NMB is defined as:

$$\begin{aligned}
\Delta\overline{\text{NMB}} &= \overline{\text{NMB}_A} - \overline{\text{NMB}_B} \\
&= \left(\gamma \cdot \overline{\text{QALY}_A} - \overline{\text{Cost}_A}\right) - \left(\gamma \cdot \overline{\text{QALY}_B} - \overline{\text{Cost}_B}\right) \\
&= \gamma\left(\overline{\text{QALY}_A} - \overline{\text{QALY}_B}\right) - \left(\overline{\text{Cost}_A} - \overline{\text{Cost}_B}\right) \\
&= \gamma \times \Delta\overline{\text{QALY}} - \Delta\overline{\text{Cost}}.
\end{aligned}$$

Thus,

$$P(\text{A is cost-effective} \mid \gamma) \;=\; P(\gamma \times \Delta\overline{\text{QALY}} - \Delta\overline{\text{Cost}} \geq 0).$$

Plotting this probability against γ yields a curve shown in Figure 6.9(B), which approaches zero at very low γ and one at very high γ. The CEAC helps decision makers understand how likely an intervention is to be cost-effective across different WTP thresholds.

6.3.4 Prevalence costs

Prevalence costs represent the economic burden associated with individuals of a defined population living with a disease at a given time. These costs capture ongoing treatment, monitoring, and supportive care expenses, and are essential for understanding the economic impact of disease on healthcare systems and society. While prevalence costs can be applied to any chronic condition, the applications in this thesis focus specifically on cancer. Therefore, the methodological framework and examples presented below are within applications to cancer epidemiology and health economics.

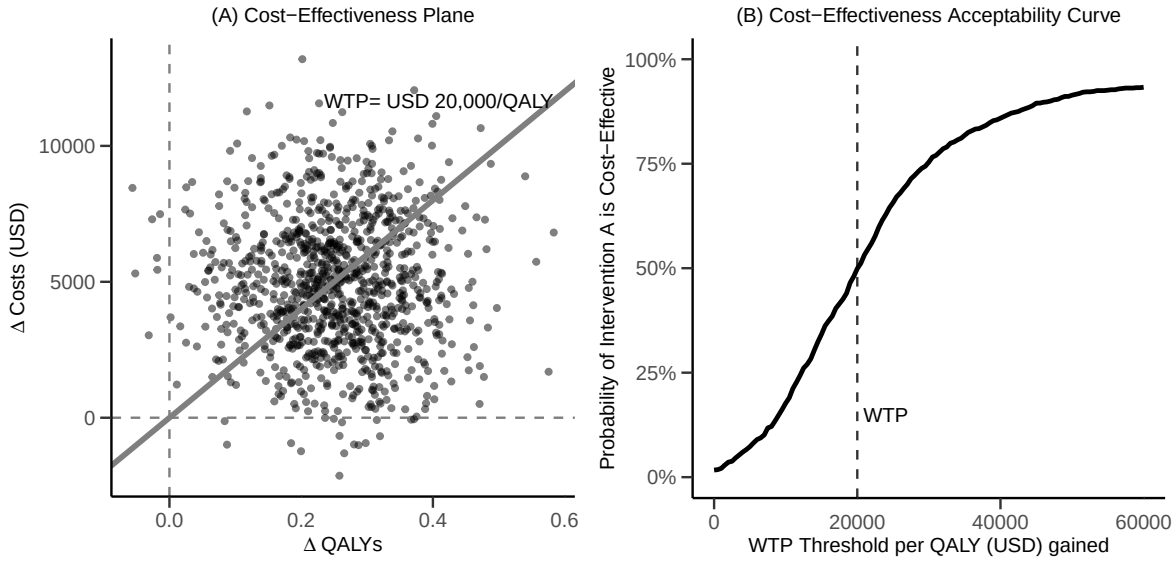


Figure 6.9. Example of an cost–effectiveness analysis of intervention A versus intervention B. (A) Cost–effectiveness plane based on 1,000 Monte Carlo simulations. The solid line indicates a WTP threshold of USD 20000 per QALY gained. Points falling below (and to the right of) this line are considered cost-effective. (B) Cost–effectiveness acceptability curve shows that the probability that intervention A is cost-effective at thresholds ranging from USD 0 to USD 60000 per QALY gained. WTP, willingness-to-pay; QALY, quality-adjusted life year.

Estimating cancer prevalence. For a given birth cohort, the cancer prevalence at age i, denoted v_i, is the probability of having been diagnosed with cancer by age i, conditional on surviving to that age.

Using the Prevalence, Incidence, Analysis MODel (PIAMOD) approach [92], prevalence is calculated by summing, over all ages up to i, the proportion still cancer-free at age j, multiplied by the probability of being diagnosed with cancer at that age, and the probability of surviving cancer-specific mortality from age j to i:

$$v_i = \sum_{j=0}^{i} (1 - v_j)\, \mu_j\, R_{ji}, \tag{6.17}$$

where:

- $(1 - v_j)$ is the proportion of individuals without cancer at age j;
- μ_j is the probability of being diagnosed with cancer within the age interval $[j, j + 1)$;
- R_{ji} is the net survival probability from age j to i following cancer diagnosis at age j.

The incidence probability μ_j is estimated using age-period-cohort (APC) modelling. Let m_{it} be the number of new cancer cases at age i and calendar year t, P_{it} the corresponding population size at risk, and μ_{it} the incidence rate (cases per person-year) at age i and year t. Since the incidence rate is modelled at one-year intervals, the incidence probability and incidence rate are numerically equivalent. Assume that

$$m_{it} \sim \text{Poisson}(\mu_{it} P_{it}).$$

We model the logit of the incidence probability, μ_{it}, as a polynomial function of age, period, and cohort:

$$\text{logit}(\mu_{it}) = a_0 + \sum_{k=1}^{K_1} a_k i^k + \sum_{k=1}^{K_2} b_k t^k + \sum_{k=1}^{K_3} c_k (t-i)^k, \tag{6.18}$$

where a_0 is the intercept, i denotes age, t denotes calendar year, and $(t-i)$ denotes birth cohort. K_1, K_2, K_3 are the polynomial degrees for age, period, and cohort effects, with a_k, b_k, c_k as the corresponding coefficients.

The net survival R_{ji} can be estimated using an RSF (see Section 2.3 for details on net survival). The estimation can use either non-parametric methods (for example, the Pohar-Perme estimator [30]) or parametric approaches (for example, mixture cure models or FPMs). For projections beyond the available follow-up data, one can either assume net survival remains constant at levels observed in the most recent data, or extrapolate to the future using parametric models fitted to the observed data.

Equation (6.17) yields the age-specific prevalence for each birth cohort, given known age-specific incidence μ_j and net survival R_{ji}. To translate these cohort-specific prevalences into calendar year estimates, one must account for the fact that individuals of the same age in a given calendar year belong to different birth cohorts.

For calendar year t, let z denote the birth cohort satisfying $t = z+i$, where i represents age. The average prevalence at age i during year t, denoted $p_i(t)$, is approximated by averaging the prevalences of the two cohorts that contribute to age i in year t:

$$p_i(t) = \frac{v_i^{(z)} + v_{i+1}^{(z-1)}}{2},$$

where:

- $v_i^{(z)}$ is the cohort-specific prevalence at age i for cohort z (individuals who turn age i during year t);
- $v_{i+1}^{(z-1)}$ is the cohort-specific prevalence at age $i+1$ for cohort $(z-1)$ (individuals who were already age i at the beginning of year t).

This approach accounts for the continuous flow of individuals through each age during the calendar year. The total number of prevalent cases $V_i(t)$ at age i during year t is then obtained by multiplying the prevalence proportion, $p_i(t)$, by the corresponding population size at age i during year t, $N_i(t)$:

$$V_i(t) = p_i(t) \times N_i(t).$$

Estimating prevalence costs. The prevalence cost at a given year represents the economic burden of all individuals living with a given disease during a specific calendar year. To calculate this cost, one approach is to stratify the prevalent population into disease states that reflect different phases of the cancer treatment and their associated costs.

Let S denote the total number of disease states, and let $C_s(t)$ represent the average annual cost per patient in disease state s during calendar year t. The prevalence cost for a given cancer type in calendar year t, denoted $K(t)$, is calculated as:

$$K(t) = \sum_{s=1}^{S} C_s(t) \times V_s(t), \tag{6.19}$$

where:

- $C_s(t)$ is the average annual cost per patient in state s during year t;
- $V_s(t)$ is the number of prevalent patients in state s during year t;
- S is the total number of transient (non-absorbing) states considered in the model.

For various cancers, the average annual cost per patient would rise preceding death due to cancer. One approach to capturing this increase is to separate the last year of life as a distinct phase in the analysis [93]. Our CML natural history model (see Section 6.2.2) did not explicitly model the costs for the last year of life. Nevertheless, CML patients who die from the cancer itself usually progress to the progression state (defined in Figure 6.5), which therefore implicitly captures the increasing costs at the end of life.

Chapter 7 Overview of the studies

The subsequent sections present key insights from each study, following a structure that includes the study title, knowledge gap, aim, study population, methods, main results, discussion, and conclusions. These sections summarise each study and critically evaluate its methodological contributions, strengths, and limitations. As the first author of all studies included in this thesis, I use "we" throughout this chapter to acknowledge the collaborative contributions of my co-authors.

7.1 Study I

Title

Comparing Survival Extrapolation within All-Cause and Relative Survival Frameworks by Standard Parametric Models and Flexible Parametric Spline Models Using the Swedish Cancer Registry

Knowledge gap

Previous studies have assessed how well parametric models extrapolate survival using cancer registry data. Gray et al. [94] compared SPMs and FPMs for projecting 10-year survival within an ASF using the Surveillance, Epidemiology, and End Results (SEER) data; they recommended the use of FPMs. However, their analysis was limited to the ASF and short-term survival outcomes, without explicitly assessing LE. Although the 10-year RMST may approximate LE in some cohorts, this aspect was not thoroughly examined. In contrast, Andersson et al. [90] focused on lifetime survival extrapolation using FPMs within an RSF using the Swedish Cancer Register, but their primary interest was estimating LLE rather than shorter-term measures like 10-year RMST. A comprehensive evaluation across two survival frameworks, different parametric model families, and varying time horizons remains lacking. Table 7.1 summarises the knowledge gap by comparing previous studies by Gray et al. [94] and Andersson et al. [90] with Study I.

Aim

In Study I, we aim to evaluate the accuracy of survival extrapolation methods. We compare standard and flexible parametric models within all-cause and relative survival frameworks, using data from the Swedish Cancer Register to project 10-year RMST, LE, and survival proportions at both 10 years and lifetime.

Table 7.1. Comparison of previous studies and the Study I by survival framework (ASF versus RSF), model family (SPMs versus FPMs) and outcome (10-year RMST versus LE).

Studies	Outcome		ASF		RSF	
	10-year RMST	LE	SPMs	FPMs	SPMs	FPMs
Gray et al. [94]	X	-	X	X	–	–
Andersson et al. [90]	-	X	–	X	Only Weibull	X
Study I	X	X	X	X	X	X

ASF, all-cause survival framework; RSF, relative survival framework; LE, life expectancy RMST, restricted mean survival time; SPMs, standard parametric models; FPMs, flexible parametric models.

Study population

We identified patients from the Swedish Cancer Register diagnosed between 1981 and 1990 at ages 18 to 99 years, with follow-up until December 31, 2020. The study population included five cancer types classified according to ICD version 7: colon cancer (code 153.x), breast cancer (code 170.x), malignant melanoma (code 190.x), prostate cancer (code 177.x), and CML (code 205.1). We further stratified the patients of each cancer into three age groups: 18–59, 60–69, and 70–99 years to produce a total of 15 cancer cohorts.

Data on date of death were obtained through linkage with the Cause of Death Register using Swedish unique personal numbers. Patients were censored at the date of first emigration, where applicable, based on data from the Total Population Register. For patients with multiple primary tumours of the same cancer type, we included only the first diagnosis. Autopsy-diagnosed cases were excluded from the analysis. For breast cancer, male patients (<1% of cases) were excluded.

Figure 7.1 summarises the study population for Study I, illustrating the linkage of registers, inclusion period for cancer diagnoses (1981–1990) and the follow-up until the end of 2020.

Methods

Evaluating survival extrapolation. For observed survival, we used K–M estimates from 40-year follow-up data from the Swedish Cancer Register on 15 cancer cohorts (five cancer sites by three age groups). To predict 10-year outcomes (10-year RMST and 10-year survival), follow-up was right-censored at 2, 3, or 5 years and extrapolated to 10 years (corresponding to fitting a parametric model to t_{last} and projecting to $t_\tau = 10$, as in Figure 6.1(B)). For lifetime outcomes (LE or 40-year RMST, and survival proportions at lifetime, or 40 years), follow-up was right-censored at 2, 3, 5, or 10 years and extrapolated to lifetime or 40 years ($t_{\max} = 40$). If the curve did not reach zero by 40 years (particularly for younger patients),

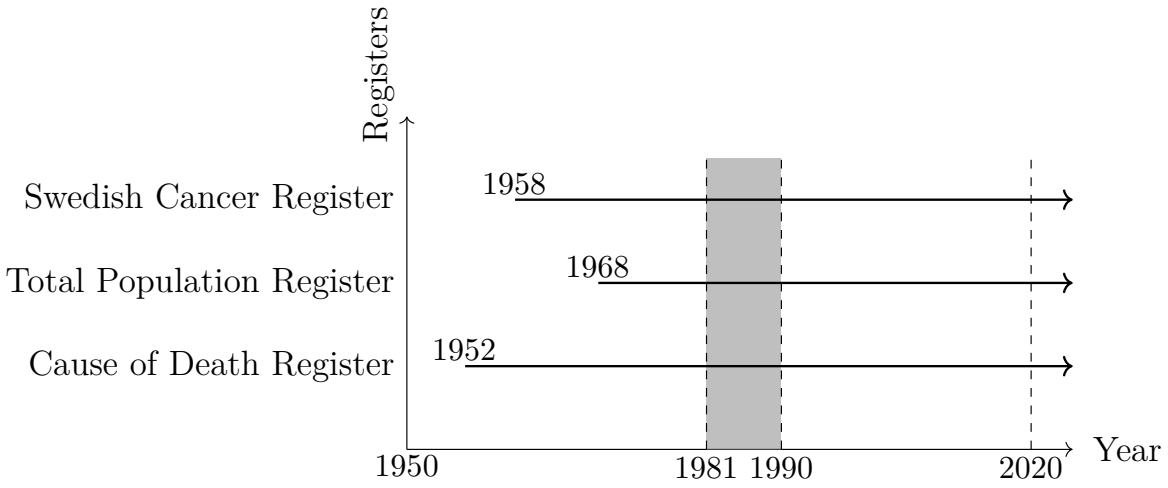


Figure 7.1. Study population for Study I. The shaded region (1981–1990) represents the inclusion period for cancer diagnoses, with follow-up until 2020.

we reported 40-year RMST instead. Nine parametric models (six SPMs and three FPMs) were fitted under an all-cause or a relative survival framework. We used box plots to show the distribution of differences (extrapolated − observed) across all cohorts and models. A summary table listed how many models achieved differences within 0.1 years for RMST/LE or within 1 percentage point for survival probabilities. We also provided both survival and hazard plots for visual assessment of extrapolation performance.

Main results

Baseline characteristics for cancer cohorts. We analysed survival data for 122 875 individuals diagnosed between 1981 and 1990 with one of five cancer types: colon (n = 24 683), breast (n = 43 769), malignant melanoma (n = 11 494), prostate (n = 41 980) and CML (n = 949), identified from the Swedish Cancer Register. Patients were followed until 31 December 2020, providing up to 40 years of observed survival. Prognosis varied across cancers. Adults aged 18–59 years with melanoma showed the best survival outcomes (10-year RMST = 8.70 years; 10-year survival = 78.79 %), whereas CML patients aged 70–99 years had the poorest (10-year RMST = 1.88 years; 10-year survival = 1.08 %).

In total, we evaluated 1 890 survival extrapolation scenarios: 5 cancer types × 3 age groups × 9 parametric models (six SPMs versus three FPMs) × 2 modelling frameworks (ASF versus RSF) × (3 follow-up cutoffs for 10-year extrapolations, plus 4 cutoffs for lifetime or 40-year extrapolations) .

Evaluating 10-year survival extrapolation. When projecting 10-year survival outcomes from only two years of follow-up data, FPMs overall demonstrated superior performance compared to SPMs. FPMs predicted 10-year RMST within ±1 year of difference from the observed across both ASF and RSF. In contrast, SPMs showed a higher frequency of

predictions exceeding ±1 year difference.

When predicting 10-year survival proportion from two years of follow-up, all models produced substantial differences, ranging from -67.42% to 26.68%. However, extending the follow-up period to five years substantially improved prediction accuracy. Specifically, FPMs achieved notably better performance with 10-year survival proportion differences narrowed to -4.88% to 4.61%, while SPMs continued to show greater prediction differences.

Evaluating lifetime or 40-year survival extrapolation. Prediction differences at the lifetime horizon varied systematically by survival framework. With only two years of follow-up, none of the models, regardless of survival framework, extrapolated accurately for either LE or survival proportion at lifetime. Whenever discrepancies between observed and predicted arose, models within the RSF tended to underestimate survival, whereas those within the ASF tended to overestimate. The most evident example was the Gompertz model within an ASF extrapolated from three years of data for colon cancer patients aged 60–69 years, which overestimated the lifetime survival proportion by 46.00%.

Accuracy improved substantially once follow-up was extended to ten years, particularly for models within an RSF. Compared to SPMs, FPMs within an RSF generated predictions aligning more closely with observed survival outcomes. The highest difference of survival proportion here was −16.16% for melanoma patients aged 18–59 years extrapolated with an exponential distribution in an RSF. In contrast, models within an ASF overestimated survival, with larger differences across multiple cancer types and age groups.

Discussion

Survival extrapolation. For 10-year predictions, the RSF provided limited improvement over the ASF, as the population mortality contributed minimally to the all-cause hazard during this period. The choice of parametric model, particularly the ability to flexibly represent the hazard shape, was more influential than the survival framework itself. In this context, FPMs overall projected more accurately than SPMs.

In contrast, for lifetime (or 40-year) extrapolations, the RSF consistently predicted closer survival to the observed data than the ASF. This can be explained by the RSF's incorporation of background mortality rates, which increasingly contribute to the all-cause hazard as the patient cohort ages. As a result, the choice of survival framework became more critical than the choice of parametric model. Models using an ASF tended to overestimate all-cause survival by underestimating all-cause hazards in the extrapolated period, while RSF models often underestimated all-cause survival due to overestimating excess hazards.

Uncertainty. In this study, we primarily addressed structural uncertainty in survival extrapolation, which refers to uncertainty arising from the choice of model structure and functional form (see Section 2.4.5 for details). We compared the extrapolation accuracy of the

RSF and ASF models under the same parametric distribution, follow-up length, cancer type, and age group. Across 1 890 scenarios, with long-term K-M estimates, we evaluated performance as the difference between predicted versus observed estimates for the area under the survival curve, and survival proportions at times.

Generalisability. We used data from 15 cancer cohorts from the Swedish Cancer Register (five cancer types, three age groups). Censoring survival data at multiple timepoints generated a large number of scenarios with known true survival outcomes. This design enabled assessment of extrapolation performance across a wide range of hazard shapes, which enhanced the generalisability to other diseases with varied survival patterns.

The sample sizes in our study were generally larger than those typically in clinical trials submitted for HTA appraisals. However, we also considered CML cohorts with fewer than 400 patients in each age group, which more closely resemble the sample sizes in clinical trial data. This variation in sample sizes strengthens the generalisability of our findings to a different study setting.

Previous research by Kearns [95] demonstrated that the length of follow-up has a greater impact on extrapolation performance than sample size. By evaluating varying hazard patterns and population characteristics, our study produced findings that may be generalised to other settings even with limited sample sizes.

Conclusions

For extrapolation to 10 years, FPMs overall predicted more accurately than SPMs, and there was no distinct difference between all-cause and relative-survival frameworks given the same parametric model. Over a lifetime horizon, extrapolation within an RSF agreed more closely with the observed survival than an ASF, especially when FPMs were used. When extrapolating survival from limited follow-up, models within an ASF tended to overestimate lifetime survival proportions, whereas those within an RSF were more likely to underestimate them.

7.2 Study II

Title

A Multistate Model Incorporating Relative Survival Extrapolation and Mixed Time Scales for Health Technology Assessment

Knowledge gap

Relative survival extrapolation has been established as a suitable method for predicting long-term survival in cancer patients [90, 96–98], but its integration into multistate models for

health economic evaluation remains limited. Current multistate models used in HTA predominantly employ SPMs that do not account for background mortality, which potentially leads to implausible long-term projections under a lifetime horizon.

Although several studies have explored the incorporation of relative survival into multistate frameworks through non-parametric [99], semi-parametric [100–102], and parametric [103] approaches, existing parametric methods have not been specifically developed to meet the methodological requirements of CEA. These requirements include: (i) parametric survival extrapolation beyond follow-up periods, (ii) the ability to model transitions using mixed time scales (expected and excess hazards are typically modelled on different time scales), and (iii) the application to PSA.

To our knowledge, no published multistate framework that combines extrapolation from an RSF and mixed time scales satisfies all requirements for CEA. This represents a methodological gap in modelling practice for HTA, which is especially crucial for accurately predicting long-term survival in order to estimate ICER.

Aim

In Study II, we aim to develop a multistate model that integrates relative survival extrapolation with mixed time scales, in particular for applications in CEA.

Study population

We utilised the CLL-8 trial data as an illustrative example to demonstrate the multistate model with relative survival extrapolation (detailed in Section 6.2.1), building upon and extending the cost-effectiveness analysis by Williams et al. [79].

The CLL-8 trial [75] was a phase 3, open-label, randomised clinical trial conducted across 190 centres in 11 countries. It included treatment-naive patients aged 30–81 years with immunophenotypically confirmed chronic lymphocytic leukaemia: either Binet stage C, or active disease in stages A or B. Eligible patients had Eastern Cooperative Oncology Group performance status 0–1, low comorbidity (cumulative illness rating scale ≤ 6), and a creatinine clearance ≥ 1.17 mL/s. The CLL-8 trial compared rituximab in combination with fludarabine and cyclophosphamide (RFC) versus fludarabine and cyclophosphamide alone (FC) as the first-line treatment of chronic lymphocytic leukaemia. It enrolled 810 individuals between 2003 and 2006, randomising 403 patients to the RFC arm and 407 to the FC arm. Baseline characteristics were well-balanced between treatment arms, with a median age of 61 years and 74% male participants [75]. Over the follow-up period, clinical outcomes differed between arms: the RFC arm experienced 106 progressions, 23 deaths after progression, and 21 deaths without progression, whilst the FC arm had 148 progressions, 27 deaths after progression, and 26 deaths without progression.

The original trial paper reported outcomes with a maximum follow-up of approximately 4

years [75]. To evaluate the performance of our survival extrapolation methods, we projected survival outcomes from 4 to 50 years using the developed multistate models and compared these long-term projections with subsequently published 8-year follow-up data [76].

Implementation of relative survival extrapolation within the multistate modelling framework required corresponding general population mortality rates for the CLL-8 trial population. Since the trial was conducted across 11 countries, we constructed mortality data using general population mortality rates from the HMD [27]. These rates were stratified by calendar year and age, and then weighted by the sex distribution of the trial population across the 11 participating countries, using country-specific mortality rates.

Methods

Evaluating survival extrapolation. We evaluated parametric model extrapolations by comparing their predicted transition-specific hazards against observed non-parametric hazards. All parametric models were fitted to the full 4-year CLL-8 data [75] and extrapolated to 50 years. We compared extrapolated survival curves from the multistate model used as the base-case analysis in William et al. [79] with our proposed multistate models using FPMs with survival extrapolation within an ASF or an RSF. For each model, we calculated sojourn times and RMSTs and survival proportions at 4, 8, 15, and 50 years (or lifetime) across treatments. We also overlaid hazards and survival functions for visual assessment. We evaluated 4-, 8-, 15-, and 50-year RMST and LE for participants in the CLL-8 trial [75]. We also calculated QALYs over a 15-year horizon to serve as the effectiveness measure in the subsequent CEA.

Extended multistate model with relative survival extrapolation and mixed time scales. The multistate framework described in Section 6.2.1 was implemented in `R` utilising the `microsimulation` [104] and `hesim` [105] packages. During the method development process, both packages were extended to simulate event times from transitions with mixed time scales. To carry out relative survival extrapolation, we extrapolated the excess and expected hazard separately. Specifically, for the excess hazard, event times were simulated from a fitted parametric relative survival model, while for the expected hazard, event times were simulated from a piecewise-constant exponential distribution using either the `C++` class `Rpexp` function in the `microsimulation` package [104] or the `rpexp` function in the `msm` package [106].

All code for the extended multistate model with relative survival extrapolation and mixed time scales is available at https://github.com/enochytchen/ChenEYT_microsim. This GitHub repository contains the full reproducible analysis for Study II.

Estimating incremental cost-effectiveness ratio. We calculated the ICER of RFC versus FC for the CLL-8 trial [75]. In doing so, we applied the results estimated using the

proposed multistate model incorporating relative survival extrapolation compared to the base case analysis by Williams et al. [79].

We calculated the ICER based on the total costs and QALYs over a 15-year horizon from the UK healthcare sector perspective. We conducted PSA using 10,000 Monte Carlo simulations. Finally, the CEAC was plotted to illustrate the probability that RFC is cost-effective compared to FC across WTP thresholds from £0 to £100 000 per QALY gained.

Main results

Comparing hazard functions. We estimated observed and extrapolated hazards for the three transitions of an irreversible illness–death model in the CLL-8 trial [75]: progression-free → progression, progression-free → death, and progression → death:

1. Progression-free → progression: modelled with Gompertz and FPMs;
2. Progression-free → death: modelled with generalised gamma models and FPMs within an ASF or an RSF;
3. Progression → death: modelled with Gompertz models and FPMs within ASF or RSF.

Comparing survival extrapolations. Three multistate specifications were fitted to 4-year data and projected to 50 years: (i) SPMs in an ASF (base case of Williams et al. [79]), (ii) FPMs in an ASF, and (iii) FPMs in an RSF (proposed in Section 6.2.1). Both FPM multistate models predicted the observed 8-year overall survival more closely than the SPM–ASF model. Short-term projections for multistate models with FPMs were nearly identical, but beyond 10 years the RSF model yielded modestly lower survival, reflecting background mortality.

Comparing cost-effectiveness analyses. With the FPM–RSF model, the incremental cost per QALY gained for RFC versus FC was £34 511, compared with £37 665 using the SPM–ASF model [79]. The PSA showed a 50.7 % chance that RFC is cost-effective at a £30 000 threshold, versus 36.3 % under the SPM–ASF framework. Although the FPMs showed higher uncertainty on the cost-effectiveness plane due to flexible spline functions, the RSF models considered background mortality and may provide more plausible long-term survival estimates than the ASF models.

Discussion

Survival extrapolation. In Study II, we extended the application of relative survival extrapolation into a multistate framework using an CEA of the CLL-8 trial as an example. To carry out relative survival extrapolation in a multistate model, one needs to predict from mixed time scales, as the excess and expected hazards are estimated using different time scales. Our results showed that FPMs, whether in an ASF or an RSF, overall agreed better with the observed data than the SPMs within an ASF, applied by Williams et al. [79]

However, none of the models (including ours) accurately projected the 8-year overall survival for the RFC arm.A subject matter expert suggested that a turning point should have appeared in the hazard function for the progression-free to progression transition after the 4-year observed period. However, the fitted hazard functions failed to capture this change. Both Williams et al.'s Gompertz model and our FPM may have been unable to reflect the turning point(s) in the long-term hazard shape.

Uncertainty. In this study, we primarily addressed structural uncertainty in health economic modelling by comparing alternative survival extrapolation frameworks in multistate models. We demonstrated that the selection of multistate models, including the choice of SPMs and FPMs, and between an ASF and an RSF, can substantially impact ICER estimates and their uncertainty.

Parameter uncertainty was partially addressed through PSA, which involved Monte-Carlo simulations from the joint distributions of transition-specific hazard parameters. However, this study had limitations in fully characterising parameter uncertainty, as survival data were reconstructed by digitising published K-M curves rather than by using individual-level patient data. As a result, we were not able to assess the parameter uncertainty introduced by original sample sizes and individual-level covariates in the CLL-8 trial.

Stochastic uncertainty was handled through individual-level simulations of patient trajectories. Heterogeneity was examined only with respect to treatment groups (RFC versus FC).

Overall, the primary contribution of Study II lies in (i) reducing structural uncertainty in multistate modelling by incorporating background mortality into long-term extrapolation and (ii) demonstrating how this proposed framework can accommodate parameter uncertainty through PSA.

Generalisability While only one case study (the base case analysis of Williams et al. [79] for the CLL-8 trial [75]) was analysed, the generalisability of the results to other clinical contexts remains limited. However, the main contribution of this study lies in its methodological innovation. The proposed framework demonstrates how relative survival extrapolation can be integrated within a multistate structure. This integration enables accounting for background mortality in a multistate modelling framework, which is particularly preferred when extrapolating survival outcomes over a lifetime horizon in an CEA.

The framework proposed in this study has been extended and applied to other disease areas and multistate models. For example, Study III adopted this proposed framework and extended it into an eight-state model. Future applications of this method across different disease and study populations would provide more evidence for this approach in health economic evaluations.

Conclusions

Integrating relative survival extrapolation and mixed time scales into multistate models introduces a novel approach to incorporate long-term external data when extrapolating from short-term follow-up. It thereby offers an alternative framework for cost-effectiveness studies using multistate models in HTA.

7.3 Study III

Title

Loss in Overall and Quality-Adjusted Life Expectancy for Patients with Chronic-Phase Chronic Myeloid Leukemia

Knowledge gap

Previous studies in Sweden [60] and the Netherlands [107] have shown that the LE for patients with CML has recently approached that of the general population. The improved survival has been mainly attributed to the introduction of TKIs [108]. However, these studies have focused on patient survival (quantity of life), without considering the potential impact on quality of life, which is also an important component of health outcomes.

Multistate models have been used to estimate QALYs or relevant survival outcomes for CML treatments [109–111], but most have extrapolated survival within an ASF rather than an RSF. Currently, patients with CML generally have a good long-term prognosis under TKI treatment [62]. Consequently, as patients age, background mortality rates become increasingly relevant for survival prediction, making extrapolation within an RSF preferable [96, 97].

To date, no study has used population-based data and a multistate model with relative survival extrapolation to estimate both LE and QALE, or their losses compared to the general population, for patients with CML. This gap limits our understanding of the impact of CML and its treatment on patients' long-term survival and quality of life.

Aim

In Study III, we aim to (i) develop a natural history model for CML treatment by applying and extending the multistate model from Study II, (ii) estimate overall LE and QALE among CP-CML patients in Sweden, and (iii) quantify losses in LE and QALE compared to the general population.

Study population

We included all patients aged 18–99 years diagnosed with CP-CML between 1 January 2007 and 31 December 2017, from the Swedish CML register [56]. Patients were followed from

diagnosis until death, emigration, or the end of follow-up (31 December 2018), whichever happened first.

Disease progression was defined as a transformation to accelerated or blastic phase, identified through subsequent diagnoses of acute myeloid leukaemia or acute lymphoblastic leukaemia in the Swedish Cancer Register (ICD-O-3 codes C920 and C910, respectively). Complete data on TFR were only available for patients diagnosed between 2007 and 2012 [65].

Using each patient's unique personal identity number, records were linked to multiple national registers: the Cause of Death Register (to obtain vital status and dates of death); the Total Population Register (migration status); the Prescribed Drug Register (TKI prescription data); and the National Patient Register (to identify allogeneic stem cell transplant recipients through procedure codes). Data from both the Swedish CML Register and Prescribed Drug Register were used to ensure comprehensive records of TKI treatment.

Figure 7.2 summarises the study population for Study III, illustrating the inclusion period for CP-CML diagnoses (2007–2017) and the follow-up until the end of 2018.

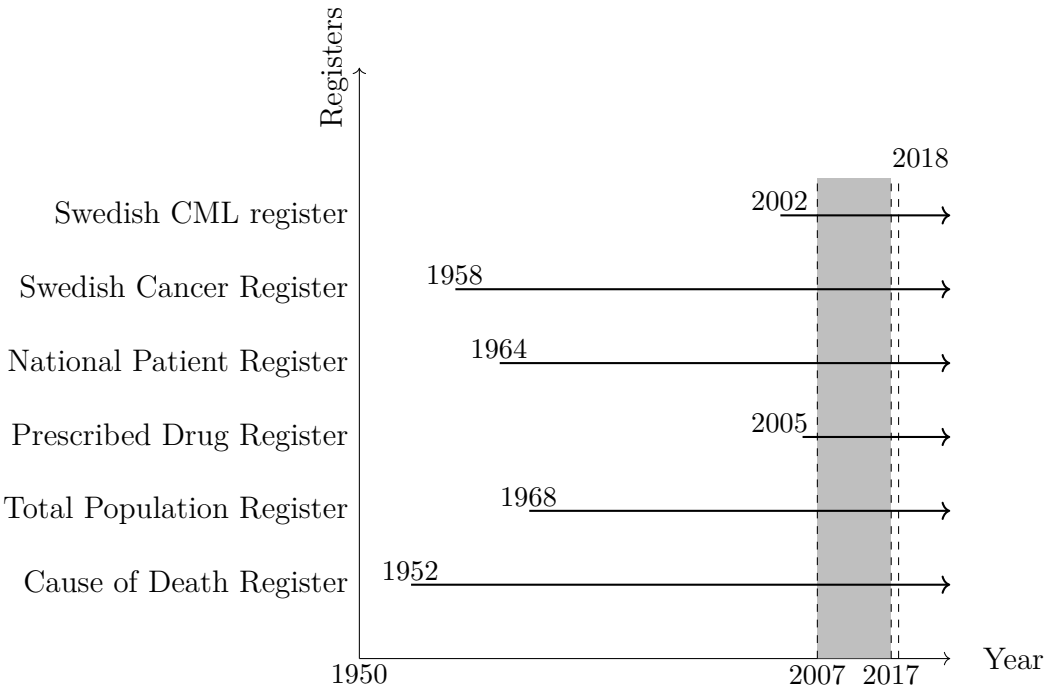


Figure 7.2. Study population for Study III. The shaded region (2007–2017) represents the inclusion period for patients with chronic-phase chronic myeloid leukaemia diagnoses in Sweden, with follow-up until 2018.

Methods

Natural history model for chronic myeloid leukaemia treatment. We applied the extended multistate model developed under Study II to the natural history model for CML treatment introduced in Section 6.2.2, where the death state was partitioned into excess and expected deaths to allow for relative survival extrapolation. This multistate natural history model was employed to estimate both LE and QALE.

LLE was calculated as the difference between the expected LE, based on Swedish population mortality rates matched by age, sex, and calendar year, and LE estimated with our multistate model. For LQALE estimation, we also used HSUVs from CML patients (Foulon et al. [66], Szabo et al. [81]) and Swedish population norms (Teni et al. [82]) to derive quality-adjusted survival.

Main results

Baseline characteristics of the patient cohort. We analysed 991 adults diagnosed with CP-CML in Sweden between 2007 and 2017 (median age 61 years; 44% women) with follow-up until 2018. During the follow-up period, 19.6% died, 4.5% progressed to AP/BP, and 4.9% received allSCT. Imatinib was the 1L TKI for 85.4% of patients. TFR records were available only for patients diagnosed between 2007 and 2012: among the 124 patients who achieved TFR, 68 (55%) subsequently re-initiated TKI therapy.

Loss in life expectancy and loss in quality-adjusted life expectancy. Using 65-year-old women as an illustrative example, patients with CP-CML experienced a LLE of 3.3 years and a LQALE of 6.6 QALYs compared to age-, sex-, and calendar-year-matched individuals without CP-CML from the general population. Across ages 45–85 years, both LLE and LQALE declined with increasing age, with LQALE consistently exceeding LLE across all age groups. Losses were slightly greater in women than men at younger ages. The largest loss was observed in 45-year-old women (LLE = 5.7 years; LQALE = 12.0 QALYs). Proportions of LLE and LQALE remained stable across age groups, with PLLE ranging from 9–15% and PLQALE ranging from 29–33% across all age–sex strata.

Discussion

Survival extrapolation. In Study III, we demonstrated the extended multistate framework with relative survival extrapolation to model the natural history of CML treatment using Swedish register data. This illustrated how excess and expected mortality can be partitioned in each health state to estimate LE, QALE, and their respective losses. Integrating population mortality into each state enabled the model to avoid implausible LE projections, particularly in states where background mortality may explain a great proportion of the long-term all-cause transition-specific hazards. An important methodological question is how this multistate formulation compares to simpler models. In particular, whether a two-state model (that is, standard survival analysis) that estimates QALYs by multiplying the survival function with longitudinally measured HSUVs could yield more accurate QALY estimates [98] than a multistate model. This remains an area for further investigation.

Uncertainty. Stochastic uncertainty was addressed through individual-level simulations to reflect random variation in disease trajectories. Parameter uncertainty was via Monte-

Carlo simulations to estimate 95% CI for LE, and QALE. Heterogeneity was considered, as the results were presented by age and sex strata. Overall, the main methodological contribution lies in reducing structural uncertainty by developing and applying a multistate model that incorporates relative survival extrapolation for CML treatment using Swedish registers.

Generalisability The generalisability of these results is limited by changes in clinical practice after the end of follow-up. Several newer TKIs became therapies for CML after 2018 [112, 113], so our projections may not fully reflect outcomes under current guidelines. In addition, the multistate model was specifically developed to align with the structure of the Swedish CML register and established multistate modelling practices, utilising HSUVs derived from Foulon et al. [66], Szabo et al. [81], and Swedish population norms [82]. These results are most generalisable to countries with similarly structured population registers and comparable treatment patterns, particularly where data infrastructure supports the application of the developed multistate model. In settings where access to these drugs is limited or where different treatment practices are common, researchers would need to adapt both the multistate model and the input parameters, such as HSUVs, before applying our findings.

Conclusions

Patients with CP-CML in Sweden experienced only a modestly low LLE but a considerably larger LQALE. This highlights the need for improved CML management, including the development of more effective or safer TKIs, improvement of predictive models for treatment, and continued lifelong monitoring to evaluate impacts on survival and quality of life.

7.4 Study IV

Title

Empirical and Projected Economic Burden of Chronic Myeloid Leukemia in Sweden from 2015 to 2030: a Population-Based Study

Knowledge gap

To estimate and project the economic burden of CML, defined here as prevalence costs, it is essential to understand how changes in disease trends, treatment patterns, and healthcare utilisation influence costs. While researchers have examined CML costs [114–116], several key limitations existed in the previous studies. First, these studies did not use whole-population data that follow patients across their full disease course. Second, they did not consider temporal changes in treatment, such as the use of generic TKIs, the practice of TFR, and the introduction of newer-generation TKIs. Third, there was a lack of methods that combine real-world registry data with suitable statistical models to forecast future costs in evolving clinical practice. These gaps highlight the need for a more comprehensive analysis using

population-based data and suitable statistical models to estimate and predict the prevalence cost of CML.

Aim

We aim to estimate and project the prevalence costs of CML in Sweden from 2015 to 2030 by applying the natural history model developed in Study III.

Study populations

Cancer incidence data (1973–2019) and relative survival estimates were obtained and estimated from the Swedish Cancer Register and Cause of Death Register. These inputs were integrated into a PIAMOD approach [117] to project CML prevalence for the period 2015–2030.

Treatment patterns for CML were primarily based on the report by Ohm et al. [83], supplemented by input from haematology experts. Costs were derived from multiple sources: individual-level TKI prices from the Prescribed Drug Register for patients diagnosed 2007–2017, aggregate price data from the TLV's Price and Decision Database for 2019–2025, and procedure costs from official healthcare price lists [86, 87] as well as cost estimates from Hernlund et al. [84]

Figure 7.3 summarises the study population for Study IV, showing the inclusion period for CML diagnoses (1973–2019), follow-up through 2020, and the estimation and projection period for prevalence and prevalence costs from 2015 to 2030. Table 7.2 provides an overview of the cost data sources and the original study periods they cover.

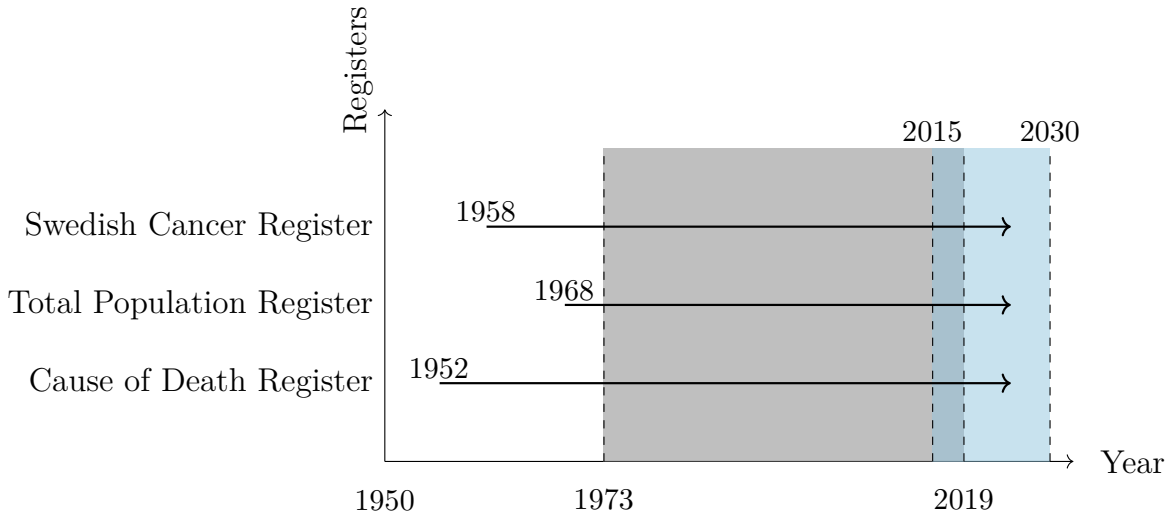


Figure 7.3. Study population for Study IV. The grey-shaded region (1973–2019) represents the inclusion period for patients with CML diagnoses, with follow-up until 2020. The blue-shaded region (2015–2030) indicates the estimation and projection period for prevalence and prevalence costs.

Table 7.2. Cost data sources and time periods for Study IV.

Item	Data source	Time period
TKI costs (individual-level)	Swedish CML register, National Prescribed Drug Register	Diagnosed 2007–2017 with follow-up until 2018
TKI costs (aggregate-level)	TLV Price and Decision Database [85]	2019–2025 (projected constant after)
Costs for alloSCT and progression	Hernlund et al. [84]	Diagnosed 2007–2015
Costs for laboratory testing	National Price List for Laboratory Medical Services [86]	2024 (projected constant before and after)
	Extra-regional Pricing List for EU/EEA and Switzerland [87]	2015–2025 (projected constant after)

Methods

In Study IV, the multistate model from Study III (see Section 6.2.2) was used to project the distribution of CML patients at disease states over time. By combining these state-occupancy probability estimates with estimated CML prevalence, we predicted the expected numbers of prevalent cases in each disease state by year, which were further used to estimate and project the prevalence costs of CML in Sweden from 2015 to 2030. We employed the PIAMOD method [117] to estimate and project CML prevalence throughout this period.

We calculated the annual cost per patient for each disease state using data from multiple sources described in Section 4. The analysis adopted a healthcare sector perspective [51] for cost estimation, focusing on direct medical expenditures on the healthcare system. All costs were inflated to 2024 SEK (Swedish krona) using the Consumer Price Index [118] and converted to USD (U.S. dollar) using the 2024 exchange rate (1 USD=10.56 SEK) [119].

The prevalence cost for each calendar year was obtained by multiplying the estimated number of prevalent patients in each disease state by the corresponding annual cost per patient, then summing across all states according to Equation (6.19). This calculation therefore reflects yearly changes in both CML prevalence and prevalence costs.

Main results

Incidence and prevalence. CML incidence was estimated to be similar between 2015 and 2030 (around 1.0–1.3 per 100,000 person-years), but the number of prevalent CML patients increased due to increasing survival over time. Under the fixed-survival scenario, prevalent cases increased from 1 205.4 (95% CI, 1 014.3–1 396.5) in 2015 to 2 120.3 (95% CI, 1 916.1–2 324.6) in 2030. Using the dynamic-survival scenario, the numbers were slightly higher, estimated to be 2 211.9 (95% CI, 2 007.7–2 416.2) cases by 2030.

Distribution of patients across disease states. The increase in the number of prevalent patients was mainly due to larger groups of patients receiving first- and second-line TKIs. Meanwhile, the numbers of patients experiencing progression, achieving TFR, or undergoing alloSCT remained relatively stable over time.

Cost projections. Despite prevalence nearly doubling from 2015 to 2030, the prevalence costs reduced from USD 40.04 million (95% CI, 33.70–46.40) in 2015 to USD 26.04 million (95% CI,23.13–28.97) in 2025, with projections of USD 30.67 million (95% CI, 27.70–33.64) by 2030. This decline was largely driven by reduced TKI prices once generics entered the market. Average 1L TKI costs dropped dramatically from USD 35 244 per patient-year in 2015 to USD 8 063 by 2025, while second- and third-line costs fell by similar levels over the same period. First-year alloSCT remained the most expensive treatment (over USD 150 000 per patient per year) but affected fewer than ten patients annually. TFR was consistently the lowest costs, approximately USD 6 000 to 8 000 per patient per year.

Sensitivity analysis. We explored the impact of alternative assumptions, such as percentage changes in TKI prices at different time points, through a series of one-way sensitivity analyses. Switching from the fixed-survival to the dynamic-survival scenario increased total costs by only USD 0.43 million in 2025 and USD 0.57 million in 2030. In contrast, various assumptions on drug prices were more influential. A 25% increase in 3L TKI prices (reflecting potential uptake of asciminib) increased 2025 costs up by USD 1.62 million, while a 25% price cut lowered the costs by the same amount. When we increased first-, second-, and third-line TKI prices by ±25% in 2029, it shifted 2030 costs by USD ±1.46, ±2.06, and ±2.00 million, respectively. An interactive R Shiny application was developed to reproduce and extend the sensitivity analyses, available at https://enochytchen.shinyapps.io/CMLEcoBurdenSE/.

Discussion

Survival extrapolation We applied the multistate model developed in Study III to project the distribution of prevalent CML cases across disease states in Sweden. The interplay between stable incidence, improved survival, evolving treatment strategies, and changing drug prices makes forecasting the future prevalence costs of CML inherently complex. Our projections showed a steady increase in the number of individuals living with CML between 2015 and 2030 under both fixed and dynamic survival scenarios. The dynamic scenario employed a flexible parametric relative survival model, representing a novel implementation of net survival projection within the PIAMOD framework. Despite increasing CML prevalence, total prevalence costs were projected to decline from USD 40.04 million (95% CI: 33.70–46.40) in 2015 to USD 30.67 million (95% CI: 27.70–33.64) in 2030. This trend was primarily driven by reductions in the average annual cost of TKI therapy, demonstrating that improved survival does not necessarily translate into increased prevalence costs.

Uncertainty. We addressed structural uncertainty by applying two survival scenarios: fixed and dynamic, based on the Pohar-Perme estimator [30] and flexible parametric relative survival models [120], respectively. This allowed us to explore how different assumptions about future survival affect prevalence and cost projections. Incorporating background mortality into the multistate model may reduce structural uncertainty and improve accuracy in estimating prevalent CML patient distributions across states over time, compared to extrapolation within an ASF that does not consider background mortality.

Parameter uncertainty was partly addressed by presenting 95% CI around our estimates of incidence, prevalence, and costs. However, for cost estimates, we treated the projected distribution of disease states as fixed inputs and did not account for the uncertainty around those estimates when calculating total costs. Instead, we conducted one-way sensitivity analyses and developed an interactive R Shiny application, which allows users to change inputs to evaluate how projected prevalent cases and total prevalence costs may change accordingly.

Stochastic uncertainty was considered through Monte-Carlo simulations from the multistate model. We accounted for heterogeneity by stratifying survival and prevalence models by age and sex. However, we presented the prevalence costs by disease state, since the primary aim was estimating and predicting total direct health expenditures for CML.

Generalisability Several methodological limitations constrain the generalisability of the cost estimates from our study. Individual-level price data were available only up to 2018. To capture later price changes, we adjusted costs for 2019–2025 using relative price ratios from the TLV's Price and Decision Database [85], where 2018 was the baseline. Additionally, the cost estimates for patients treated with TKIs included only direct healthcare expenditures and did not capture the potential costs associated with adverse events, such as cardiovascular complications from nilotinib use [121], or costs related to hospitalisation [122]. These limitations suggest that the cost estimates may not fully capture current treatment costs, particularly given the introduction of newer, potentially more expensive therapies and evolving clinical practices since 2018. The analysis adopted a healthcare sector perspective, thereby excluding informal care costs, productivity losses, and other indirect costs [51]. Consequently, the findings are most relevant to decision-makers who adopt the same payer perspective, such as national health services or insurance providers focused on direct medical costs. Healthcare systems considering a broader societal perspective would need to incorporate additional cost components, such as productivity loss, beyond those included in our study.

Conclusions

The number of individuals living with CML in Sweden is expected to continue to rise, but declining treatment costs have led to an overall reduction in prevalence costs and will likely mitigate the economic burden on the Swedish healthcare system.

Chapter 8 Overall summary and conclusions

In this thesis, I present a methodological advancement in survival extrapolation. The work begins with an evaluation of survival extrapolations using SPMs and FPMs, applied within both ASF and RSF frameworks. These models are assessed using population-based cancer registry data with known survival outcomes, providing a solid foundation for comparison. An innovation of this thesis is the integration of relative survival extrapolation into a multistate modelling framework. Critical challenges in HTA using trial data and population-based studies using registry data are examined. The developed approaches may improve predictions by combining short-term observed data with long-term external data sources, such as general population mortality data, enabling more credible survival projections under a lifetime horizon.

The methodological contributions show clear findings. For survival extrapolation to 10 years, FPMs in general provided more accurate survival estimates than SPMs, with minimal difference between ASF and RSF models at this time horizon. However, when extrapolating survival over a lifetime horizon, survival extrapolations within an RSF demonstrated superior accuracy compared to an ASF given sufficient follow-up, particularly when using FPMs. Notably, with limited follow-up, ASF models tended to overestimate survival, whereas RSF models showed more underestimates.

The integration of relative survival extrapolation into multistate models is a methodological breakthrough. This framework improves the credibility of long-term survival projections and leverages general population mortality data to enhance the plausibility of ICER calculations when extrapolating survival from short-term trial data to a lifetime horizon.

The application of these methods to CML in Sweden provides real-world clinical insights, and the results (Studies III and IV) serve as important inputs for future CEA of CML in Sweden. The analysis showed that there is currently a modestly low LLE but a more considerable LQALE among patients with CML.

From a health economic perspective, this work also presents information which may support healthcare planning and policy decisions for CML in Sweden. While the prevalence of CML is projected to increase, declining treatment costs will likely help mitigate the economic burden on the Swedish healthcare system. The 15-year cost projections (2015 to 2030) offer quantitative economic evaluations for optimising CML care management in the future.

In conclusion, this thesis provides a comprehensive evaluation of survival extrapolation, and pushes the multistate framework forward to improve long-term survival extrapolation with applications in HTA. By applying these models to the CML studies, this work shows the value in applying advanced statistical modelling into practical evidence for both clinical and health economic decisions. This thesis thereby contributes to the fields of biostatistics, epidemiology, health economics, and medicine.

Chapter 9 Future perspectives

This thesis has explored and applied methods for extrapolating survival with applications to HTA and other modelling in HEOR. Several important directions emerge for future research as follows:

Methodological innovation. Promising research areas related to developing and applying novel methodological approaches have the potential to improve the accuracy of survival extrapolation, including:

- Alternative approaches for model evaluation could improve the assessment of survival extrapolation. These include calculating the log-likelihood for out-of-sample predictions versus observed data, and in simulation studies the Kullback–Leibler divergence [123] can be used to quantify differences between the true and estimated survival distributions.
- Bayesian approaches, which offer the advantage of incorporating prior knowledge and uncertainty quantification.
- Machine learning methods, which may have the potential to predict complex patterns in survival data that traditional parametric models do not have.
- Large language models, which are beginning to have more applications in survival analysis and HTA.

Multistate model. Within a multistate modelling framework, key questions remain regarding which transitions can be appropriately modelled and whether it is justifiable to assume specific hazard shapes for different types of transitions. Current relative survival extrapolation methods utilise long-term general population mortality rates to predict all-cause hazards over extended time horizons. These mortality rates are readily available through databases such as the HMD [27]. However, when the event of interest is not death, applying relative survival extrapolation becomes considerably more challenging, as background rates for non-mortality events are not as easily accessible as life tables. One potential solution is to construct background rates using population registers [103], decompose the transition-specific hazards, and model them separately. However, this approach requires further investigation before it can be applied reliably for survival extrapolation.

Use of population-based registers. Population-based registers are valuable resources for understanding disease dynamics in a population and have potential for informing HTA, particularly due to their longer follow-up periods. Longer follow-up can enhance the accuracy of predictions for survival, mortality, incidence, and prevalence; even a few additional months of data can meaningfully improve model performance on survival extrapolation. To maximise the utility of registers, three key areas require continued attention: (i) ensuring high data

quality, (ii) maintaining timely and regular updates, and (iii) facilitating efficient data delivery processes. Optimising the procedures for cleaning, linking, and delivering register data would improve both the quality and timeliness of analyses. For example, in COI studies, access to more recent follow-up data could reduce the need to rely on assumptions about future prevalence and costs, thereby strengthening the estimates of future economic burden.

Measuring health-related quality of life. Research related to measuring HRQoL, such as identifying ways to improve or streamline routine HRQoL data collections, would have valuable implications for using register data in HTA.

While LLE serves as an important measure for monitoring health service effectiveness, focusing on quantity of life provides only part of the picture for population health. From a public health perspective, the goal is to maximise population health benefits, which should consider not only longevity but also quality of life. LQALE represents an important extension beyond quantitative measures, offering a more comprehensive assessment of health outcomes incorporating both the quantity and quality of life lost.

Good statistical practice. A common criticism of health economic models is that they operate as "black boxes" (models where the inputs and outputs are known, but the internal structure, assumptions, and calculations remain hidden or unexplained) [124]. However, this critique is not unique to complex or code-based models; it applies to any model that lacks adequate documentation and transparency. Even health economic models developed in `Microsoft Excel` can be difficult to interpret without clear reporting. The core issue lies in promoting good scientific practices that ensure analyses are repeatable, replicable, and reproducible [125]. Guidance documents by HTA regulators and future research should emphasise that, while data may remain confidential, the methods and analytical code should be well-documented and openly accessible (or at least available upon request). This approach would help address the "black box" concern in modelling and demonstrate that scientific rigour is compatible with the transition from spreadsheet-based softwares (such as `Microsoft Excel` and `TreeAge Software`) to code-based softwares (such as `R`, `Stata`, `C++`, and `Python`).

Other statistical practice. The PIAMOD framework requires updating, particularly in how it selects the degrees of freedom for the APC model used to estimate incidence (currently using likelihood ratio statistics [117], which is computationally inefficient). Optimising this step could lead to more accurate prevalence estimates and, in turn, improve the estimation of total healthcare expenditures for the disease of interest.

Future work could also emphasise developing user-friendly visualisation tools, such as `R Shiny` applications, to improve communication of results and allow decision-makers or other stakeholders to conduct their own uncertainty analyses. These interactive tools can bridge the gap between complex statistical analyses and practical decision-making needs.

Research integration and knowledge transfer. HTA operates within a triangular ecosystem consisting of academia, pharmaceutical companies/consultancies, and regulatory bodies. While advanced statistical methods are typically developed in academic settings (though increasingly also within research departments of private sectors and regulatory agencies), there is often a time lag before these methods are adopted by industry and regulators. Future efforts should focus on maintaining methodological standards and facilitating knowledge transfer through: (i) developing guidance documents, (ii) providing targeted training programs, (iii) fostering collaborative partnerships (while avoiding conflicts of interest), and (iv) providing user-friendly software to support implementations of new methods.

Final words. Supporting an effective healthcare system requires reliable scientifically generated evidence to guide clinical and policy decisions. To contribute to that goal, in this thesis I evaluate, advance, and apply methods that aim to improve the accuracy of survival extrapolation, reduce unrealistic estimates, estimate patients' LLE and LQALE, and demonstrate the application of these methods in an economic context. Ongoing efforts to utilise such multifaceted approaches that consider the interplay between survival, quality of life, and health economics are essential for healthcare systems to sustainably maximise patient care–both today and for future generations.

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[3] Henry Adams. *The Education of Henry Adams.* 1907.
[4] Antoni Gaudí (1852–1926).

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Declaration About the Use of Generative AI

In writing this comprehensive summary, I made limited use of generative AI tools, including `ChatGPT o3` and `ChatGPT 4o` (OpenAI), and `Claude Sonnet 4` (Anthropic), to support the refinement of academic English and improve clarity. These tools were used primarily to explore alternative phrasings and improve complex sentences. All outputs were carefully evaluated, revised, and adapted to align with my own understanding. I take full responsibility for the final content of this thesis and its accuracy.

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